The Teichmüller tower: arithmetic images, normalizers, and stratified completions

Development of *La Longue Marche*, §§19, 21, 26–30, 32–36

AI-produced by GPT-6 Astra (OpenAI, ChatGPT Pro mode)

11 November 2026. No priority is claimed.

Notation fixed before any comparison. We write n for the manuscript's ν. A type is hyperbolic if $2g-2+n>0$. All profinite homomorphisms are continuous; quotients of profinite groups are by closed normal subgroups. An outer homomorphism to H is a homomorphism modulo conjugation by H, unless a smaller conjugating group is explicitly specified. An identification obtained from a geometric base point is invariant only up to the indicated inner conjugations. We never infer injectivity of a profinite completion map from injectivity of the discrete map.

Our notation	Manuscript notation and scan	Meaning
$A_{g,n}$, $A^{!}_{g,n}$	$A_{g,\nu}$, $A^{!}_{g,\nu}$; 204–205	Homeomorphisms preserving the puncture set, respectively each puncture.
$M^{\pm}_{g,n}$, $M^{+}_{g,n}$	$\Gamma_{g,\nu}$, $\Gamma^{+}_{g,\nu}$, later $\mathcal{T}$, $\mathcal{T}^{+}$; 205, 226, 328	Extended mapping class group; orientation-preserving subgroup; punctures may be permuted.
$P_{g,n}$	$\Gamma^{!+}_{g,\nu}$; 205, 226	Pure orientation-preserving mapping class group. Part F uses $G_n = P_{0,n}$.
$\Pi_{g,n}$, $\pi_{g,n}$	$\Pi_{g,\nu}$, $\widehat{\Pi}_{g,\nu}$; 221, 328	Discrete surface fundamental group and its full profinite completion, with peripheral data.
$O_{g,n}$	$\widehat{\widehat{\mathcal{T}}}$; 321–324	$\mathrm{Out}_{\mathrm{lac}}(\pi_{g,n})$, not a second application of completion.
$\Sigma_{g,n}$	$\widehat{\mathcal{T}}^{+} = \Sigma$; 323–324	Geometric image in $O_{g,n}$; distinguished from the abstract completion $\widehat{M^{+}_{g,n}}$.
N^{norm}, $\mathfrak{G}^{\mathrm{norm}}$	$\mathcal{N}_{\Sigma}$, $\mathbf{\Gamma}_{\Sigma}$; 324–326	$N_O(\Sigma)$ and $N_O(\Sigma)/\Sigma$, §26 version.
N^{arith}, $\mathfrak{G}^{\mathrm{arith}}$	$N_{g,\nu}$, $\Gamma_{g,\nu}$; 370–371	Arithmetic monodromy image and its quotient by Σ, §29 version.
$\mathscr{M}_{g,[n]}$	$M_{g,\nu}$; 326–329	Moduli stack of curves with an unordered reduced degree-n puncture divisor. $\mathscr{M}_{g,n}$ has labelled punctures.
C_n, $T_n = C_n^{\mathrm{op}}$	No manuscript symbol; STRATA F.7	Exit category, respectively degeneration category. Completion retains these variances.
G_K, χ	$\mathbb{I}\Gamma_K$, χ; 327, 323	$\mathrm{Gal}(\overline{K}/K)$ and the cyclotomic multiplier.

Later sections may use locally declared synonymous letters when quoting a source. The labels **Known**, **Proved**, **Refuted**, and **Open** classify the exact numbered assertion. "Open" also marks an exact assertion left unverified by this audit; we distinguish that from an established open problem in its accompanying text. A conditional implication can be **Proved** while its antecedent remains an unrefereed claim. "Verified" in the bibliography means that the cited source was opened, not that its entire proof was independently certified.

Abstract

We distinguish the normalizer quotients introduced in §26 from the arithmetic-image quotients introduced in §29. The distinction changes the mathematical content of the conjecture fondamentale: faithfulness modulo geometric monodromy identifies the latter quotients with the absolute Galois group, whereas a description of all compatible profinite tower automorphisms by Galois automorphisms is a separate problem. We give typed comparison maps, audit the claims concerning change of type and reconstruction, and construct a profinite category retaining the boundary groups and their crossing bimodules. We distinguish verified theorems, proofs given here, counterexamples, and unresolved statements, including the exact dependence on the attached unrefereed manuscripts of 2026. A statement register, an editorial reading register, reproducible computations, and a verification report accompany this paper.

Contents

1 A. The tower and the manuscript's changing definitions **4**
1.1 A.1. Surfaces, punctures, and the two finite characters . . . 4
1.2 A.2. Teichmüller space and the modular topos . . . 6
1.3 A.3. Completion, discretification, and arithmetization . . . 7
1.4 A.4. Exceptional types and editorial limits . . . 9
1.5 A.5. The universal lift in §21 . . . 10

2 Congruence, normalizers, and discretizations **12**
2.1 The congruence problem and the size of the ambient group . . . 12
2.2 Actual discretizations under a finite covering . . . 15
2.3 Passing to prediscretizations and the remaining comparisons . . . 18

3 Filling punctures and reconstructing the complete curve **20**
3.1 The quotient and the two geometric exact sequences . . . 20
3.2 The arithmetic image and the revised quotient . . . 22
3.3 What the sequence marked "??" requires . . . 23
3.4 What it means to reconstruct the entire étale topos . . . 25

4 Arithmetic curves, inertia, sections, and faithfulness **28**
4.1 What the arithmetic extension determines . . . 28
4.2 The section manuscript: an audited conditional chain . . . 31
4.3 Relative and absolute open homomorphisms . . . 35
4.4 Centres, restriction to covers, and the faithfulness questions . . . 37
4.5 Small types and the exact remaining qualifications . . . 40

5 The fundamental conjecture: three different comparison problems **44**
5.1 The change of definition in §29 . . . 44
5.2 The genus-zero comparison, including the isolated tripodal case . . . 45
5.3 The two elliptic types and the meaning of an all-genus tower . . . 48
5.4 Arithmetic reconstruction uses more morphisms . . . 50
5.5 The rational comparison and the maps to the pro-ℓ tower . . . 51

5.6 Audit of the new all-weight upper bound 55
5.7 Consequences for the requested categorical formulation 57

6 F. One stratified object, its completion, and its automorphisms 58
6.1 F.1. Which category is being completed 58
6.2 F.2. Transporters, closed stabilizers, and an explicit model 60
6.3 F.3. Geometric comparison and the Galois action 62
6.4 F.4. Boundary-category automorphisms and the two levels 64
6.5 F.5. Positive genus, isotropy, and the rational comparison 66

A Verification protocol and limits of the deliverable 69
A.1 The three attached 2026 proof families 69
A.2 Computations and proof boundaries 69
A.3 What remains unresolved 70

1 A. The tower and the manuscript's changing definitions

1.1 A.1. Surfaces, punctures, and the two finite characters

Statement A.1 (Proved). The permutation and orientation conventions.
Let X_g be a closed connected orientable topological surface and $S_n \subset X_g$ a specified n-element set. Give its homeomorphism group the compact-open topology on a homeomorphism and its inverse, and put $U_{g,n} = X_g \setminus S_n$. Then

$$1 \longrightarrow A^!_{g,n} \longrightarrow A_{g,n} \longrightarrow \mathfrak{S}_n \longrightarrow 1, \qquad 1 \longrightarrow \pi_0(A^!_{g,n}) \longrightarrow M^{\pm}_{g,n} \longrightarrow \mathfrak{S}_n \longrightarrow 1.$$

The orientation character $\epsilon : M^{\pm}_{g,n} \to \{\pm 1\}$ is surjective and its kernel is $M^{+}_{g,n}$. The intersection of the permutation and orientation kernels is $P_{g,n}$. In particular the manuscript's unadorned group is neither pure nor orientation-preserving by definition. *Manuscript:* §19, folio 102r–v, scan 204–205.; the positive superscript is specified in §19, folio 113, scan 226.

Proof. Restriction to the finite set gives the first homomorphism. Every permutation is realized by an orientation-preserving homeomorphism: one realizes transpositions by interchanging two small discs along an embedded arc avoiding the other punctures, and composes. Its kernel is precisely the pointwise stabilizer. A path of homeomorphisms induces a constant permutation, so the identity components of the two groups coincide. Taking component groups therefore gives the second sequence, including surjectivity. Degree on the oriented fundamental class gives ϵ. A reflection model for a genus-g surface, followed by a point-moving homeomorphism, realizes degree -1 preserving the prescribed punctures individually. Thus both signs occur even in the pure group. No splitting of the permutation sequence is asserted.

Every homeomorphism of $U_{g,n}$ extends uniquely to its finite end compactification X_g: the ends are exactly the deleted punctures, and a homeomorphism permutes ends. This proves the abstract isomorphism in manuscript (5). The extension and restriction respect the indicated homeomorphism topologies: a compact core together with one smaller annular neighbourhood of each end controls the image and inverse image of the complementary discs. This provides uniform control on the compactification from compact-open control on the punctured surface. □

Statement A.2 (Known). Surface groups with peripheral structure.
Choose an orientation, a base point $b \in U_{g,n}$, and paths to small positively oriented puncture circles. Then

$$\Pi_{g,n} = \left\langle a_1, b_1, \dots, a_g, b_g, c_1, \dots, c_n \;\middle|\; \prod_{i=1}^{g} [a_i, b_i] \prod_{j=1}^{n} c_j = 1 \right\rangle.$$

For $n > 0$ this is free of rank $2g + n - 1$. For hyperbolic type it is centre-free and residually finite. Its peripheral datum consists of the conjugacy classes of the cyclic subgroups $\langle c_j \rangle$, with the coherent orientation datum; for $n = 0$ we retain the orientation line in $H_2(\Pi_{g,0}, \mathbb{Z})$. The Dehn–Nielsen–Baer identification is

$$M^{\pm}_{g,n} \xrightarrow{\sim} \mathrm{Out}_{\mathrm{lac}}(\Pi_{g,n}).$$

The target preserves the peripheral classes as a set; restricting to fixed classes and multiplier $+1$ gives $P_{g,n}$. *Manuscript:* §19, folio 111–113, scan 221–226.; see Farb and Margalit (2012, Theorems 8.1 and 8.8).

Verification and scope. The presentation follows by cutting into a polygon and applying van Kampen. Eliminating c_n proves the free-rank assertion. For a closed hyperbolic surface the universal covering is the disc and the deck group is torsion-free cocompact Fuchsian; its centre is trivial. Free groups of rank at least two have trivial centre by reduced words. Residual finiteness follows from the faithful finitely generated characteristic-zero linear realizations and reduction modulo finite residue fields. The identification with peripheral-preserving outer automorphisms is exactly the punctured or closed Dehn–Nielsen–Baer theorem cited above; this is an imported theorem, not an assertion that an arbitrary group automorphism preserves omitted punctures. Changing b or paths conjugates the resulting fundamental-group maps. Completing yields $\Pi_{g,n} \hookrightarrow \pi_{g,n}$; it does not yet prove faithfulness of $\widehat{M^{+}_{g,n}} \to \mathrm{Out}(\pi_{g,n})$. □

Statement A.3 (Known). Filling one point and the precise Birman group.
For a hyperbolic target $U_{g,n}$ and one extra point b, let $M^{\pm}_{g,n+1,*}$ be the subgroup fixing that extra point, while allowing the old punctures to be permuted. There are exact sequences of discrete groups

$$1 \longrightarrow \Pi_{g,n} \xrightarrow{\text{push}} M^{\pm}_{g,n+1,*} \xrightarrow{\text{forget}} M^{\pm}_{g,n} \longrightarrow 1, \qquad 1 \longrightarrow \Pi_{g,n} \longrightarrow P_{g,n+1} \longrightarrow P_{g,n} \longrightarrow 1.$$

There is no forgetting homomorphism on the full group that permits the extra point itself to be permuted. On fundamental groups the inclusion of opens instead gives the surjection $\Pi_{g,n+1} \to \Pi_{g,n}$ killing the normal closure of the last peripheral loop. These are different maps between different groups. *Manuscript:* §19, folio 103–111, scan 206–221.; Farb and Margalit (2012, Theorem 4.6).

Source and comparison maps. Evaluation at b is a locally trivial fibration of homeomorphism groups, by isotopy extension. Its homotopy sequence, and contractibility of the identity component in hyperbolic type, give the first exact sequence. The connecting map pushes b along a based loop. Its conjugation action on $\Pi_{g,n}$ is the point-pushing action and its injectivity uses the centre-free hyperbolic hypothesis. Restricting to homeomorphisms fixing the old punctures gives the second sequence. Surjectivity follows by isotoping the image of b back to b through an arc missing the punctures. For the surface-group map, the filled disc attaches along the last boundary circle, so van Kampen gives exactly the indicated normal quotient. Section C distinguishes their profinite analogues and their kernels. □

Statement A.4 (Proved). A covering gives a correspondence with specified lifts.
Let $p : U' \to U$ be a connected finite unramified covering of orientable finite-type surfaces, with U hyperbolic. A choice of points above b identifies $\pi_1(U')$ with a finite-index subgroup $H < \Pi = \pi_1(U)$. Put $L_H = \mathrm{Stab}_{M^{\pm}(U)}([H])$, where $[H]$ denotes conjugacy in Π. The group of isotopy classes of pairs of commuting lifts (f', f) fits into

$$1 \longrightarrow N_{\Pi}(H)/H \longrightarrow \mathrm{Lift}(p) \longrightarrow L_H \longrightarrow 1, \qquad \mathrm{Lift}(p) \longrightarrow M^{\pm}(U').$$

Thus the general covering operation in the tower is a correspondence, not a canonical homomorphism from all of $M^{\pm}(U)$ to $M^{\pm}(U')$. It preserves the sign of orientation and the induced peripheral subgroups. The degree $d = [\Pi : H]$ satisfies $2g' - 2 + n' = d(2g - 2 + n)$. *Manuscript:* §19 and 28, folio 102–111 and 177–180, scan 204–221 and 350–356..

Proof. A base homeomorphism lifts if and only if its outer action preserves $[H]$, by the covering-space lifting criterion. Any two lifts differ by a deck transformation, and the deck group is $N_{\Pi}(H)/H$. Lifting an isotopy gives the asserted exact sequence on isotopy classes. A deck transformation isotopic to the identity on a hyperbolic surface is trivial: it has finite order,

whereas a nontrivial periodic orientation-preserving mapping class cannot be represented by an identity-isotopic map; equivalently use its faithful action on the hyperbolic universal cover and the centre-free surface group. This also verifies that the kernel has not been accidentally collapsed. Forgetting the lower homeomorphism gives the displayed map to the upstairs mapping class group. The restriction of a representative automorphism to H changes, when the lift changes, by the deck action. Even for a characteristic subgroup this is not generally an inner automorphism of H. The correspondence records that ambiguity. Peripheral annuli lift to peripheral annuli, and Euler characteristic multiplies under a finite covering; both assertions follow by lifting a finite cell decomposition. The same subgroup inclusion completes to an open embedding $\widehat{H} \hookrightarrow \widehat{\Pi}$, since a finite-index subgroup inherits the full profinite topology. □

1.2 A.2. Teichmüller space and the modular topos

Statement A.5 (Known). The modular object retains isotropy.

For a stable labelled type, the analytic moduli stack is $[\mathcal{T}_{g,n}/P_{g,n}]$, with $\mathcal{T}_{g,n}$ contractible; the unordered stack is $[\mathcal{T}_{g,n}/M^{+}_{g,n}]$. Consequently its geometric *stack* fundamental group is respectively $\widehat{P}_{g,n}$ or $\widehat{M^{+}_{g,n}}$. The manuscript's $M_{g,n}$ in (19)–(25), scans 326–329, corresponds to the unordered version: (20) explicitly has $\widehat{\mathcal{T}}^{+}_{g,n}$, not $\widehat{\mathcal{T}}^{!+}_{g,n}$. In particular $\mathscr{M}_{0,3} = \operatorname{Spec}\mathbb{Q}$ but $\mathscr{M}_{0,[3]} \simeq B\mathfrak{S}_3$. *Manuscript:* §21 and 26, folio 128–131 and 165–166, scan 255–261 and 326–329..

Verification. The complex structure on an oriented surface gives the usual Teichmüller parameter; quotient by the identity component of the diffeomorphism group is the manuscript's $E_g/A^{\circ}_{g,n}$. Teichmüller contractibility, proved by Fenchel–Nielsen coordinates in Farb and Margalit (2012, Theorem 10.6 and §10.6.3), and the mapping class action give the stated quotient stack, retaining automorphism groups at its points. Analytic Riemann existence for these Deligne–Mumford stacks identifies finite covering spaces with finite étalecovers, hence gives their completed fundamental groups. One can prove this stack version by applying ordinary Riemann existence to an étaleatlas and its twofold and threefold intersections, then descending finite covers. This is the stack extension of the comparison used in Barwick et al. (2020, §13.9). A labelled three-pointed projective line has one isomorphism class and no automorphisms; allowing permutations gives precisely $\mathfrak{S}_3$. A coarse-space quotient would erase this group, so it cannot satisfy the manuscript's sequence (20). The convention in STRATA F.16 is correct only with this unordered/labelled distinction retained explicitly. □

Statement A.6 (Known). Arithmetic fundamental-group and universal-monodromy sequences.

Fix $\overline{\mathbb{Q}} \hookrightarrow \mathbb{C}$ and compatible geometric base points on the unordered moduli stack and its universal punctured curve $\mathscr{U}$. For hyperbolic type,

$$1 \longrightarrow \widehat{M^{+}_{g,n}} \longrightarrow A^{\mathrm{arith}}_{g,n} := \pi_1^{\text{ét}}(\mathscr{M}_{g,[n]}) \longrightarrow G_{\mathbb{Q}} \longrightarrow 1,$$

$$1 \longrightarrow \pi_{g,n} \longrightarrow \pi_1^{\text{ét}}(\mathscr{U}) \longrightarrow A^{\mathrm{arith}}_{g,n} \longrightarrow 1.$$

Conjugation on the normal geometric fibre defines a commutative diagram to

$$1 \longrightarrow \pi_{g,n} \longrightarrow \mathrm{Aut}_{\mathrm{lac}}(\pi_{g,n}) \longrightarrow O_{g,n} \longrightarrow 1.$$

The last sequence uses $Z(\pi_{g,n}) = 1$. The induced map $\rho_{g,n} : A^{\mathrm{arith}}_{g,n} \to O_{g,n}$ is genuine after these choices; its change under base-point transport is conjugation in $O_{g,n}$. Its geometric restriction need not be identified with an injection before the congruence problem is settled. *Manuscript:* §26, folio 165–166, scan 326–329..

Source and exactness. The first sequence is the homotopy sequence of a geometrically connected stack over $\mathbb{Q}$. For the second sequence, the universal punctured curve is smooth with a smooth proper compactification and relative normal-crossings puncture divisor; its geometric fibres have the fixed hyperbolic type. The homotopy sequence is exact and the geometric fibre embeds; analytically it is the Birman sequence, and the profinite fibre injection is the assertion justified in Section C. Passing to an étaleatlas and descent gives the same sequence for the unordered stack. Normality makes conjugation a homomorphism into automorphisms, and centre-freeness identifies inner automorphisms with the fibre. Images of peripheral loops are determined by the puncture divisor, with the cyclotomic multiplier; this proves that the maps land in the indicated groups. Restricting to the algebraic closure recovers the completed topological monodromy. The labelled sequences are displayed explicitly in Hoshi and Mochizuki (2011, Corollary 6.4); the finite permutation quotient gives the unordered versions just proved. These are precisely (19)–(25) with all base points and orientation conventions stated. □

1.3 A.3. Completion, discretification, and arithmetization

Statement A.7 (Proved). A completion is not its image.
The discrete action induces a continuous surjection

$$\widehat{M^+_{g,n}} \twoheadrightarrow \Sigma_{g,n} := \overline{\rho_{g,n}(M^+_{g,n})} \subset O_{g,n}.$$

The kernel is the congruence kernel. The double hat in the manuscript denotes the ambient automorphism group $O_{g,n}$, not the profinite completion of $\widehat{M^+_{g,n}}$. In all formulae for stabilizers or normalizers inside $O_{g,n}$, it is the image Σ that can be used without the congruence hypothesis. *Manuscript:* §26, folio 162r–163v, scan 320–323..

Proof. The group $\pi_{g,n}$ is topologically finitely generated. Characteristic open subgroups are cofinal among its open normal subgroups, so $\mathrm{Aut}(\pi_{g,n})$ and $\mathrm{Out}(\pi_{g,n})$ have their usual profinite topologies. The discrete mapping class action is continuous for the topology pulled back from the finite outer automorphism groups, hence extends from its full profinite completion. Compactness makes its image closed, and density identifies that image with the closure of the discrete action. The universal property alone says nothing about the kernel. The manuscript explicitly writes on scan 320 that the natural surjection is being assumed injective. Section B supplies the exact known ranges and implications. □

Statement A.8 (Proved). The quotient sets behind the terminology.
Let π be a profinite group with the loop structure of $\widehat{\Pi}_{g,n}$. Write $I(\pi) = \mathrm{Isom}_{\mathrm{lac}}(\widehat{\Pi}_{g,n}, \pi)$ and $I^{\mathrm{out}}(\pi) = I(\pi)/\operatorname{Inn}(\pi)$. Actual discretifications are the images of $\Pi_{g,n}$ under elements of $I(\pi)$, with their transported loop structures. Their based parameter set is

$$\mathrm{Discr}(\pi) = I(\pi)/\operatorname{Aut}_{\mathrm{lac}}(\Pi_{g,n}).$$

After taking inner-conjugacy classes, the denominator is $M^{\pm}_{g,n}$. Pre-discretifications replace this discrete denominator by its closure. If $\Sigma^{\pm} = \overline{M^{\pm}_{g,n}} \subset O_{g,n}$, the outer quotient sets are

$$\mathrm{Pre}(\pi) = I^{\mathrm{out}}(\pi)/\Sigma^{\pm}, \qquad \mathrm{Pre}^{+}(\pi) = I^{\mathrm{out}}(\pi)/\Sigma.$$

The maps from discretifications to pre-discretifications are quotient maps, not identifications of dense subgroups. Oriented pre-discretifications map two-to-one to unoriented ones for hyperbolic type. *Manuscript:* §26, folio 162–163, scan 320–323.; §29, folios 188–189, scans 371–374.

Proof. Two identifications give the same embedded discrete group and transported structure exactly when their quotient restricts to an automorphism of the standard discrete group. An automorphism of the discrete group extends uniquely to its completion, which proves the first orbit description. Taking target inner conjugacy gives the outer description. The closure of the based discrete automorphism group contains $\mathrm{Inn}(\pi)$, because the discrete surface group is dense in π; thus the pre-discretification quotient is already outer. The sign character on the discrete mapping class group has image $\{\pm 1\}$, and remains so on its closure. Its positive subgroup is closed of index two. This proves the last assertion without imposing the congruence property. It also explains the harmless change of notation from classes of discretifications in §26 to pre-discretifications in §29. □

Statement A.9 (Proved). Two different groups called N, and two different conjectures.

For a chosen geometric subgroup $\Sigma \subset O$, define

$$N^{\mathrm{norm}} = N_O(\Sigma), \qquad \mathfrak{G}^{\mathrm{norm}} = N^{\mathrm{norm}}/\Sigma.$$

These are the §26 definitions. In §29 the manuscript defines instead

$$N^{\mathrm{arith}} = \rho(A^{\mathrm{arith}}), \qquad \mathfrak{G}^{\mathrm{arith}} = N^{\mathrm{arith}}/\Sigma.$$

There is an inclusion $N^{\mathrm{arith}} \subseteq N^{\mathrm{norm}}$, and a canonical continuous surjection $G_{\mathbb{Q}} \twoheadrightarrow \mathfrak{G}^{\mathrm{arith}}$ once the algebraic closure and the arithmetic marking have been fixed. The statement that this surjection is injective does not assert that $N^{\mathrm{arith}} = N^{\mathrm{norm}}$. *Manuscript:* §26 and 29, folio 164–165 and 188, scan 324–326 and 370–371..

Proof. The geometric subgroup of the arithmetic fundamental group is normal. Taking its image under ρ proves the inclusion in the normalizer. The image is compact and therefore closed. Quotienting the surjective map from A^{arith} to its image by the geometric subgroup gives the displayed surjection. No element outside the image is produced by this quotient operation. Hence replacing the arithmetic image by the full normalizer is an additional assertion, and changes an injectivity problem into a larger realization problem. Sections D and E settle the arithmetic injectivity and retain the other question separately. □

The dictionary now has an economical form:

Manuscript object	Precise modern object	Qualification
$\widehat{\mathcal{T}}^{+}$	$\widehat{M^{+}}$ or its image Σ	These agree exactly when the relevant congruence kernel vanishes.
$\widehat{\widehat{\mathcal{T}}}$	$\mathrm{Out}_{\mathrm{lac}}(\pi)$	Full continuous loop-preserving outer automorphisms.
Discrét	Dense discrete surface models	Actual models and their inner-conjugacy classes are distinguished.
Prédiscrét^{+}	I^{out}/Σ	A homogeneous space, not a group without additional normality.
Strict geometric structure	Conjugate of Σ	Orbit parametrized by O/N^{norm}.
Arithmetization a	Class in $I^{\mathrm{out}}/N^{\mathrm{arith}}$	Its oriented pre-discretifications form a $\mathfrak{G}^{\mathrm{arith}}$-torsor.
Arithmetic Teichmüller group	$\pi_1^{\text{ét}}(\mathscr{M}_{g,[n]})$	Its image N^{arith} can be a congruence quotient.

Statement A.10 (Proved). The arithmetization fibre is a torsor.
The map $I^{\mathrm{out}}/\Sigma \to I^{\mathrm{out}}/N^{\mathrm{arith}}$ has fibres that are torsors for $N^{\mathrm{arith}}/\Sigma$. This assertion needs normality of Σ in the arithmetic image, already proved, but not normality in all of O. *Manuscript:* §29, folio 189–190, scan 373–376..

Proof. For a representative f of an arithmetization, its fibre consists of the right cosets $fn\Sigma$, for $n \in N^{\mathrm{arith}}$. Multiplication identifies it with a simply transitive $N^{\mathrm{arith}}/\Sigma$-set. Replacing f changes this identification by translation, which is precisely why the fibre is canonically a torsor and has no canonical origin. After the faithfulness theorem in D, the acting group is $G_{\mathbb{Q}}$, subject to the fixed arithmetic identification. □

1.4 A.4. Exceptional types and editorial limits

Statement A.11 (Proved). The non-hyperbolic types do not satisfy the hyperbolic assertions.
The following table refers to punctures, not boundary circles fixed pointwise:

Type	Π	M^+	P
$(0,0)$	1	1	1
$(0,1)$	1	1	1
$(0,2)$	$\mathbb{Z}$	$\mathbb{Z}/2$	1
$(0,3)$	F_2	$\mathfrak{S}_3$	1
$(1,0)$	$\mathbb{Z}^2$	$\mathrm{SL}_2(\mathbb{Z})$	$\mathrm{SL}_2(\mathbb{Z})$
$(1,1)$	F_2	$\mathrm{SL}_2(\mathbb{Z})$	$\mathrm{SL}_2(\mathbb{Z})$

The extended groups are obtained by including degree -1 maps; for $(0,2)$ the result is $(\mathbb{Z}/2)^2$, and for both torus types it is $\mathrm{GL}_2(\mathbb{Z})$. The surface group F_2 in the last line has peripheral element $[a,b]^{-1}$; the F_2 in type $(0,3)$ has three peripheral classes. They are not the same group with loops. Forgetting the marked point on a torus gives an isomorphism of mapping class groups, not a Birman sequence with an injected $\mathbb{Z}^2$ kernel. *Manuscript:* §19, folio 111–115, scan 222–230.; §21, folios 129–131, scans 257–261.

Proof. A sphere with three ordered points has trivial orientation-preserving mapping class group, by cutting into discs; allowing their permutations gives $\mathfrak{S}_3$. Forgetting points gives the first three entries, with the remaining transposition for two unordered points. Complex conjugation fixes the real punctures and supplies the orientation-reversing involution; for two punctures it is independent of the permutation. A torus homeomorphism acts on its first homology by an integral matrix, and the linear maps realize every such matrix. Cutting along a basis proves that the action on homology detects the mapping class. A point can be fixed by translation, and translations are isotopic to the identity, so the pointed and unpointed groups coincide. For the cylinder, write ρ_{perm} for the permutation sign and ρ_π for the sign on $\pi_1 \simeq \mathbb{Z}$. Then $\rho_\pi = \rho_{\mathrm{perm}}\epsilon$. Indeed $z \mapsto z^{-1}$ has permutation sign -1 and orientation sign $+1$, while $z \mapsto \overline{z}$ has signs $+1, -1$; both induce -1 on π_1. Thus the identification of the two homomorphisms called ρ on scan 225 is false if read literally, and the determinant-one image assertion for the rank-one case on scan 227 is false as well. The orientation-preserving cylinder group acts on $\mathbb{Z}$ by both signs. These computations also exhibit why the fundamental group of the identity component, which is nontrivial in the torus case, destroys the hyperbolic point-push injection. Their finite permutation and matrix checks are reproduced in `code/af_checks.py`. For $(0,0)$ and $(0,1)$ the group alone has no orientation information; the manuscript adds an orientation line and the type, as stated explicitly at scans 229–230. □

Statement A.12 (Proved). Fields, real structures, and characteristics are part of the data.

The complex Teichmüller tower in this paper consists of orientable surfaces. A non-orientable surface is not the underlying surface of a smooth complex algebraic curve. For a real curve the relevant structure is an extension by $G_{\mathbb{R}} \simeq \mathbb{Z}/2$, together with its outer involution on the geometric group; it is not obtained by replacing the curve with the topological quotient by complex conjugation. All arithmetic comparison statements here are in characteristic zero. At primes 2 and 3, prime-to-characteristic or tame specialization cannot silently replace the full profinite group, and stack isotropy of orders 2 and 3 must be retained.

Proof. Holomorphic coordinate changes preserve the real orientation, proving the first assertion. The arithmetic fundamental-group sequence over $\mathbb{R}$ has geometric kernel and quotient $G_{\mathbb{R}}$; conjugation reverses the complex orientation and has cyclotomic value -1. Its quotient surface may have boundary or be non-orientable and is a different topological object. In characteristic p, wild inertia is pro-p and is killed by prime-to-p completion. Moreover elliptic curves in characteristic zero already have stabilizers of orders 2, 4, and 6 at suitable points. Thus erasing all 2- or 3-primary isotropy cannot preserve the stack group. The p-adic statements cited in D have their own all-prime hypotheses; no comparison made here transports them to a curve over a finite field. These are scope restrictions, not an unproved assertion of a non-orientable analogue. □

Reading record. On scan 204 the sequence of marked points is written with a positive-index convention, while (1) and the margin use indices beginning with zero. We use an arbitrary nested enumeration: the mathematics supports a reindexing, not a settled palaeographic correction. On scan 320 the assumption of injectivity is visible in the image and must not be suppressed. On scan 328 the transcription's folio alternatives 166|168 do not alter any argument; the numerical sequence favours 166, without settling the reading. The phrase "classes of discretifications" at scan 321 is explicitly broadened by the insertion about pre-discretifications. At scan 261 the transcribed chain with changing genus cannot be literally correct by dimension: the intended zero-genus chain $\mathcal{T}_{0,3} \to \mathcal{T}_{0,2} \to \mathcal{T}_{0,1} \to \mathcal{T}_{0,0}$ fits the following sentence. That scan image was not supplied, so we record mathematical support only, not a visual emendation.

1.5 A.5. The universal lift in §21

Statement A.13 (Proved). The three lift extensions and the mixed fundamental group.

Assume hyperbolic type and choose the additional base point b. Write $A = A^{+}_{g,n}$, A_0 for its identity component, $M = \pi_0 A$, and $M_* = \pi_0(A^{+}_{g,n+1,*})$. The topological pullback

$$\widetilde{A} = A \times_M M_*$$

has the three exact sequences

$$\begin{aligned} &1 \longrightarrow \Pi_{g,n} \longrightarrow \widetilde{A} \longrightarrow A \longrightarrow 1, \\ &1 \longrightarrow A_0 \longrightarrow \widetilde{A} \longrightarrow M_* \longrightarrow 1, \\ &1 \longrightarrow A_0 \times \Pi_{g,n} \longrightarrow \widetilde{A} \longrightarrow M \longrightarrow 1. \end{aligned} \tag{A.1}$$

The first extension splits canonically over the subgroup A_* fixing b, by lifting maps to fix a chosen point of the universal cover. If $E = EA$ is a universal principal A-space (for example

the Milnor construction), the associated bundle $\mathcal{X} = (E \times U_{g,n})/A_0$ has base E/A_0 contractible up to homotopy and fibre $U_{g,n}$. Its residual M-action has mixed fundamental group

$$\pi_1(\mathcal{X} /\!\!/ M) \simeq M_*, \qquad 1 \to \pi_1(U_{g,n}) \to M_* \to M \to 1. \tag{A.2}$$

Here the double slash is the homotopy quotient. These identifications use compatible base points; changing them gives inner conjugation. The same pullback construction applies to the extended groups. *Manuscript:* §21, folio 126–132, scan 251–263., especially (74)–(75), scans 252–254.

Proof. A.3 gives $1 \to \Pi \to M_* \to M \to 1$. Pull it back along $A \to M$. The first two kernels in (A.1) follow immediately. Over the identity of M the pullback is the *direct* product $A_0 \times \Pi$: its multiplication is componentwise. This proves the third sequence and supports the reading $\times$, rather than a nontrivial semidirect product, in (75). Every $f \in A_*$ has a unique lift fixing the chosen lift of b; composition of such lifts has the same property. This gives the canonical splitting over A_*.

Hyperbolicity implies contractibility of A_0, as in the proof of the Birman theorem. Thus E/A_0 has the homotopy type of BA_0 and is contractible. The associated bundle has fibre U, so its fibre inclusion is a homotopy equivalence. Quotienting the residual action gives $\mathcal{X}//M \simeq E \times_A U$. Evaluation identifies the homogeneous space U with A/A_*; local sections are supplied by isotopy extension. Therefore $E \times_A U \simeq E/A_* \simeq BA_*$. Its fundamental group is $\pi_0 A_* = M_*$. The fibration over BM gives exactly (A.2), including the point-pushing kernel. This is the mixed-group construction sketched in §21, not a genuine action of M on a single marked surface chosen without additional data. □

2 Congruence, normalizers, and discretizations

We use the conventions of Part A. Thus $P_{g,n}$ is the pure, orientation-preserving mapping class group, $M_{g,n}^{+}$ allows permutations of the punctures, and $M_{g,n}^{\pm}$ also allows reversal of orientation. We write $\Pi_{g,n}$ for the discrete surface group and $\pi_{g,n} = \widehat{\Pi}_{g,n}$ for its full profinite completion. Unless an exception is explicitly stated, $2g - 2 + n > 0$. All automorphisms of profinite groups are continuous. In

$$O_{g,n} = \mathrm{Out}_{\mathrm{lac}}(\pi_{g,n})$$

the peripheral conjugacy classes may be permuted and their generators may be raised to a common profinite unit. For a proper surface the corresponding multiplier is its action on the orientation class. We set

$$\check{P}_{g,n} = \overline{P_{g,n}}, \qquad \Sigma_{g,n} = \overline{M_{g,n}^{+}}, \qquad H_{g,n} = \overline{M_{g,n}^{\pm}}, \qquad N_{g,n}^{\mathrm{norm}} = N_{O_{g,n}}(\Sigma_{g,n}).$$

These are images or closures in $O_{g,n}$, not, by definition, abstract profinite completions. This distinction is indispensable in reading §26, folios 162–165, scans 320–326. The arithmetic image N^{arith} introduced in §29, folio 188, scan 371 is a different group; its behavior under filling is treated in Part C.

2.1 The congruence problem and the size of the ambient group

Statement B.1 (Proved). The precise congruence question.
There is a canonical continuous epimorphism

$$c_{g,n} : \widehat{P}_{g,n} \longrightarrow \check{P}_{g,n}, \qquad C_{g,n} = \ker(c_{g,n}).$$

The following assertions are equivalent: $C_{g,n} = 1$; the profinite and congruence topologies on $P_{g,n}$ coincide; and every subgroup of finite index in $P_{g,n}$ contains the kernel of $P_{g,n} \to \mathrm{Out}(\Pi_{g,n}/K)$ for some characteristic subgroup K of finite index in $\Pi_{g,n}$. None of these assertions asserts normality of $\check{P}_{g,n}$, $\Sigma_{g,n}$, or $H_{g,n}$ in $O_{g,n}$.

Proof. The kernels in the statement form a neighborhood basis for the induced congruence topology. Indeed, a finitely generated group has only finitely many subgroups of a prescribed finite index; intersecting all subgroups of index at most a prescribed integer gives cofinal characteristic subgroups. The completion for this topology is exactly the closure of the image in $\mathrm{Out}(\pi_{g,n})$, since the latter has the topology determined by its characteristic finite quotients. The universal property of profinite completion gives $c_{g,n}$, whose image is closed and contains a dense subgroup, hence is the whole closure. If its kernel is trivial, compactness makes it an isomorphism of topological groups; pulling back open subgroups gives cofinality of the congruence neighborhoods. The converse follows from the same inverse limit description. A kernel calculation for a map into an ambient group makes no assertion about conjugation by elements outside its image. Even for finite groups, an injective map $C_2 \hookrightarrow \mathfrak{S}_3$ need not have normal image. Thus the replacement of all the normality questions in scans 322–326 by the congruence problem, as suggested in Lead 4, is invalid. □

Statement B.2 (Known). The established congruence cases.
For full profinite completion, $C_{g,n} = 1$ when $g = 0$, $n \geq 3$; when $g = 1$, $n \geq 1$; and when $g = 2$, $n \geq 0$. The same faithfulness holds for the finite extensions allowing permutations and orientation reversal. In particular, all genus-zero groups used in Part F have their full profinite topology induced by their surface-group action.

The first two assertions are Asada's results, in the precise forms recorded and proved in Boggi (2009, Proposition 2.6 and Theorem 2.8). The same attached paper proves the hyperelliptic result for punctured surfaces, hence the genus-two result for $n \geq 1$, in its Theorem 3.5. The closed genus-two case requires the subsequent Boggi (2020, Theorem 1.4(ii), with $\mathscr{C} = \mathscr{F}$): the hyperelliptic mapping class group equals the full mapping class group in genus two. The distinction matters: the 2009 paper expressly restricts its main theorem to open surfaces and records a gap in an earlier proof. The more recent primary source Boggi (2026b, §1.1 and Corollary B) confirms the genus-at-most-two range and establishes additional centralizer cases, without solving the problem in arbitrary genus.

To justify the finite-extension assertion, the pure orientation subgroup is congruence-open: permutation of the finitely many peripheral classes is continuous, and orientation reversal is distinguished from orientation preservation by the multiplier modulo 3. Within the geometric image the multiplier takes only the values 1 and -1. The kernel of the completed action on the finite extension therefore lies in $\widehat{P}_{g,n}$, where it is trivial. This is not the ordinary congruence-subgroup property of $\mathrm{SL}_2(\mathbb{Z})$ acting on $\mathbb{Z}^2$. For the punctured torus the decisive representation acts on $\widehat{F}_2$ and remembers more than homology.

Statement B.3 (Open). The remaining full congruence problem.

For $g \geq 3$, determine whether

$$\widehat{P}_{g,n} \longrightarrow \mathrm{Out}(\pi_{g,n})$$

is injective. The smallest genus and number of punctures not covered by B.2 is $(g, n) = (3, 0)$. Part C proves $C_{g,n+1} \simeq C_{g,n}$ whenever the target type is hyperbolic, so adding punctures does not remove this unresolved case. The exact missing step is cofinality of characteristic-quotient levels among all finite-index subgroups; neither the normality of a geometric image in an arithmetic image nor faithfulness of the Galois action supplies it. This is a literature-open problem, not merely a gap in the manuscript. Our status check includes the 2026 primary source cited in B.2.

Statement B.4 (Proved). Properness of the geometric image and the exceptional types.

For every hyperbolic type,

$$H_{g,n} \neq O_{g,n}.$$

The assertion "never an isomorphism" in §26, folio 163, scan 322 is correct in this range. Read as an assertion about bare fundamental groups outside that range, it is false; adding an independent orientation line changes the data and must be specified.

Proof. The continuous multiplier $\lambda : O_{g,n} \to \widehat{\mathbb{Z}}^\times$ takes values in $\{1, -1\}$ on $H_{g,n}$. Choose a smooth marked curve of the specified type over some number field L; all required markings can be made rational after a finite extension. The arithmetic outer action has multiplier the cyclotomic character of G_L. Its image is an open, hence infinite, subgroup of $\widehat{\mathbb{Z}}^\times$, since $[L \cap \mathbb{Q}(\mu_\infty) : \mathbb{Q}] < \infty$. Consequently an arithmetic element with multiplier outside $\{1, -1\}$ lies in $O_{g,n} \setminus H_{g,n}$. No faithfulness theorem and no congruence conjecture are needed for this argument.

For $(0, 0)$ and $(0, 1)$ the geometric fundamental group is trivial, so its automorphism group and every image in it are trivial. For $(0, 2)$ it is $\widehat{\mathbb{Z}}$, and the geometric image is $\{1, -1\} \subset \widehat{\mathbb{Z}}^\times$; forgetting the two labels does not change this image. For $(1, 0)$ it is $\widehat{\mathbb{Z}}^2$, and the ambient group is $\mathrm{GL}_2(\widehat{\mathbb{Z}})$. The extended geometric image is the subgroup with determinant in $\{1, -1\}$, whereas the pure oriented image is $\mathrm{SL}_2(\widehat{\mathbb{Z}})$. Here the action factors through the usual homological congruence completion, and its kernel on the abstract profinite completion of $\mathrm{SL}_2(\mathbb{Z})$ is nontrivial. Thus the punctured torus and the unpunctured torus must not be merged. The auxiliary orientation

line that the manuscript adds in the trivial-group cases, noted in A.11, is not contained in the bare fundamental group; the preceding triviality statement does not erase that extra datum.

For completeness, this last nontriviality has a finite witness. The presentation $\mathrm{PSL}_2(\mathbb{Z}) = C_2 * C_3$ maps onto $\mathrm{SL}_2(\mathbb{F}_8)$ by sending its generators to

$$\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathbb{F}_8 = \mathbb{F}_2[t]/(t^3 + t + 1).$$

The code verifies their orders, that they generate all 504 determinant-one matrices, and that the group is simple: its conjugacy-class sizes are $1, 56, 56, 56, 56, 63, 72, 72, 72$, and no proper union containing the identity has cardinality dividing 504. A finite quotient of $\mathrm{SL}_2(\widehat{\mathbb{Z}})$ has only cyclic simple factors and groups $\mathrm{PSL}_2(\mathbb{F}_p)$ for primes $p \geq 5$. This follows by reduction modulo p in each factor; the reduction kernel is pro-p, and the groups for $p = 2, 3$ are solvable. None of those projective groups has order 504: the order formula gives 168 at $p = 7$ and 660 at $p = 11$, with no intervening prime. The displayed simple quotient therefore cannot be a congruence quotient, which proves the claimed nontrivial kernel. □

Statement B.5 (Refuted). Normality of the full extended geometric group.
The assertion that $H_{g,n}$ is normal in $O_{g,n}$ for every hyperbolic type is false already at $(0, 3)$.

Proof. The three-punctured sphere has $P_{0,3} = 1$, $M^+_{0,3} = \mathfrak{S}_3$, and $M^\pm_{0,3} = \mathfrak{S}_3 \times C_2$. For the rational curve $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ the outer Galois action is faithful, as proved in D.16 from Belyi's theorem. Its elements fix the three peripheral labels. The six rational fractional linear transformations permuting $0, 1, \infty$ commute with this action in the outer automorphism group. Hence

$$\mathfrak{S}_3 \times G_{\mathbb{Q}} \hookrightarrow O_{0,3}.$$

The intersection of the two factors is trivial because the first factor acts faithfully on the labels and the second fixes them. If $c \in G_{\mathbb{Q}}$ is complex conjugation, its image is the orientation-reversing class fixing those labels. Thus $H_{0,3} = \mathfrak{S}_3 \times \langle c \rangle$ inside this subgroup. Normality of $H_{0,3}$ would imply normality of $\langle c \rangle$ in $G_{\mathbb{Q}}$.

The splitting field of $t^3 - 2$ has Galois group $\mathfrak{S}_3$: Eisenstein's criterion gives irreducibility and its discriminant is -108, which is not a square. Complex conjugation acts as a transposition on its three roots. Conjugating this transposition by a three-cycle gives another transposition outside its order-two subgroup. Lifting the three-cycle to $G_{\mathbb{Q}}$ contradicts normality. The finite calculation, including the discriminant and the conjugation, is reproduced by `code/bc_counterexamples.py`. □

This answers the normality question attached to the extended group in scan 322. It does not answer the different question about $\Sigma_{g,n} = \overline{M^+_{g,n}}$. Nor does an author's annotation "ultra-faux" by itself provide a mathematical counterexample to a different self-normalizer assertion: the conjecture so annotated is recorded as withdrawn, and we do not silently substitute B.5 for it.

Statement B.6 (Proved). Conjugacy of geometric structures and their normalizers.
Fix an identification of a surface-group completion with a profinite group with loops π, and let $\Sigma \subset O = \mathrm{Out}_{\mathrm{lac}}(\pi)$ be its oriented geometric image. The conjugates of Σ are parameterized by $O/N_O(\Sigma)$. The natural map

$$O/\Sigma \longrightarrow O/N_O(\Sigma)$$

has fibers which are torsors under $N_O(\Sigma)/\Sigma$, after transport by a representative. In particular, uniqueness of this conjugate subgroup means $N_O(\Sigma) = O$, equivalently normality of Σ in O.

Proof. Two conjugates $a\Sigma a^{-1}$ and $b\Sigma b^{-1}$ agree exactly when $b^{-1}a$ normalizes Σ. For a fixed representative a, the fiber over $aN_O(\Sigma)$ is $aN_O(\Sigma)/\Sigma$. This is the asserted torsor. All subgroups used for the displayed quotients are closed: the normalizer of a closed subgroup of a profinite group is closed. These are therefore profinite coset spaces, although the first need not be a group. This proves the assertions in §26, folios 164–165, scans 324–326, with the normalizer definition kept explicit. □

Statement B.7 (Open). The unrestricted normalizer comparison.

The unrestricted questions

$$\Sigma_{g,n} \triangleleft O_{g,n}, \qquad N_{g,n}^{\mathrm{norm}} = O_{g,n}$$

are equivalent to each other by B.6 and are separate from B.3. We do not resolve this unrestricted comparison. At the smallest test type $(0, 3)$ it asks whether every inertia-preserving outer automorphism normalizes the geometric $\mathfrak{S}_3$. E.7 identifies the normalizer quotient using the additional symmetry condition; that identification does not show that every element of $O_{0,3}$ satisfies the condition. Our searches of the primary sources on quasi-special symmetric automorphisms and procongruence automorphisms did not supply either this implication or a counterexample. Here the label Open records the precise unresolved comparison in this audit; unlike B.3, we do not assert that each individual case has been proved open by a literature classification. The earlier self-normalizer conjecture for H, explicitly withdrawn in scan 322, is not used as a premise.

2.2 Actual discretizations under a finite covering

A based discretization of π will mean a dense subgroup $D \subset \pi$, equipped with the oriented surface peripheral structure, such that completion induces an isomorphism $\widehat{D} \simeq \pi$ preserving that structure. In particular it is more than an arbitrary dense subgroup. Forgetting orientation gives the corresponding unoriented notion. Quotienting by inner automorphisms of π gives strict prediscretizations; quotienting a marking torsor by the closure of the geometric automorphism group gives the manuscript's non-strict prediscretizations. The adjective "strict" must not be dropped in these comparisons.

Statement B.8 (Proved). A surface lattice is self-normalizing in its completion.

If D is a nonabelian free group or the fundamental group of a closed orientable surface of genus at least two, then

$$N_{\widehat{D}}(D) = D.$$

Proof. Free groups and closed orientable surface groups are conjugacy separable and have Grossman's Property A: every automorphism taking each element to a conjugate of itself is inner. For the exact conjugacy-separability statements we use Paris (2009, Theorems 3.2 and 4.1; Property A is stated in §1 immediately before Theorem 1.7). These theorems are stronger, giving conjugacy p-separability. If $h \in \widehat{D}$ normalizes D, conjugation by h induces an automorphism α of D. For every $d \in D$, $\alpha(d)$ and d are conjugate in $\widehat{D}$; hence they are conjugate in D. Property A gives $\alpha = \mathrm{int}(d_0)$ for some $d_0 \in D$. Then $d_0^{-1}h$ centralizes the dense subgroup D and therefore centralizes $\widehat{D}$. The latter is center-free. For example, this follows from the slimness theorem in D.14, or from the centralizer calculation for two independent simple loops in Boggi and Zalesskii (2019, Proposition 3.6). Consequently $h = d_0$. The reverse inclusion is immediate. □

Statement B.9 (Proved). Injectivity of restriction to a finite cover.
Let π have hyperbolic orientable surface type and let $V \subset \pi$ be an open subgroup with its induced peripheral structure. The map

$$\mathrm{Discr}(\pi) \longrightarrow \mathrm{Discr}(V), \qquad D \longmapsto D \cap V$$

is well defined and injective, for based discretizations, and the same holds with orientation retained. No congruence-subgroup property is assumed. This proves the corollary in §28, folio 178, scans 352–353.

Proof. The intersection $D \cap V$ is of finite index in D. The topology induced on a finite-index subgroup by the full profinite topology is its full profinite topology: the core in D of a subgroup of finite index in $D \cap V$ is of finite index in D. Therefore $\widehat{D \cap V} \simeq V$. The ordinary finite covering of surfaces determines the asserted peripheral and orientation data.

First suppose that V is normal. If $D_1 \cap V = D_2 \cap V = \Lambda$, both D_1 and D_2 normalize Λ. B.8 applied inside $V = \widehat{\Lambda}$ gives

$$N_\pi(\Lambda) \cap V = \Lambda.$$

Each D_i maps onto the finite group π/V, since it is dense. For $a \in N_\pi(\Lambda)$ choose $d \in D_i$ with the same image in π/V. Then $ad^{-1} \in \Lambda \subset D_i$, proving $N_\pi(\Lambda) = D_i$. Thus $D_1 = D_2$. For arbitrary V, replace it by its core $W = \bigcap_{a \in \pi} aVa^{-1}$. This is open and normal, and $D_1 \cap W = D_2 \cap W$ follows by intersecting the given equality with W. The normal case applies. Orientation is unchanged under these restrictions. □

Statement B.10 (Proved). The image criterion for a normal finite cover.
Suppose $V \triangleleft \pi$ is open, $G = \pi/V$, and $\Lambda \subset V$ is an oriented discretization. It extends to an oriented discretization of π if and only if the faithful deck outer action

$$G \longrightarrow \mathrm{Out}_{\mathrm{lac}}(V)$$

factors through the image of $\mathrm{Out}^+_{\mathrm{lac}}(\Lambda)$ in $\mathrm{Out}_{\mathrm{lac}}(V)$. The extension, when it exists, is unique as a subgroup of π.

Proof. Necessity is immediate by conjugating $D \cap V$ inside an extending lattice D. For sufficiency, the discrete outer action on Λ is unique because the completion map on its outer automorphism group is injective, by the same conjugacy-separability and Property A argument as B.8. The deck action is faithful: an element of π acting on V by an inner automorphism differs from an element of V by an element centralizing V, and $C_\pi(V) = 1$. The last equality is the slimness of a hyperbolic profinite surface group.

Since $Z(\Lambda) = 1$, the specified outer action determines a unique extension $1 \to \Lambda \to E \to G \to 1$. Explicitly, it is the pullback of $\mathrm{Aut}(\Lambda) \to \mathrm{Out}(\Lambda)$ along that action. Completing automorphisms maps it into the analogous pullback for V; the latter is π, since $Z(V) = 1$. The resulting map $E \to \pi$ is injective, induces the given inclusion on Λ, and is the identity on G. Its image is dense. Because Λ has finite index in E, completion of this extension is exact, and the comparison with $1 \to V \to \pi \to G \to 1$ gives $\widehat{E} \simeq \pi$.

The profinite group π is torsion-free. A precise source is Boggi and Zalesskii (2019, Lemma 3.1(ii)), applied to a Fuchsian surface group without orbifold points: there are no decomposition groups in which a nontrivial finite subgroup could lie. Thus E is torsion-free. The finite extension is allowable in the Fuchsian sense: its outer action preserves the peripheral classes and orientation. Kerckhoff (1983, Theorem 7, p. 258) says that an allowable finite extension of a Fuchsian group is Fuchsian when its outer action is injective. Hence E is the fundamental group

of an orientable surface of finite type, and the covering defined by Λ is a genuine unramified surface covering.

It remains to check the loops, rather than merely the underlying group. Each peripheral cyclic subgroup of E meets Λ in a finite-index peripheral subgroup. Its closure is the centralizer in π of the corresponding peripheral subgroup of V. Conversely, the original peripheral groups of π are obtained from those of V by exactly this centralizer operation. Both assertions follow from Boggi and Zalesskii (2019, Proposition 3.6): a nonzero power of an algebraically simple element has normalizer, and hence centralizer, equal to the maximal procyclic group it generates. The two peripheral structures therefore agree. The allowable orientation agrees on the finite cover and hence on the base. Uniqueness is B.9. □

In scan 352 the displayed quotient for the deck group is transcribed as π'/π, although the surrounding text has $\pi' \subset \pi$. The image inspection confirms this apparent slip. B.10 requires $G = \pi/\pi'$; it does not assert that the handwriting has thereby been editorially settled. The same page only sketches the loop-compatibility step, supplied above using the centralizer theorem.

Statement B.11 (Refuted). Surjectivity for raw discretizations of a finite cover. Restriction of based discretizations need not be surjective. A counterexample is the degree-two covering

$$\mathbb{P}^1(\mathbb{C}) \setminus \{0, \infty, 1, -1\} \longrightarrow \mathbb{P}^1(\mathbb{C}) \setminus \{0, \infty, 1\}, \qquad z \longmapsto z^2.$$

Proof. Let $V = \widehat{\Lambda}$ be the subgroup defined by this cover in $\pi = \widehat{F}_2$, and let s be the deck involution in $\mathrm{Out}_{\mathrm{lac}}(V)$. It fixes the punctures $0, \infty$ and interchanges $1, -1$. In the discrete pure mapping class group $P_{0,4} \simeq F_2$, conjugation by s is nontrivial: it takes the twist about a curve separating $\{0, 1\}$ from the other two punctures to the distinct twist about a curve separating $\{0, -1\}$. The permutation of these two partitions is checked in `code/bc_counterexamples.py`.

The centralizer of s in $P_{0,4}$ has infinite index. Otherwise the induced automorphism would fix a subgroup of finite index. Each element of F_2 has a positive power in that subgroup, and the unique-root property in a free group would force the automorphism to fix every element, a contradiction. Let $B = \overline{P_{0,4}} \subset \mathrm{Out}_{\mathrm{lac}}(V)$. Its closed subgroup $C_B(s)$ is not open, since openness would make its intersection with the dense group $P_{0,4}$ of finite index. Thus $B/C_B(s)$ is uncountable: a compact Hausdorff space cannot be a countable union of closed cosets with empty interior.

The discrete group $\mathrm{Out}^+_{\mathrm{lac}}(\Lambda)$ is countable. We may therefore choose $u \in B$ for which $u^{-1}su$ is not in that discrete group. For definiteness this choice can be specified by enumerating the discrete group and choosing successively nested basic clopen neighborhoods which avoid the closed conjugator set for the next element; require also containment in the next characteristic finite-quotient neighborhood. Each avoidance is possible because the nonempty conjugator sets are cosets of the nonopen centralizer. Compactness and the cofinal neighborhood condition determine an element u. Choose a genuine automorphism representing u, and put $\Lambda' = u(\Lambda)$. The deck outer action does not factor through $\mathrm{Out}^+_{\mathrm{lac}}(\Lambda')$. By B.10, Λ' does not extend to π.

The counterexample is necessarily profinite: all discretizations are dense and their difference cannot be certified by distinguishing their images in a single finite quotient. The nested construction above specifies the required element, rather than relying on a purported finite-quotient witness. Notice also that $u \in B$: Λ and Λ' have the same non-strict oriented prediscretization. This counterexample does not disprove the prediscretization assertion in scan 355. □

2.3 Passing to prediscretizations and the remaining comparisons

Statement B.12 (Proved). Restriction descends to both prediscretization quotients. For every finite covering $V \subset \pi$ as in B.9, restriction of discretizations descends to strict prediscretizations and to the non-strict oriented and unoriented prediscretizations. In particular, the well-definedness doubts in §28, folios 179–181, scans 355–358 have an affirmative answer.

Proof. First let $D_2 = aD_1a^{-1}$, $a \in \pi$. Density gives $\pi = VD_1$: every coset of the open subgroup V meets D_1. Write $a = vd$ with $v \in V$, $d \in D_1$. Then $D_2 = vD_1v^{-1}$, and $D_2 \cap V = v(D_1 \cap V)v^{-1}$. This is precisely the correction already made in scan 357.

For the closure quotient, fix $D = D_1$ and let A_D be its discrete loop-preserving automorphism group, with orientation restriction if required, inside $\mathrm{Aut}_{\mathrm{lac}}(\pi)$. Suppose $D_2 = u(D)$ with $u \in \overline{A_D}$. The group π is topologically finitely generated, so it has finitely many open subgroups of index $[\pi : V]$. The action on this finite set is continuous. Density of A_D in its closure permits a choice $a \in A_D$ with $a(V) = u^{-1}(V)$. Thus ua stabilizes V and $ua(D) = D_2$.

The intersection of $\overline{A_D}$ with the open stabilizer of V is the closure of $A_D \cap \mathrm{Stab}(V)$. Restriction of automorphisms stabilizing V to V is continuous; for instance, restriction is a continuous map between the closed automorphism spaces in the spaces of maps on the corresponding finitely generated profinite groups. It takes the discrete stabilizer into $\mathrm{Aut}_{\mathrm{lac}}(D \cap V)$. Therefore the restrictions of D and $u(D)$ differ by an element in the closure of that latter group. The same proof preserves orientation, because orientation restricts compatibly along a surface covering. It proves descent, not bijectivity. □

Statement B.13 (Proved). The exact coset criteria behind conditions (a)–(e). Let $f : A \to A'$ be a continuous homomorphism of profinite groups, and let $B \subset A$, $B' \subset A'$ be closed subgroups with $f(B) \subset B'$. The map of right-coset spaces $A/B \to A'/B'$ is injective if and only if

$$f^{-1}(B') = B,$$

and is surjective if and only if $A' = f(A)B'$. If in addition f is surjective and $\ker(f) \subset B$, then

$$B = f^{-1}(B') \quad \Longrightarrow \quad N_A(B) = f^{-1}(N_{A'}(B')), \qquad N_A(B)/B \simeq N_{A'}(B')/B'.$$

Proof. The fiber over the identity coset is $f^{-1}(B')/B$, and translating by an element of A gives all other nonempty fibers. The image consists exactly of cosets meeting $f(A)$. For the last assertion, if $B = f^{-1}(B')$ and f is onto, an element conjugates B onto itself exactly when its image conjugates B' onto itself. The map on normalizers is therefore onto and its kernel is $\ker(f) \subset B$; quotienting gives the isomorphism. This proves the precise formal mechanism, and also specifies the extra hypotheses, behind §28, folio 181, scans 358–359. In a restriction map to a characteristic open subgroup one need not have surjectivity of f; the displayed last assertion cannot be used without verifying it or an appropriate replacement. □

Statement B.14 (Open). Bijectivity for arbitrary profinite prediscretizations. For a characteristic open subgroup $V \subset \pi$, put

$$A = \mathrm{Aut}_{\mathrm{lac}}(\pi), \quad A' = \mathrm{Aut}_{\mathrm{lac}}(V), \quad B = \overline{\mathrm{Aut}^{+}_{\mathrm{lac}}(D)}, \quad B' = \overline{\mathrm{Aut}^{+}_{\mathrm{lac}}(D \cap V)}.$$

Restriction gives a continuous injection $f : A \to A'$, and B.12 gives $f(B) \subset B'$. The unrestricted bijectivity proposed in scan 355 requires both

$$f^{-1}(B') = B, \qquad A' = f(A)B'.$$

We have not proved these two assertions for arbitrary loop-preserving automorphisms. A concrete characteristic test, already indicated in §28, folio 183, scan 363, is

$$\pi = \widehat{F}_2, \qquad V = \ker(\widehat{F}_2 \to (\mathbb{Z}/2)^2),$$

the degree-four covering of the three-punctured sphere of type $(0, 6)$. Among noncharacteristic coverings, the degree-two cover in B.11 is the first basic test of the set map of B.12; its raw failure of surjectivity disappears in the closure quotient and does not decide this test. The congruence property is known for both surfaces in these tests, so restating these conditions as CSP does not supply the missing argument. The analogous filling conditions are made exact in C.8. As in B.7, Open here records the unresolved unrestricted statement in this audit, without claiming a literature classification of every individual cover. For completeness, injectivity of the restriction homomorphism uses slimness, not CSP: if an automorphism is the identity on V, then for each $a \in \pi$ the element $a^{-1} f_0(a)$ centralizes the normal open subgroup V, and hence is trivial by D.14–D.15. Here f_0 denotes that automorphism, to distinguish it from the restriction homomorphism f.

Statement B.15 (Refuted). Raw discretization injectivity does not transfer to filling.

If the proposed injection of raw discretizations in item 3 of the prompt is applied also to filling a puncture, it is false. It already fails for $(0, 4) \to (0, 3)$.

Proof. Let $D = F(x, y, z)$ be the fundamental group of a four-punctured sphere with peripheral representatives $x, y, z, (xyz)^{-1}$, and let $\pi = \widehat{D}$. Filling the puncture represented by x gives

$$q : \pi \longrightarrow \widehat{F}(y, z).$$

Choose $\alpha \in \widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$ with $\alpha_2 = 0$ and $\alpha_p = 1$ for every odd prime. This is not an integer: an integer equal to zero in $\mathbb{Z}_2$ is zero, whereas its 3-adic component here is one. Set $h = x^\alpha$. The retraction $D \to \langle x \rangle$, extended to completions, shows that $D \cap \overline{\langle x \rangle} = \langle x \rangle$; hence $h \notin D$. B.8 gives $hDh^{-1} \neq D$. Both are oriented discretizations, because inner automorphisms preserve all peripheral conjugacy and orientation data. But $q(h) = 1$, so

$$q(hDh^{-1}) = q(D) = F(y, z).$$

Thus the raw filling map is not injective. The code records compatible finite Chinese-remainder approximations to α; the proof of nonintegrality is the argument just given, not a claim that finitely many residues certify it. The two discretizations are inner-conjugate, so this counterexample leaves both versions of the prediscretization question untouched. □

All statements in this part concern orientable surfaces with the specified orientation option. Allowing orientation reversal does not turn them into statements about fundamental groups of nonorientable surfaces. A real curve is handled by its arithmetic extension over $\mathbb{R}$, whose quotient is C_2; the image of complex conjugation is exactly why the full extended group in B.5 behaves differently from the oriented geometric image. The full profinite results retain the primes 2 and 3; no argument has inverted either prime. The use of reduction modulo 3 in B.2 only distinguishes the two orientation signs.

3 Filling punctures and reconstructing the complete curve

The three rows in §32, folio 202, scan 399 must be read separately. They concern an arithmetic image, a geometric completion, and the full group of loop-preserving outer automorphisms. The first two have exact filling sequences. The unrestricted third row, marked "??", requires a further assertion about the full automorphism group. It is not a formal consequence of either of the first two. Throughout this part a filling map from type $(g, n+1)$ has hyperbolic target (g, n) unless the exceptional case is explicitly discussed.

3.1 The quotient and the two geometric exact sequences

Statement C.1 (Proved). Filling is quotient by the closed normal inertia subgroup. Let $U = X \setminus S$ be a smooth geometrically connected curve over a field K of characteristic zero, with smooth proper completion X, and let $S_0 \subset S$ be a Galois-stable subset. Put $U' = X \setminus (S \setminus S_0)$. After choosing compatible geometric base points, the inclusion $U \hookrightarrow U'$ induces a continuous epimorphism

$$\pi_1^{\text{ét}}(U) \longrightarrow \pi_1^{\text{ét}}(U')$$

whose kernel is the closed normal subgroup generated by the geometric inertia groups at the points of S_0. The same statement holds for the geometric groups, and the quotient is compatible with the maps to G_K. In particular, the quotient classifies precisely the finite *étale* coverings extending across the filled punctures.

Proof. Normalize U' in the function field of a connected finite étale cover of U. Normalization is finite for curves of finite type over a field. The resulting finite morphism is étale at a point of S_0 exactly when its inertia action on the associated finite set is trivial; in characteristic zero no wild-inertia qualification is needed. Thus finite étale covers of U' identify with the full subcategory of finite continuous $\pi_1(U)$-sets on which all these inertia groups act trivially. The fundamental group of this Galois category is the quotient by their closed normal closure. The same argument after base change to $\overline{K}$ gives the geometric assertion. The structural maps to $\operatorname{Spec} K$ commute with the inclusion, giving compatibility with G_K. A change of geometric base point or connecting path changes the homomorphisms by the corresponding inner automorphisms of the target. This is the group-theoretic filling operation used in §28, folios 175–176, scans 347–348, and §32, folios 200–202, scans 395–400. □

Statement C.2 (Proved). The full profinite Birman sequence. For $2g - 2 + n > 0$, forgetting the last marked point gives a short exact sequence of profinite groups

$$1 \longrightarrow \pi_{g,n} \longrightarrow \widehat{P}_{g,n+1} \longrightarrow \widehat{P}_{g,n} \longrightarrow 1. \tag{C.1}$$

The first arrow is the completed point-pushing homomorphism. Allowing permutations of the retained punctures replaces the two mapping class groups by $M^+_{g,n+1,i}$ and $M^+_{g,n}$, where the source fixes the puncture being forgotten. The same version holds with orientation reversal allowed.

Proof. The discrete Birman sequence is $1 \to \Pi_{g,n} \to P_{g,n+1} \to P_{g,n} \to 1$. We justify the potentially delicate passage to full completions. The action by conjugation on its normal subgroup $\Pi_{g,n}$ gives $P_{g,n+1} \to \operatorname{Aut}(\pi_{g,n})$, hence a map from its profinite completion. The composite

$$\pi_{g,n} \longrightarrow \widehat{P}_{g,n+1} \longrightarrow \operatorname{Aut}(\pi_{g,n})$$

is the inner-automorphism map, which is injective because $\pi_{g,n}$ is center-free. Thus the first arrow of (C.1) is injective. Completion is right exact in the following precise sense: $\widehat{P}_{g,n+1}$ modulo the closure of the point-pushing subgroup is $\widehat{P}_{g,n}$, because both classify the same finite quotients trivial on that subgroup. The image of $\pi_{g,n}$ is compact and contains the dense discrete point-pushing group, so it is exactly that closure. This proves exactness. The finite permutation and orientation quotients give the other versions by the identical argument. The exact sequence is also explicitly recorded in Boggi (2009, §2, p. 4). □

The hyperbolic target hypothesis is essential. Filling the sole puncture of a torus induces the isomorphism $P_{1,1} \simeq P_{1,0} \simeq \mathrm{SL}_2(\mathbb{Z})$; the alleged kernel $\pi_{1,0} = \widehat{\mathbb{Z}}^2$ would be wrong. Point-pushing is killed by translations isotopic to the identity on the unmarked torus. Similarly, $P_{0,3} \to P_{0,2}$ is a map of trivial groups, not an extension with kernel $\pi_{0,2} = \widehat{\mathbb{Z}}$. For targets $(0,0)$ and $(0,1)$ the pure oriented groups and geometric fundamental groups are trivial, so the corresponding trivial sequence carries no anabelian content. For hyperbolic types the *surface* group kernel is center-free, even in cases where a *mapping class* group has center: the elliptic involution at $(1,1)$ and the hyperelliptic involution at $(2,0)$ are retained. For the pure procongruence groups the exact center statement is Boggi (2020, Theorem 4.4): the center equals the discrete center, of order two in these two cases and trivial in the other hyperbolic pure cases. Additional centers in variants permitting permutations must not be suppressed by this pure-group statement.

Statement C.3 (Known). The procongruence Birman sequence.
For every hyperbolic target type there is an exact sequence

$$1 \longrightarrow \pi_{g,n} \longrightarrow \check{P}_{g,n+1} \longrightarrow \check{P}_{g,n} \longrightarrow 1. \tag{C.2}$$

It is compatible with (C.1), and the map on the left-hand kernels is the identity under their natural identifications. The point-pushing subgroup therefore has its full profinite topology inside the procongruence completion.

This is Boggi (2020, Corollary 4.7, with $\mathscr{C} = \mathscr{F}$, and Proposition 2.4(ii)). Its Remark 4.8 specifies the history: Matsumoto's theorem treated open targets, and Hoshi–Mochizuki extended it to closed targets. The earlier attached Boggi (2009, Theorem 2.5) alone has the extra condition $n \geq 1$ and cannot be cited for filling the last puncture of a higher-genus surface. Corollary 4.7 also has a pro-p congruence form, with kernel the pro-p surface group, for every prime p, including 2 and 3. This does not identify an arbitrary pro-p congruence completion with the pro-p completion of the whole discrete mapping class group.

Statement C.4 (Proved). The congruence kernel is unchanged by puncturing.
The natural map

$$C_{g,n+1} \longrightarrow C_{g,n}$$

is an isomorphism for every hyperbolic target (g,n).

Proof. Compare (C.1) and (C.2). If an element of $C_{g,n+1}$ maps to the identity of $\widehat{P}_{g,n}$, it lies in $\pi_{g,n}$; this subgroup maps injectively to the lower middle group, so the element is trivial. Conversely, take $a \in C_{g,n}$ and lift it to $b \in \widehat{P}_{g,n+1}$. Its image in $\check{P}_{g,n+1}$ lies in the left-hand copy of $\pi_{g,n}$. Multiply b by the inverse of the corresponding element of the upper copy. The resulting lift lies in $C_{g,n+1}$. The map is therefore a continuous bijection between profinite groups and hence a topological isomorphism. This proves exactly which congruence property transfers in either direction under filling; it does not assume its truth. □

3.2 The arithmetic image and the revised quotient

Let $\mathcal{M}_{g,n}$ now denote the stack of smooth curves with *ordered* markings over $\mathbb{Q}$. With compatible geometric base points put

$$\mathcal{A}_{g,n} = \pi_1^{\text{ét}}(\mathcal{M}_{g,n}), \quad \rho_{g,n} : \mathcal{A}_{g,n} \to O_{g,n}, \quad N_{g,n}^{\text{arith},!} = \text{im}(\rho_{g,n}).$$

Its geometric image is $\check{P}_{g,n}$. The image is closed because its source is profinite. This is the ordered version of the revised definition in §29, folio 188, scan 371, not the normalizer definition of §26.

Statement C.5 (Proved). The revised arithmetic quotient is the absolute Galois group.

For every hyperbolic type, the structural quotient induces a canonical isomorphism of profinite groups

$$G_{\mathbb{Q}} \xrightarrow{\sim} N_{g,n}^{\text{arith},!}/\check{P}_{g,n}. \tag{C.3}$$

Moreover, the kernel of the universal monodromy representation $\rho_{g,n}$ is exactly $C_{g,n}$, viewed in its geometric subgroup $\widehat{P}_{g,n}$. With unordered markings the corresponding assertion is $N_{g,n}^{\text{arith}}/\Sigma_{g,n} \simeq G_{\mathbb{Q}}$.

Proof. The standard geometric/arithmetic exact sequence of the moduli stack gives

$$1 \to \widehat{P}_{g,n} \to \mathcal{A}_{g,n} \to G_{\mathbb{Q}} \to 1.$$

Consequently its quotient map to the right-hand side of (C.3) is surjective by definition. Let its kernel be J. Choose a marked curve of this type over a finite extension $L/\mathbb{Q}$. Such a curve exists by choosing a geometric curve and its finitely many markings over $\overline{\mathbb{Q}}$ and descending their finite defining data; no assertion about arbitrarily many rational points over $\mathbb{Q}$ is needed. The associated point of the moduli stack supplies a section over G_L. Hoshi and Mochizuki (2011, Corollary 6.4, pp. 66–67) says that the preimage in G_L of the geometric monodromy image is trivial. Hence $J \cap G_L = 1$, so J is a finite normal subgroup of $G_{\mathbb{Q}}$.

There is no nontrivial finite normal subgroup of $G_{\mathbb{Q}}$. Indeed, a finite normal subgroup is centralized by an open subgroup, whereas every open subgroup of $G_{\mathbb{Q}}$ is slim and has trivial centralizer in $G_{\mathbb{Q}}$; see Mochizuki (2006, Corollary 2.2 in the published version; Corollary 1.3 in the author preprint) and D.14. Thus $J = 1$. The kernel of $\rho_{g,n}$ is now contained in the geometric subgroup, where it is by definition $C_{g,n}$. The unordered version follows by the finite permutation extension; alternatively its pure subgroup and geometric image give the same quotient, precisely as in §32, folio 200, scan 395, equations (1)–(3). □

The required arithmetic input is a published anabelian theorem. No section conjecture and no conjectural identification of $G_{\mathbb{Q}}$ with $\widehat{\text{GT}}$ is used. Corollary 6.5 of the same Hoshi–Mochizuki paper states the useful consequence that total universal monodromy is faithful if and only if its geometric restriction is faithful. Thus C.5 does not silently solve the high-genus congruence problem.

Statement C.6 (Proved). The arithmetic filling sequence and invariance of the quotient.

For every hyperbolic target there is a continuous short exact sequence

$$1 \longrightarrow \pi_{g,n} \longrightarrow N_{g,n+1}^{\text{arith},!} \longrightarrow N_{g,n}^{\text{arith},!} \longrightarrow 1. \tag{C.4}$$

The quotient map on the groups in (C.3) is the identity of $G_{\mathbb{Q}}$. The corresponding sequence with retained markings unordered uses the stabilizer of the forgotten marking in the source. In particular, the faithfulness condition (f) and the filling isomorphism of the revised arithmetic quotients in §32, folios 201–202, scans 397–400 hold for every hyperbolic type.

Proof. The universal punctured curve over $\mathcal{M}_{g,n}$ is $\mathcal{M}_{g,n+1}$. The fundamental-group homomorphism fits into the arithmetic exact sequence

$$1 \to \pi_{g,n} \to \mathcal{A}_{g,n+1} \to \mathcal{A}_{g,n} \to 1,$$

whose geometric row is (C.1) and whose quotient row is the identity of $G_{\mathbb{Q}}$. One may obtain exactness directly by this diagram: lift an element over a prescribed Galois element, then adjust by geometric surjectivity; the kernel lies in the geometric row. Universal monodromy is compatible with filling, so quotient this sequence by its monodromy kernels. By C.5 they are $C_{g,n+1}$ and $C_{g,n}$, and by C.4 the map between them is an isomorphism. Their intersection with $\pi_{g,n}$ is trivial by C.3. It follows that the quotient sequence is precisely (C.4). The arithmetic quotients in the resulting diagram are induced by the same structural morphism to $\operatorname{Spec}\mathbb{Q}$, hence their comparison is canonically the identity on $G_{\mathbb{Q}}$.

The same argument over a finite extension L gives the identity on G_L, compatibly with restriction from $G_{\mathbb{Q}}$. Changing the base points conjugates the group homomorphisms by geometric elements. It has no effect on their arithmetic quotients, which is the precise sense in which their isomorphism is canonical. For coverings defined over a number field the arithmetic quotients likewise compare through that same field group. This last observation concerns arithmetic images; it does not identify the full normalizers of arbitrary geometric structures under every abstract finite covering. □

On the scan of page 399, the first sequence has N, rather than the based automorphism group M, and its arrow is directed towards the type with one fewer puncture. Equation (C.4) is its correct arithmetic interpretation. On scan 395, equation (4) omits the pure-marking superscript on the geometric denominator in its left-hand quotient; equation (3) immediately above supplies the mathematically consistent version. We record both observations as evidence for the editors, not as silent changes to the source.

3.3 What the sequence marked "??" requires

Statement C.7 (Proved). The canonical based map supplied by a distinguished cusp.

Fix a representative inertia subgroup $I_i \subset \pi_{g,n+1}$ and put $\pi' = \pi_{g,n+1}/\overline{\langle I_i \rangle}^{\text{normal}}$ of type (g,n). There is a continuous homomorphism

$$\theta_i : O_{g,n+1,i} \longrightarrow \mathrm{Aut}_{\mathrm{lac}}(\pi'), \tag{C.5}$$

where $O_{g,n+1,i}$ stabilizes the conjugacy class of I_i. Its composite with the outer quotient is the usual filling map $f_i : O_{g,n+1,i} \to O_{g,n}$.

Proof. Represent an outer automorphism by an actual automorphism preserving I_i, which is possible by the definition of its stabilizer. It induces an automorphism of the quotient π'. Two such representatives differ by conjugation by an element of $N_{\pi_{g,n+1}}(I_i) = I_i$; the equality follows from Boggi and Zalesskii (2019, Proposition 3.6). That conjugation is killed in π'. Thus the induced genuine automorphism is independent of the representative. Composition and continuity follow by passing from the closed stabilizer of I_i in the automorphism group to its compact quotient. Replacing I_i by a conjugate conjugates the construction by the corresponding inner automorphism of π'. This is the based ambiguity. On the point-pushing subgroup $\pi_{g,n} \subset \check{P}_{g,n+1} \subset O_{g,n+1,i}$, the map is the inner-automorphism embedding of $\pi_{g,n}$ in $\mathrm{Aut}(\pi')$, as is seen first in the discrete Birman sequence and then by continuity. □

Statement C.8 (Open). The unrestricted filling assertion and its exact equivalent forms.

With the choices of C.7, the unrestricted third row of scan 399 would assert

$$1 \longrightarrow \pi_{g,n} \longrightarrow O_{g,n+1,i} \xrightarrow{f_i} O_{g,n} \longrightarrow 1. \tag{C.6?}$$

This is equivalent to θ_i in (C.5) being an isomorphism. It is also equivalent to bijectivity of the induced filling map on the unrestricted oriented prediscretization spaces

$$O_{g,n+1,i}/\Sigma_{g,n+1,i} \longrightarrow O_{g,n}/\Sigma_{g,n}. \tag{C.7?}$$

Here the source is canonically the corresponding marking quotient with the chosen cusp filled: the geometric permutation group permits the cusp to be fixed before taking the quotient. For individually labeled punctures use the pure variants in both numerator and denominator. We have not proved (C.6?) for the full O groups. Its smallest hyperbolic target instance is $(0,4) \to (0,3)$. The remaining conditions in this instance, and in general, are explicitly

$$f_i(O_{g,n+1,i}) = O_{g,n}, \qquad \ker(f_i) = \pi_{g,n}.$$

Proof of the stated equivalences. Since π' is center-free, the lower row of

$$\begin{array}{ccccccccc} 1 & \to & \pi_{g,n} & \to & O_{g,n+1,i} & \xrightarrow{f_i} & O_{g,n} & \to & 1 \\ & & \| & & \downarrow \theta_i & & \| & & \\ 1 & \to & \mathrm{Inn}(\pi') & \to & \mathrm{Aut}_{\mathrm{lac}}(\pi') & \to & O_{g,n} & \to & 1 \end{array}$$

is exact, with the left identification supplied by C.7. Exactness of the upper row is consequently equivalent to θ_i being an isomorphism, by elementary lifting and kernel arguments. C.3 with permutations gives $f_i(\Sigma_{g,n+1,i}) = \Sigma_{g,n}$ and kernel $\pi_{g,n}$ on that geometric subgroup. By B.13, (C.7?) is bijective if and only if f_i is onto and $f_i^{-1}(\Sigma_{g,n}) = \Sigma_{g,n+1,i}$. Under geometric surjectivity, the latter is equivalent to $\ker(f_i) \subset \Sigma_{g,n+1,i}$, and then the known geometric kernel makes it equivalent to the second condition above. This proves all the equivalences without assuming any of them. □

Thus the two geometric rows and the arithmetic row have been settled, while the unrestricted double-hat row has been reduced to its exact missing content. The map to genuine automorphisms in scan 348 and the map to outer automorphisms in scan 399 are not interchangeable: the former would be an isomorphism; the latter would have the point-pushing kernel. The manuscript's "peut-être entièrement faux" in folio 176, scan 349 is a question, not a counterexample. As in B.7 and B.14, the label Open here specifies what this audit has not established; no claim is made that a comprehensive literature proof of independence has been supplied. In particular, we do not rename this unresolved lifting problem the congruence subgroup problem: both involved genus-zero congruence assertions are known.

Statement C.9 (Known). The established combinatorial cuspidalization theorem.

Let X be a hyperbolic curve over an algebraically closed field of characteristic zero, and let X_m be its configuration space of m ordered distinct points. Let Λ be either all primes or a singleton, and let π_m be the maximal pro-Λ quotient of $\pi_1(X_m)$. Define $\mathrm{Out}^{\mathrm{FC}}(\pi_m)$ by preservation of the fiber subgroups of all configuration projections and of the cuspidal inertia classes in their surface-group subquotients, as in the cited definition. Then forgetting one point gives

$$\mathrm{Out}^{\mathrm{FC}}(\pi_{m+1}) \longrightarrow \mathrm{Out}^{\mathrm{FC}}(\pi_m)$$

which is injective for all $m \geq 1$, bijective for $m \geq 3$ if X is affine, and bijective for $m \geq 4$ if X is proper. The factor-permutation subgroup centralizes these FC-admissible outer automorphisms.

These are the exact assertions of Hoshi and Mochizuki (2011, Theorem 6.1, pp. 61–63), extending Mochizuki (2010, Theorem 4.1). In the original definition, F-admissibility preserves each fiber subgroup, rather than merely the collection up to an unspecified permutation; FC-admissibility adds cuspidal compatibility on the relevant subquotients. The image of an F-admissible automorphism from one higher configuration level is FC-admissible, by Hoshi and Mochizuki (2012, Theorem A and Corollary 1.9). These extra data are a substantive hypothesis. A single filled surface group is a fiber in this construction, not the whole configuration-space group, and an arbitrary member of $O_{g,n+1,i}$ has not thereby been supplied with such a lift. Consequently this theorem does not by itself remove the question mark in (C.6?). Its full and single-prime forms include the primes 2 and 3 in characteristic zero; its positive characteristic version requires the characteristic to lie outside the chosen set of primes.

3.4 What it means to reconstruct the entire étale topos

Statement C.10 (Refuted). The finite-cover topos is not the full étale topos.
For a connected positive-dimensional smooth curve X, the topos of continuous discrete $\pi_1^{\text{ét}}(X)$-sets need not, and in fact cannot, be its full small étale topos. Thus C.1 alone does not reconstruct "tout le topos étale".

Proof. In the classifying topos of any group, a subobject of the terminal object is an invariant subset of a one-element set. There are exactly two such subobjects. In the small étale topos of a curve, every Zariski open subset represents a subterminal object. A nonempty proper open subset gives a third one, distinct from the empty and the whole scheme. An equivalence of topoi preserves subterminal objects, so the topoi cannot be equivalent. One particularly transparent case is $X = \mathbb{P}^1_{\mathbb{Q}}$: its geometric fundamental group is trivial, while its étale topos still contains all its distinct open subobjects. The finite two-versus-three subobject count is also recorded in `code/bc_counterexamples.py`. The argument applies to the classifying topos of all continuous discrete sets, so it does not rely on confusing finite sets with all such sets. □

Statement C.11 (Proved). Reconstruction of the complete curve and its full étale topos.
Fix a sub-p-adic field K of characteristic zero, in particular a number field or a finite extension of $\mathbb{Q}_p$. On the essential image of hyperbolic curves over K, the arithmetic extension

$$1 \longrightarrow \pi_1(U_{\overline{K}}) \longrightarrow \Pi_U = \pi_1(U) \longrightarrow G_K \longrightarrow 1 \tag{C.8}$$

reconstructs the smooth proper completion X of U and its entire small étale topos, functorially for K-isomorphisms. The group datum includes the projection to the specified G_K; isomorphisms of it are taken modulo inner automorphisms from the geometric kernel. The resulting topos is determined up to the equivalence induced by the reconstructed K-isomorphism, with the usual natural-isomorphism ambiguity.

Proof. The relative anabelian Isom theorem in D.4 follows from Mochizuki (1999, Theorem A, equivalently Theorem 16.5 and its following Remark): for hyperbolic curves over a sub-p-adic field, the map from dominant K-morphisms to continuous open homomorphisms of arithmetic fundamental groups over G_K, modulo geometric inner automorphisms of the target, is bijective. Applied to an isomorphism and its inverse, this reconstructs a unique K-isomorphism $U \simeq U'$ inducing the prescribed outer class. Consequently the functor on isomorphism groupoids is fully faithful, and an inverse on its essential image reconstructs U with precisely this uniqueness. Existence is a hypothesis on the input datum, not a claim that every extension of G_K is geometric.

A smooth geometrically connected curve has a unique smooth proper completion up to unique isomorphism extending the identity on U. It may be constructed as the nonsingular proper model of its one-variable function field $F = K(U)$. An isomorphism of open curves extends uniquely to these proper models. It remains to show explicitly that this reconstructs the whole topos rather than just its finite coverings.

For every finite separable field extension L/F, let X_L be the normalization of X in L. It is finite over X. Consider the site whose objects are finite disjoint unions of opens $V \subset X_L$ on which $V \to X$ is étale. Morphisms are X-morphisms, equivalently the compatible generic field embeddings and their extensions on the chosen opens. Covering families are jointly surjective étale families. All these data are determined by the recovered curve X, and hence by (C.8).

This site is a basis for the small étale site of X. Indeed, a connected separated étale scheme Y of finite presentation over X is normal and integral, and its function field is a finite separable extension L/F. Zariski's main theorem identifies Y with an open subscheme of X_L. One can use the precise factorization of The Stacks Project Authors (2026, Lemmas 37.43.2 and 37.43.3, Tags 02LR and 05K0): a separated quasi-finite morphism factors as an open immersion followed by a finite morphism; normalizing and using the same generic field gives the specified normalization. Every étale scheme over X is covered by separated affine étale opens, and the quasi-compact ones are of finite presentation. Hence these objects form a covering basis even when a general object of the small site is not separated or quasi-compact. The comparison lemma for sites now identifies its sheaf topos with $\mathrm{Sh}(X_{\text{ét}})$. Transport of normalized extensions and opens along K-isomorphisms proves functoriality. □

This theorem gives a complete affirmative formulation of the reconstruction sought in §32, folios 206–217, scans 407–430. The exact phrase quoted in the prompt was not located as a literal sentence in scans 395–420; the subsequent construction of varying open groupoids and coverings occurs in scans 421–430. We cite that development rather than manufacture a quotation. The indispensable input in C.11 is the *arithmetic* group over its field, and the proof uses relative anabelian reconstruction before constructing the site. A bare geometric $\pi_{g,n}$ is the same for every complex curve of that type and cannot recover its algebraic moduli. Nor does retaining only the finite covers unramified over the original U produce the displayed site: finite extensions branched at newly removed points are also needed. We do not claim that the manuscript's direct construction from an independently characterized class of admissible splittings has thereby been completed without anabelian reconstruction.

Statement C.12 (Proved). The correct point-filling statement over a field.
For a smooth curve U' over a field K, a K-rational point $x \in U'(K)$, and $U = U' \setminus \{x\}$,

$$U(K) = U'(K) \setminus \{x\}.$$

Under the anabelian point maps this gives the corresponding bijection of *geometric point-induced* conjugacy classes of splittings, compatibly with filling. If the deleted set is merely Galois-stable, the correct statement uses its K-rational points; a point of that set need not itself be rational. The statement does not assert that every group section is point-induced.

Proof. A morphism $\operatorname{Spec} K \to U'$ factors through the open complement precisely when its image is not x. The equality follows. The maps on arithmetic fundamental groups commute with these morphisms, so the same equality holds for their images in the set of section classes; injectivity of the point map in the stated anabelian setting is the theorem recorded in D.5. Nothing in this argument requires surjectivity of the section map. This supplies the field-valued form of the expectation in §32, folios 211–212, scans 417–420, while keeping the word "admissible" tied to actual geometric points. □

Statement C.13 (Refuted). Deleting one section is not deleting one element of a section set.

The analogue of C.12 for sections over an arbitrary base is false: distinct sections of a family can intersect. Thus the assertion in scan 418 cannot be imposed on arbitrary profinite acting groups by treating all sections as a set of disjoint geometric points.

Proof. Over $\mathbb{Q}$ put $S = \mathbb{P}^1 \setminus \{0, 1, \infty\}$ and consider the constant hyperbolic family $S \times S \to S$, projection to the second factor. The sections

$$s(t) = (t, t), \qquad e(t) = (2, t)$$

are distinct and intersect at $t = 2$. Delete the image of e from the family. Its fibers now have type $(0, 4)$, but s does not lift to a section of the deleted family: at 2 its value is exactly the deleted point. It is therefore false that the new section set is the old one minus the single section e. All objects and morphisms are defined over $\mathbb{Q}$, and the intersection is checked by substituting $t = 2$ in $t - 2$ in the accompanying code. This also gives a geometric example for the acting group $\pi_1(S)$, so the defect is not an artifact of inventing unrelated group actions.

The manuscript itself recognizes this obstruction later: §32, folios 219–220, scans 433–435 withdraws the overgeneralized formulation because sections of families can meet. We preserve that withdrawal. C.12 over a field and C.11 via the reconstructed curve are the valid replacements needed for the present task. □

4 Arithmetic curves, inertia, sections, and faithfulness

Throughout this section a hyperbolic curve is a smooth geometrically connected curve U/K with smooth proper completion X, reduced complement $S = X \setminus U$, and type $(g, n) = (g(X), \#S(\overline{K}))$ satisfying $2g - 2 + n > 0$. Unless a different coefficient set is explicitly specified, all fundamental groups are full profinite étale groups. We choose compatible geometric base points and write

$$1 \longrightarrow \Delta_U \longrightarrow \Pi_U \xrightarrow{p_U} G_K \longrightarrow 1, \qquad \Delta_U = \pi_1(U_{\overline{K}}), \quad G_K = \mathrm{Gal}(\overline{K}/K).$$

Changing a geometric base point changes maps by geometric inner conjugacy. An inertia subgroup is the image of the geometric fundamental group of the punctured strict henselian local scheme at a point of $S_{\overline{K}}$; it is a conjugacy class of closed subgroups $I_s \simeq \widehat{\mathbb{Z}}(1)$. We write $\mathrm{Sec}(U/K)$ for continuous sections of p_U, modulo conjugation by Δ_U, and $\rho_U : G_K \to \mathrm{Out}(\Delta_U)$ for the outer action. The topology on automorphism groups is the topology of uniform agreement on finite characteristic quotients.

The distinction between an extension, a rational-point section of that extension, and a section of a moduli-stack extension is essential in reading §34. We first settle the statements that do not depend on the three 2026 proof packs.

4.1 What the arithmetic extension determines

Statement D.1 (Proved). A centre-free extension is its outer-action pullback.

Let Δ be a topologically finitely generated centre-free profinite group and let $1 \to \Delta \to E \xrightarrow{p} G \to 1$ be an extension of profinite groups. If $\rho : G \to \mathrm{Out}(\Delta)$ is its associated outer action, then

$$E \longrightarrow \mathrm{Aut}(\Delta) \times_{\mathrm{Out}(\Delta)} G, \qquad e \longmapsto (\mathrm{conj}_e \,|_\Delta, p(e))$$

is an isomorphism of extensions. Sections of p, modulo Δ-conjugacy, therefore correspond exactly to homomorphic lifts of ρ to $\mathrm{Aut}(\Delta)$, modulo $\mathrm{Inn}(\Delta)$-conjugacy. This justifies the equivalence asserted in §34, folio 245, scan 483.

Proof. The displayed map is a continuous homomorphism. Its kernel lies in Δ and centralizes Δ, hence is trivial. Given (a, g) in the fibre product, choose $e \in E$ above g. Equality of outer classes says $a = \mathrm{conj}_d \mathrm{conj}_e$ for some $d \in \Delta$; thus (a, g) is the image of de. A continuous bijection between compact Hausdorff groups is an isomorphism. Both assertions about sections follow from this isomorphism. Notice that $E \to \mathrm{Aut}(\Delta)$ by itself need not be injective: it is the pair with p that is always injective. □

Statement D.2 (Refuted). The literal open-normalizer criterion in scan 483.

It is false that the cuspidal inertia subgroups are the maximal closed subgroups $L \subseteq \Delta_U$ for which $p_U(N_{\Pi_U}(L))$ is open in G_K. It is also false that every nontrivial closed subgroup with this property is contained in one inertia subgroup, as suggested in the margin of scan 483. These failures already occur for $U = \mathbb{P}^1_{\mathbb{Q}} \setminus \{0, 1, \infty\}$.

Proof. The subgroup $L = \Delta_U$ is normal in Π_U, so its normalizer is all of Π_U and its projected normalizer is G_K. It is maximal among subgroups with the stated property simply because it contains every subgroup under consideration. Here $\Delta_U \simeq \widehat{F}_2$ has quotient $(\mathbb{Z}/2)^2$, whereas every inertia subgroup is procyclic. Thus it cannot be an inertia subgroup or be contained in one. This is an example in the arithmetic group of the curve, not merely an abstract profinite extension.

Even adding the word "proper" would not repair the marginal assertion. The characteristic subgroup $[\Delta_U, \Delta_U]$ is proper, is normalized by Π_U, and is not procyclic. For the last assertion map the two free generators onto (12345) and (123) in A_5. They generate A_5: conjugates give adjacent 3-cycles, which generate all even permutations. Since A_5 is perfect, the image of $[\Delta_U, \Delta_U]$ is A_5, which is not cyclic. The attached computation `code/d_counterexamples.py` verifies these finite calculations and records its output.

Finally, if the condition were instead imposed on subgroups of Π_U, every rational-point section $s(G_K)$ would qualify because its normalizer contains $s(G_K)$. Such a subgroup projects isomorphically onto G_K and is not geometric inertia. For instance $2 \in U(\mathbb{Q})$ gives such a section. This explains why rational-point decomposition groups must not be silently identified with cuspidal inertia. □

The scan actually says subgroups "dans π"; no doubtful glyph supplies the missing procyclic or cyclotomic hypotheses. The alternatives Γ^+ and Γ in the projected-normalizer phrase make no difference: G_K is open in $G_{\mathbb{Q}}$.

Statement D.3 (Known). The cyclotomic recognition theorem for affine inertia. Let U/K be an affine hyperbolic curve over a number field. A nontrivial element $z \in \Delta_U$ belongs to a cuspidal inertia subgroup if and only if there is a closed subgroup $N \subseteq \Pi_U$ such that N normalizes $\overline{\langle z \rangle}$, $p_U(N)$ is open in G_K, and

$$hzh^{-1} = z^{\chi_K(p_U(h))} \qquad (h \in N),$$

where $\chi_K : G_K \to \widehat{\mathbb{Z}}^\times$ is the full cyclotomic character. In particular the action factors through $p_U(N)$; this condition includes triviality of the conjugation action of $N \cap \Delta_U$. The maximal procyclic subgroups having this property are precisely the cuspidal inertia subgroups. The collection is intrinsic to the arithmetic extension together with its cyclotomic character.

Proof. The equivalence is Nakamura (1994, Theorem (2.1.1)), applied to the full class of finite groups. The source assumes $n \geq 1$, characteristic zero, and a number field; all these hypotheses have been retained here. For a cusp the branch-cycle description of its decomposition group gives the displayed conjugation law. Conversely Nakamura's weight argument identifies z as peripheral. If a procyclic subgroup satisfying the condition is generated topologically by z, it lies in the inertia subgroup supplied by the theorem. The inertia subgroup itself satisfies the condition and is therefore the maximal one. For completeness, distinct maximal peripheral inertia subgroups intersect trivially: if $1 \neq x$ belongs to two of them, Boggi and Zalesskii (2019, Proposition 3.6) gives $C_{\Delta_U}(x) = I$ for the first subgroup I, since a peripheral generator is algebraically simple. The second subgroup centralizes x, hence is contained in I and equals I by maximality. This also gives uniqueness of the containing inertia subgroup for a nontrivial cyclotomic subgroup. □

We do not erase "cyclotomic" from this theorem. The literal assertion refuted in D.2 and the published assertion in D.3 have different hypotheses. An isomorphism of arithmetic extensions over G_K preserves the latter hypotheses, so it recovers the inertia conjugacy classes without their being supplied as extra data. The conclusion for arbitrary open homomorphisms follows more cleanly from the Hom theorem below than by treating a normalizer calculation as if it commuted with a noninjective homomorphism.

Statement D.4 (Known). The relative Hom theorem, including its exact ambiguity. Let K be sub-p-adic, meaning a subfield of a finitely generated extension of $\mathbb{Q}_p$, and let U, V be hyperbolic curves over K. There is a bijection

$$\mathrm{Hom}_K^{\mathrm{dom}}(U, V) \xrightarrow{\sim} \mathrm{Hom}_{G_K}^{\mathrm{open}}(\Pi_U, \Pi_V)/\mathrm{Inn}(\Delta_V).$$

On the right the maps are continuous homomorphisms with open image whose composite with p_V is p_U. In particular this applies over every number field and every finite extension of $\mathbb{Q}_p$, including $p = 2, 3$. A homomorphism in this set takes a cuspidal inertia subgroup either to the trivial group or into a conjugate of a target cuspidal inertia subgroup.

Proof. The bijection is Mochizuki (1999, Theorem A; Theorem 16.5 and its following Remark). We use the full profinite version. The paper also has a geometrically pro-p version, whose arithmetic quotient is still the full G_K; it is not a theorem about replacing the whole arithmetic group by its maximal pro-p quotient.

For clarity we verify the peripheral consequence, which is the assertion of §34, folios 245–245 verso, scans 483–484. The associated nonconstant morphism $f : U \to V$ extends uniquely to the smooth proper completions $\bar{f} : X \to Y$. If a deleted point of X maps to a point of V, its local inertia maps trivially in $\pi_1(V)$. If it maps to a deleted point of Y, choose local parameters t, u at the source and target. The equality $\bar{f}^* u = t^e a$ with a a local unit shows that tame geometric inertia maps into the target inertia by multiplication by the positive ramification index e. Characteristic zero makes all these horizontal inertia groups tame. Thus "respects loops" allows a loop to be killed; it does not mean a permutation of all punctures. □

The relative Isom theorem follows formally: the two homomorphisms associated to a group isomorphism and its inverse give morphisms whose composites are identities by the injectivity of this bijection. No separate compatibility with a previously chosen loop system is required.

Statement D.5 (Proved). Admissible effective lifts are exactly the point sections. Let U/K be hyperbolic with K sub-p-adic. Call a lift of ρ_U *geometrically admissible* if it is obtained from a hyperbolic curve V/K, a point $v \in V(K)$, and a G_K-equivariant outer isomorphism $\Delta_V \simeq \Delta_U$ by transporting the action defined by v. Then the admissible lifts modulo Δ_U-conjugacy are in bijection with $U(K)$. This proves the deduction in §34, scan 485, folio read 246 | 245, without assuming surjectivity of the section map.

Proof. By D.1 the given outer isomorphism produces an isomorphism $\Pi_V \simeq \Pi_U$ over G_K, with the stated geometric ambiguity. The Isom consequence of D.4 gives a unique K-isomorphism $f : V \to U$. Functoriality carries the section at v to the section at $f(v)$. Conversely the point $u \in U(K)$ supplies the required admissible datum by taking $V = U$, $v = u$, and the identity outer isomorphism. Finally two points cannot give the same section class: Mochizuki (1999, Theorem C = Theorem 19.1) proves injectivity of the point-to-section map under precisely these field and hyperbolicity hypotheses. Thus the two constructions give the asserted bijection. Changing the chosen representative of the outer isomorphism conjugates the lift by Δ_U, which is exactly the ambiguity already factored out. □

This proposition is about lifts that are already assumed admissible. It does not say that every lift is admissible. For a proper curve of genus at least two that additional assertion is the section conjecture; for an affine curve the presence of cuspidal sections changes the assertion itself.

Statement D.6 (Refuted). All affine sections need not come from interior points. For $U = \mathbb{P}^1_{\mathbb{Q}} \setminus \{0, 1, \infty\}$ there is a continuous section of $\Pi_U \to G_{\mathbb{Q}}$ which is not the section of any point of $U(\mathbb{Q})$. Thus an unqualified affine version of "every lift is the lift of a rational point of U" is false.

Proof. Take the rational tangential base point at $t = 0$ with tangent vector 1. The associated fibre functor gives a section s_0. More explicitly, embed the local punctured disk in the Puiseux

field $\bigcup_m \overline{\mathbb{Q}}((t^{1/m}))$ and let $G_{\mathbb{Q}}$ act on coefficients while fixing the compatible symbols $t^{1/m}$. On the finite covers of U this gives the tangential section; changing paths conjugates it by Δ_U.

Under the morphism $t : U \to \mathbb{G}_m$, its Kummer class in $H^1(G_{\mathbb{Q}}, \mu_m) = \mathbb{Q}^\times/(\mathbb{Q}^\times)^m$ is zero for every m, since each $t^{1/m}$ is fixed by the section. If s_0 were the section of $u \in U(\mathbb{Q})$, the same classes would be the classes of u. Hence u would belong to $\bigcap_{m\geq 1}(\mathbb{Q}^\times)^m$. Every rational prime valuation of such a number is divisible by every m, so all valuations vanish and $u = \pm 1$. Taking $m = 2$ removes -1, giving $u = 1$, which is not in U. This contradiction proves the assertion. □

The appropriate affine section conjecture includes rational cusps and their cuspidal sections. D.6 does not refute that conjecture.

4.2 The section manuscript: an audited conditional chain

Statement D.7 (Open). The 2026 proper p-adic section claim.
The assertion

$$X(k) \xrightarrow{\sim} \mathrm{Sec}(X/k) \quad (k/\mathbb{Q}_p \text{ finite},\ X/k \text{ proper},\ g(X) \geq 2)$$

for every prime p is claimed in OpenAI (2026c, Theorem 1.1). The precise new input is its Theorem 1.2: for $F = k(X)$, a fixed separable closure F^{sep}, and the subfield F^{et} generated by all connected finite étale covers of X with all their F-embeddings,

$$F^{\mathrm{sep}} \subseteq \widehat{F^{\mathrm{et}}} \quad \text{inside } \widehat{F^{\mathrm{sep}}} \tag{D.7.1}$$

for every fixed prolongation of every real-valued rank-one norm on F extending the normalized p-adic norm. The copies of the fields in this inclusion are specified; an abstract embedding of a finite extension into some unrelated completion would not suffice. We retain the label Open for the unconditional new claim in this audit: claimed, unrefereed, and not completely verified by us. We have found no counterexample. The conditional implication D.8 is proved below.

What we checked in Sections 3–6 of the pack. We read these sections in full and followed their logical order, including the distinction between horizontal ramification in characteristic zero and wild reduction in characteristic p. The patching construction in Proposition 3.2 requires a root on every component above each overlap annulus. With that hypothesis the pulled-back branched cover splits on the overlap, so gluing trivial covers over the inner disks produces a cover étale over the entire proper curve. Proper GAGA and invariance of finite étale covers under extension between algebraically closed characteristic-zero fields are used after gluing. Lemma 3.3 then evaluates at the chosen completion-valued point. We checked why this yields a field embedding rather than merely an analytic map. Lemma 3.4 uses cyclic horizontal inertia over $\overline{k}((t))$: divisibility of the branch indices by one integer is preserved under compositum and Galois closure, and an upper cover étale over a lower cover introduces no further horizontal ramification.

For Lemma 4.2 we checked the graph constructions separately. A cycle in the stable dual graph carries a tame cyclic cover of degree $\ell \neq p$, with expansion factor ℓ on each lifted edge. A separating edge is first lifted to parallel edges using an unramified cover on the two sides; stability ensures that each side has a cycle or a component of positive genus. We checked that taking fibre products and then connected components retains the required roots on all remaining sheets. The external lifting result is Amini et al. (2015, Theorem 7.7); its tame-cover definition includes absence of additional branch points when the marked puncture set is empty. For Lemma 4.3 we opened Mochizuki and Tsujimura (2024, Theorem A and Proposition 2.4(iv)). The latter gives a stable model of a finite étale cover dominating a prescribed model. Taking an

irrational valuation level excludes vertices and smooth disk tubes, so the inverse image of a sufficiently small rational-ended collar lies in node annuli. This is the specific model domination needed; semistable reduction alone would not provide it. The accessible primary version is dated June 16, 2023; we did not silently identify it with the March 30, 2024 revision cited by the pack.

In Section 5 the period lattice in a totally degenerate Jacobian is torsion-free. Consequently local lifts of a multiplication-by-m pullback can be adjusted by periods and glued without a residual torsion ambiguity. We checked that Lemma 5.1 is intended to give a root of a torus character on the whole pullback, not just one component. Proposition 5.2 uses $y^\ell = T(T-1)/(T-\alpha)$, with ℓ odd and $\ell \neq p$, whose skeleton has two rational vertices joined by ℓ edges. A cycle formed from two edges supplies a torus character with slope one on the chosen edge. If $u = cT(1+h)$ and $s = v(T)$ on an annulus of length a, the Laurent coefficient inequalities give $v(h) \geq \min(s, a-s)$. The trimming condition

$$\min(s, a-s) > v_p(m) + 1/(p-1)$$

is sufficient for an analytic mth root of $1+h$, also when $p \mid m$. We checked this numerical point; replacing it by a prime-to-p root argument would not prove Theorem 1.2. We also checked the use of normality in Section 6: a completion-valued embedding of a finite normal extension into the fixed completion forces its specified image there to be the same normal subfield.

This is a substantial audit of the proof's mechanism, not a complete independent verification of all analytic constructions. In particular we have not fully reproduced the asserted simultaneous stable-model and toric-uniformization construction of Proposition 5.2, with its identification of the chosen annular coordinate on every inverse component. The opened Abel–Jacobi skeleton comparison of Baker and Rabinoff (2015, Proposition 6.1 and Corollary 6.6) explains the slope calculation but is not by itself the whole analytic construction. This is the remaining verification obligation at which we stop promoting (D.7.1) to a theorem. All later uses of (D.7.1) below are explicitly conditional. The separate attachment *Étale covers with a prescribed exterior sheet* is not cited by the main section manuscript and is not used here; its theorem is claimed, steps not checked by us. Nor do we use Corollary 8.4, which depends on an unattached Selmer-converse manuscript.

Statement D.8 (Proved). The completion assertion implies the proper local section theorem.

Assume (D.7.1) for a fixed proper hyperbolic curve X/k, with $k/\mathbb{Q}_p$ finite. Then $X(k) \to \mathrm{Sec}(X/k)$ is bijective. Thus the group-theoretic and valuative passage in Section 7 of the pack does not require an additional affine or birational-lifting conjecture.

Proof. Put $F = k(X)$, F^{et} as above, and prolong a rank-one valuation w extending the p-adic valuation to F^{sep}. Restriction gives

$$D_{\bar{w}}(F^{\mathrm{sep}}/F) \longrightarrow D_{\widetilde{w}}(F^{\mathrm{et}}/F). \tag{D.8.1}$$

It is surjective: lift an automorphism of F^{et}/F to F^{sep}/F, then correct the lift using the transitive action of $\mathrm{Gal}(F^{\mathrm{sep}}/F^{\mathrm{et}})$ on prolongations. The infinite-extension assertion follows from the finite assertions by compactness. It is injective because an element of its kernel is an isometry fixing F^{et}, hence fixes its completion, which contains the specified F^{sep} by (D.7.1). Normalization $w(p) = 1$ ensures that stabilizing the valuation ring preserves the absolute value itself. The two decomposition groups are closed, so (D.8.1) is an isomorphism of profinite groups.

No such rank-one decomposition group in Π_X can contain the image of a section. Otherwise (D.8.1) lifts it to a birational section $\widetilde{s} : G_k \to G_F$. By Koenigsmann (2005, Proposition 2.4(2),

Definition 1.4, and Observation 1.3), the entire image Σ is also contained in a decomposition group of ord_a for some $a \in X(k)$. In the fixed field $M = (F^{\mathrm{sep}})^{\Sigma}$ the restrictions v_1, v_2 of these two selected prolongations are nontrivial rank-one henselian valuations. Henselianity follows because $G_M = \Sigma$ stabilizes the prolongation, while transitivity gives uniqueness. Algebraic extension changes the value group by a torsion quotient and hence preserves its rank. They satisfy $v_1(p) = 1$ and $v_2|_{k^\times} = 0$.

Here is the explicit contradiction, including the dyadic case. Choose a nonsquare $d \in k^\times$. It is still a nonsquare in M, because $\Sigma \to G_k$ is onto. Let $t \in F$ be a uniformizer at a, choose an integer $N > v_1(t)$, and set $z = p^N/t$. Then $v_1(z) > 0$ and $v_2(z) < 0$. For positive integers r put

$$q_r = d + (1-d)\frac{z^r}{1+z^r}.$$

The denominator is nonzero. Direct valuation computations give

$$v_1(q_r/d - 1) = v_1((1-d)/d) + rv_1(z) \longrightarrow +\infty,$$
$$v_2(q_r - 1) = v_2(d-1) - rv_2(z) \longrightarrow +\infty.$$

For large r these values exceed $2v_1(2)$ and $2v_2(2)$, respectively. Hensel's lemma at 1 gives square roots of q_r/d and q_r in M; the quotient makes d a square. The contradiction uses neither discrete value groups nor $p \neq 2$.

Now let $s : G_k \to \Pi_X$ be arbitrary. Pop and Stix (2017, Theorem 26 in the 2011 preprint) places its image in the decomposition group of a valuation of F extending the p-adic valuation. The valuation transcendence inequality gives rational rank at most two, hence ordered rank at most two. Rank one has just been excluded. For rank two, let u be the rank-one coarsening. Its prolongation is preserved by the section: the relevant convex subgroup of the value group has a unique extension under an algebraic field extension, because the quotient of value groups is torsion. If u is nontrivial on k, the preceding rank-one argument excludes it. Otherwise properness gives a closed centre $a \in X$, and a valuation dominating the discrete valuation ring $\mathcal{O}_{X,a}$ and trivial on k is equivalent to ord_a. Its residue field is $k(a)$, and the remaining refinement is the unique p-adic valuation of this finite extension. The image on constants of its horizontal decomposition group is $G_{k(a)}$. Since s projects onto G_k, we obtain $k(a) = k$. Horizontal inertia is trivial in the full étale tower of the proper smooth curve, so this decomposition group is precisely the point section at a. This is also the type-$2h$ calculation in Pop and Stix (2017, Section 6.1). Surjectivity follows.

For completeness, injectivity can be checked without (D.7.1). If $a, b \in X(k)$ give the same section, use a for the Abel–Jacobi embedding into J. The multiplication-by-m torsor is finite étale for every $m \geq 1$ in characteristic zero. Equality of section classes implies $[b-a] \in mJ(k)$ for every m. An open subgroup of the compact analytic group $J(k)$ is $\mathbb{Z}_p^{g[k:\mathbb{Q}_p]}$. If it has index q, then $\bigcap_m mJ(k) \subseteq \bigcap_r p^r \mathbb{Z}_p^{g[k:\mathbb{Q}_p]} = 0$, using $m = qp^r$. Thus $[b-a] = 0$. A nontrivial equivalence of the two distinct degree-one divisors would give a degree-one map $X \to \mathbb{P}^1$; this is impossible for positive genus. Hence $a = b$. □

Statement D.9 (Proved). The exact conditional global descent statement.

Assume the proper local section theorem in D.7 over every finite completion of a number field K. For a proper hyperbolic curve C/K, define

$$C(\mathbb{A}_K)_\bullet = \prod_{v \nmid \infty} C(K_v) \times \prod_{v \text{ real}} \pi_0(C(K_v)).$$

Complex places are omitted. Let $C(\mathbb{A}_K)_\bullet^{\text{f-cov}}$ be the locus surviving every torsor under a finite K-group scheme, with survival meaning liftability to some global twist. Then

$$\mathrm{loc}(\mathrm{Sec}(C/K)) = C(\mathbb{A}_K)_\bullet^{\text{f-cov}}, \qquad C(K) \xrightarrow{\sim} \mathrm{Sec}(C/K) \iff C(\mathbb{A}_K)_\bullet^{\text{f-cov}} = C(K). \quad \text{(D.9.1)}$$

This is the precise conditional content of Proposition 8.1 and Theorem 8.2 of OpenAI (2026c). The general equality on the right remains an additional number-theoretic assertion, not a consequence of local point-theoreticity.

Proof. The local theorem gives a unique point at each finite place; the real section theorem gives a connected component at each real place (Bresciani and Vistoli, 2020, Introduction, Theorem). Thus localization is well-defined. Apply Harari and Stix (2012, Theorem 2.1 and Remark 2.2(1), preprint numbering) with its normal subgroup $U = 1$ and omitted set of places $S = \varnothing$. Its equivalence between condition (i) and condition (iii) is exactly the equivalence between survival of all finite torsors and interpolation of the local point sections by a global section, modulo the full geometric group. Evaluations are constant on real components and complex places impose no condition. We use the corrected convention of Stoll (2017, Section 1); disconnected torsors are included. This proves the first equality.

If the descent locus is $C(K)$, the local points of a section are the localizations of one $a \in C(K)$. Its localization has finite support $\{a\}$, so Stix (2015, Corollary 5) makes the section geometric. Properness excludes cuspidal sections. Injectivity is D.5, or follows by restriction to a single finite completion. Conversely bijectivity of the point-to-section map and the first equality immediately give the descent equality. These arguments also cover empty sets. □

Statement D.10 (Proved). What the modular-curve consequence says about a moduli section.

Assume the local theorem of D.7. Let N satisfy $g(X_1(N)) \geq 2$, and let $s : G_{\mathbb{Q}} \to \pi_1(Y_1(N))$ be a section. Its composite with $\pi_1(Y_1(N)) \to \pi_1(X_1(N))$ is geometrically conjugate to the section of a unique rational cusp of $X_1(N)$. Consequently the same conclusion applies to a section of the arithmetic $N_{1,1}$ whose image is contained in the open subgroup belonging to $Y_1(N)$, under the faithful stack identification of Part E. It does not assert that $s(G_{\mathbb{Q}})$ is contained in a cuspidal decomposition subgroup of $\pi_1(Y_1(N))$.

Proof. The forgetful map from the fine modular curve $Y_1(N)$ to $\mathcal{M}_{1,1,\mathbb{Q}}$ is finite étale in characteristic zero, so its fundamental group is an open subgroup, up to conjugacy. Containment of the image of a section makes the same homomorphism a section of this open subgroup's projection onto $G_{\mathbb{Q}}$. Here $g(X_1(N)) \geq 2$ ensures that N is outside the small levels with stabilizers on $Y_1(N)$. The compactification homomorphism is over $G_{\mathbb{Q}}$ and hence its composite is a section of the proper curve.

Stoll (2007, Corollary 8.8), with Stoll (2017), proves $X_1(N)(\mathbb{A}_{\mathbb{Q}})^{\text{f-ab}}_{\bullet} = X_1(N)(\mathbb{Q})$ when the genus is positive. Since rational points are contained in the finite-cover locus and that locus is contained in the finite-abelian locus, D.9 applies. This verifies the implication used in OpenAI (2026c, Corollary 8.5) without its unrelated Corollary 8.4. Finally Mazur (1977, Introduction, Theorem (7), p. 35) states directly that all rational points of $X_1(N)$ are cusps whenever its genus is positive. The equivalent torsion-order assertion is Theorem (7'); the full torsion classification is Theorem (8). Thus the proper section comes from a cusp, and D.5 gives its uniqueness.

The kernel of compactification is generated normally by cusp inertia. Identifying the image of s after quotienting by this kernel does not identify s before quotienting. D.7 is a theorem claimed for proper curves and provides no such lifting assertion for $Y_1(N)$. Nor does the conclusion imply nonexistence of these splittings: a nonzero rational tangent vector at any rational cusp gives a cuspidal section of $Y_1(N)$, exactly as in D.6. □

4.3 Relative and absolute open homomorphisms

Statement D.11 (Proved). An absolute open homomorphism preserves the geometric kernel.

Let U/K and V/K' be hyperbolic curves over number fields and let $h : \Pi_U \to \Pi_V$ be a continuous homomorphism with open image, with no compatibility assumed. Then $h(\Delta_U) \subseteq \Delta_V$, and there is a unique continuous homomorphism with open image $\alpha : G_K \to G_{K'}$ such that $p_V h = \alpha p_U$. This conclusion does not depend on the 2026 open-Hom manuscript.

Proof. Set $J = p_V(h(\Pi_U))$. The projection p_V is an open map of profinite groups, so J is open in $G_{K'}$, and $J = G_L$ for a finite extension L/K' in the chosen algebraic closure. The subgroup $A = p_V(h(\Delta_U))$ is compact, hence closed, is topologically finitely generated because Δ_U is, and is normal in J because Δ_U is normal in Π_U. By Mochizuki (2006, Theorem 2.4), a number field's absolute Galois group has no nontrivial topologically finitely generated closed normal subgroup. Thus $A = 1$. The universal property of the quotient gives a unique continuous α, and its image is J. □

The same argument is the mechanism behind Mochizuki (2006, Proposition 3.3) for isomorphisms. Merely calling a geometric subgroup "characteristic" would not justify its preservation by an arbitrary homomorphism; the proof above supplies the missing step. Openness matters because it makes J an absolute Galois group of a number field, to which the theorem applies.

Statement D.12 (Proved). The conditional absolute Hom theorem and its known cases.

Assume OpenAI (2026b, Theorem 1.1) for absolute Galois groups of number fields. Then every h as in D.11 is induced, up to target geometric inner conjugacy after fixing the base-field map, by a unique field embedding $j : K' \hookrightarrow K$ and a unique dominant K-morphism

$$f : U \longrightarrow V \otimes_{K',j} K.$$

More invariantly there is a bijection

$$\coprod_{j:K'\hookrightarrow K} \mathrm{Hom}_K^{\mathrm{dom}}(U, V \otimes_{K',j} K) \;\simeq\; \mathrm{Hom}^{\mathrm{open}}(\Pi_U, \Pi_V)\big/ \mathrm{Inn}(\Pi_V). \tag{D.12.1}$$

Without the 2026 claim the same conclusion is known whenever the induced α of D.11 is injective, or is cyclotomic-compatible, and in particular for isomorphisms h. It is also known when the source field is $\mathbb{Q}$.

Proof. Apply the assumed theorem to α. It gives a unique compatible embedding $\bar{j} : \overline{K'} \to \overline{K}$ with $\alpha(\sigma) = \bar{j}^{-1}\sigma\bar{j}$ and $j = \bar{j}|_{K'}$ taking values in K. Between algebraic closures of number fields such an embedding is onto: its image is algebraically closed and contains $\mathbb{Q}$, so it contains every element algebraic over $\mathbb{Q}$. Consequently α is injective as well as having open image.

Base change identifies the arithmetic group of $V \otimes_{K',j} K$ with

$$P = \Pi_V \times_{G_{K'},\alpha} G_K.$$

The map $\widetilde{h}(x) = (h(x), p_U(x))$ takes values in P by D.11 and is over G_K. The first projection identifies P with the open subgroup $p_V^{-1}(\alpha(G_K))$ of Π_V, since α is injective. Thus $\widetilde{h}$ has open image in P. D.4 supplies the unique dominant f with the prescribed geometric conjugacy class.

We check the larger ambiguity in (D.12.1). Conjugating h by $e \in \Pi_V$ conjugates α by $p_V(e) \in G_{K'}$; the associated change of $\bar{j}$ does not alter its restriction to K'. Hence j is intrinsic

to the absolute outer class. Once α is fixed, an e conjugating one representative to another over the same α has $p_V(e)$ centralizing the open subgroup $\alpha(G_K)$. This centralizer is trivial by Mochizuki (2006, Corollary 2.2). The remaining conjugation is therefore geometric and D.4 gives exactly the stated uniqueness of f. Conversely a pair (j, f) gives such an open homomorphism: the relative map has open image and the base-change subgroup is open in Π_V.

For the unconditional variants, an injective α is an isomorphism onto an open subgroup G_L; the Neukirch–Uchida isomorphism theorem gives the compatible field embedding. One precise solvably-closed formulation is Uchida (1979, Theorem, p. 359), also reproduced as Mochizuki (2006, Theorem 3.1). For cyclotomic compatibility use Hoshi (2025, Theorem 3.4). If h is an isomorphism, apply D.11 also to h^{-1}, so that α is an isomorphism. If $K = \mathbb{Q}$, use Uchida (1981, Theorem 1). The rest of the proof is unchanged. □

For the original relative formulation in §34, scans 483–484, the base fields lie in one $\overline{\mathbb{Q}}$ and the maps are already over $G_{\mathbb{Q}}$. Its use of the Hom theorem is unconditional. The additional absolute assertion in D.12 is the part depending on the new claim; D.11 is not an unresolved geometric-kernel gap.

Statement D.13 (Open). The 2026 unrestricted Galois open-Hom claim.
Let E_i/F_i be solvably closed Galois extensions of number fields, meaning that E_i have no nontrivial abelian algebraic extensions. The assertion that every continuous homomorphism with open image

$$\alpha : \mathrm{Gal}(E_1/F_1) \longrightarrow \mathrm{Gal}(E_2/F_2)$$

is induced by a unique compatible field embedding $E_2 \hookrightarrow E_1$ is OpenAI (2026b, Theorem 1.1), the conjecture on p. 595 of Uchida (1981). We record it as claimed, unrefereed, with its full unconditional validity not completely verified by us. D.12 is a proved conditional consequence, not an assertion that the conjecture has thereby been settled.

Proof audit of the open-Hom manuscript. We read Sections 1–6, including the full proofs of the new lemmas rather than only the outline, and checked the following dependencies. The reduction to $P = \mathrm{Gal}(\mathbb{Q}^{\mathrm{sol}}/\mathbb{Q})$ is valid: a solvably closed algebraic field contains the roots of unity and is closed under radicals, hence contains $\mathbb{Q}^{\mathrm{sol}}$. For a number field $M \subset \mathbb{Q}^{\mathrm{sol}}$ this is also its maximal prosolvable extension; a solvable normal closure of $M/\mathbb{Q}$ followed by a compositum of conjugates proves the assertion. We opened the precise local input in Uchida (1981, Section 1, pp. 595–597 and Proposition 1): for a fixed odd prime ℓ, nontrivial pro-ℓ inertia forces the nonabelian local case and hence an injective map on that inertia into a unique target prime; all but finitely many target primes occur. The excluded torsion case is dyadic. This result gives cofinite coverage for each fixed ℓ, not a bound uniform in ℓ. The claimed uniform bound later in the pack is therefore a real new step. We also opened the uniqueness statement, Proposition 2 of Uchida, and Hoshi (2025, Theorem 3.4); the latter proves that full cyclotomic compatibility is equivalent to geometricity.

Lemma 3.1 supplies the additive-character detection that drives the proof. We followed its interpolation determinant and its separate translate/degree zero estimate. The relevant strict inequality is

$$n + n/r > n + 1 + n/m \qquad (1 \le r < n,\ m > n(n-1)).$$

It compares the nonarchimedean determinant decay with the archimedean height bound; no asserted Leopoldt rank is substituted. In Proposition 3.2 the logarithmic image of global units is the actual image, and generators of powers of distinct prime ideals give disjoint divisor supports. Thus the claimed independence argument is compatible with class-field theory without assuming Leopoldt's conjecture. We checked the distinct roles of the absolute-degree-one version,

the Frobenius-average version, and the nonzero-corestriction version. Confusing them would invalidate the later prime comparisons.

In Section 4 the map is first restricted to a normal open subgroup and then extended to the normalizer of its kernel. The regular-module enlargement in Lemma 4.2 increases subgroup orders while keeping the bound on element orders fixed. Proposition 4.3 compares characters via a cyclic subgroup and its proper subgroups. We checked that its semisimple projection is U-equivariant for the normal open subgroup U; the argument does not require the larger intermediate subgroups to be normal in the whole normalizer. We followed the explicit source and target prime choices, the residue-degree f, and the separation into those powers of Frobenius that lie in U and those that lie in the other kernel conjugates. The result is surjectivity and logarithmic cyclotomic compatibility, not yet full cyclotomic compatibility.

Section 5 passes to $\mathbb{Q}(i)$ and uses a sign character to show that almost every split target prime has only finitely many possible source partners, as the auxiliary prime varies. Lemma 5.2 is an S-unit Kummer statement; the finite class-group contribution vanishes in the relevant inverse limit, so it is not a mistaken direct-sum assertion about all primes. In Proposition 5.3 a finite cyclotomic twist is killed in a cyclic extension. The extension degree can grow with ℓ, but the multiplicity of the one specified character in the S-unit representation is bounded by the number of S-orbits. The resulting bound $2[K:\mathbb{Q}]+2$ is independent of ℓ. We checked why this independence is needed before applying the simultaneous Kummer/Chebotarev separation of Lemma 5.4; the fixed-ℓ cofiniteness from Uchida cannot replace it.

Finally Section 6 compares character pullbacks at rational primes split completely in an auxiliary finite field. An infinite subset with one fixed conjugate gives equality of the corresponding additive characters by Proposition 3.2. Faithfulness of the signed regular representation yields equality modulo each finite quotient. Compactness of the sets of possible conjugating elements, with the finite-intersection property obtained by passing to composita, supplies a single conjugation. Extending equality from a normal open subgroup uses the trivial centralizer of an open subgroup of P. The resulting full cyclotomic compatibility is the hypothesis of Hoshi's theorem, which finishes the claimed proof.

These checks identify a coherent chain and expose where an apparently weaker replacement would fail. They do not constitute an independently reproduced proof of every technical generality of the zero estimate in Lemma 3.1, the character-separation argument of Proposition 4.3, and the uniform multiplicity bound in Proposition 5.3. We therefore do not treat those three new inputs as established merely because their conclusions fit the reduction. No verified counterexample or contradictory hypothesis was found. The attached eight-lesson exposition is not evidence of independent verification; we do not use it as an authority. No Lean proof is supplied for this pack.

4.4 Centres, restriction to covers, and the faithfulness questions

Statement D.14 (Known). The centres and centralizers needed in §35.
For a hyperbolic type (g,n) in characteristic zero, the geometric profinite surface group $\Delta_{g,n}$ is torsion-free and slim: every open subgroup has trivial centralizer in $\Delta_{g,n}$. In particular it and all its open subgroups have trivial centre. The absolute Galois group of a number field is also slim, and hence has trivial centre and no nontrivial finite normal subgroup. These settle the two centre questions in §35, folio 251 verso, scan 496. The second assertion does not assert that a number field's absolute Galois group is torsion-free.

Proof. For $n>0$ the geometric group is free profinite of rank $2g+n-1\geq 2$; for $n=0$ it is the profinite completion of the closed oriented surface group of genus $g\geq 2$. Nakamura (1994, Corollaries (1.3.3)–(1.3.4)) give centre-freeness and the finite projective resolutions obtained

from goodness of these groups. Their cohomological dimensions at every prime are finite (one in the free case, two in the closed surface case), so they have no elements of prime order and are torsion-free. Every open subgroup is again a hyperbolic surface group of finite index, by the correspondence with finite covers. Its Euler characteristic is its index times the original negative Euler characteristic; thus the same centre assertion applies.

To derive slimness directly, suppose c centralizes an open subgroup H. Its normal core H_0 is open; some positive power c^m belongs to H_0. That power lies in the centre of H_0, hence is 1. Torsion-freeness gives $c = 1$. This agrees with Mochizuki (2006, Lemma 3.2). For number fields use Mochizuki (2006, Corollary 2.2). If a finite subgroup F is normal in a slim profinite group G, the kernel of the conjugation map $G \to \mathrm{Aut}(F)$ is open and centralized by F, forcing $F = 1$. Complex conjugation, when a real embedding exists, is an element of order two of G_K; its subgroup is not normal. □

Statement D.15 (Proved). Rigidity on open subgroups and the two centralizer formulas.

If G is slim and $u \in \mathrm{Aut}(G)$ is the identity on an open subgroup H, then $u = 1$. If $u(H) = H$ and $u|_H$ is conjugation by an element of H, then u is that inner automorphism on G. For a free basis element x of $\widehat{F}_r$, or a standard nonseparating simple generator x of the profinite closed surface group of genus at least two, and for every integer $m \geq 1$,

$$C_G(x^m) = C_G(\overline{\langle x^m \rangle}) = \overline{\langle x \rangle}. \tag{D.15.1}$$

Thus Lemma 5 and equations (2), (4) in §35, scans 498–501, folios 252 verso–254, are valid with their stated hyperbolic hypotheses.

Proof. Let H_0 be the normal core of H in G. For $g \in G$ and $a \in H_0$ both a and gag^{-1} lie in H, so

$$u(g)au(g)^{-1} = u(gag^{-1}) = gag^{-1}.$$

Consequently $g^{-1}u(g)$ centralizes the open subgroup H_0, hence is 1. This proves the first assertion. Compose with the inverse of the specified inner automorphism to obtain the second.

For the free basis case (D.15.1) is Nakamura (1994, Lemma (2.1.2)). For the surface case, a standard generator is an algebraically simple element in the sense of Boggi and Zalesskii (2019, Definition 1.3). Their Proposition 3.6 gives the stronger identity $N_G(\overline{\langle x^m \rangle}) = \overline{\langle x \rangle}$; the centralizer contains the right-hand side and is contained in this normalizer. Both formulas follow. Centralizing one element is equivalent to centralizing the closed subgroup it topologically generates, so the scan's alternatives L_i^m and ℓ_i^m are mathematically equivalent in this use. The subgroup $\overline{\langle x \rangle} \simeq \widehat{\mathbb{Z}}$ has injective multiplication by every positive integer, which also completes the power argument written across scans 500–501. □

Statement D.16 (Known). Faithful outer actions for every hyperbolic curve.

For every hyperbolic curve U/K over a number field or a p-adic local field, the full profinite outer representation

$$\rho_U : G_K \longrightarrow \mathrm{Out}(\Delta_U)$$

is injective. This includes affine and proper curves, type $(0, 3)$, type $(1, 1)$, and all proper types $(g, 0)$ with $g \geq 2$. It proves the expectation in §35, scan 497, rather than only the weaker assertion that a sufficiently small open subscheme is faithful.

Proof. The precise general result is Hoshi and Mochizuki (2011, Corollary 6.3(i), with one configuration point). Its full-prime-set hypothesis is essential. The source explicitly identifies

the earlier affine assertion as Matsumoto (1996, Theorem 2.1); the 2011 result supplies the proper case as well.

We also give the short Belyi argument for the particular theorem proved in §35, scans 491–495. Suppose $\sigma \in G_{\mathbb{Q}}$ acts trivially in $\mathrm{Out}(\widehat{F}_2)$ for the tripod. Then it fixes the isomorphism class of every connected finite étale cover of the tripod, since these correspond to conjugacy classes of open subgroups of $\widehat{F}_2$. For any $j \in \overline{\mathbb{Q}}$ choose an elliptic curve E_j with that invariant. By Belyi (1980, Section 4, Theorem 4) it has a finite map to $\mathbb{P}^1$ branched only over $0, 1, \infty$. Deleting the inverse images of the three points gives a cover whose σ-conjugate is isomorphic to it. An isomorphism extends uniquely to smooth proper completions, so $E_j \simeq E_j^{\sigma}$ and $j = \sigma(j)$. Since every algebraic j was allowed, $\sigma = 1$. The unnumbered corollary on p. 256 of Belyi's English paper is exactly this faithfulness consequence. This argument proves injectivity, and has no surjectivity conclusion about a separately defined normalizer or GT group. □

For fixed ℓ, the analogous pro-ℓ assertion over a number field is not a consequence of this theorem. In fact for a curve with good reduction, and a good finite place of residue characteristic different from ℓ, the geometrically pro-ℓ outer representation is unramified there. Nontrivial inertia of G_K therefore lies in its kernel. The full profinite statement cannot be replaced by a single-prime statement.

Statement D.17 (Known). Faithfulness even modulo geometric Teichmüller monodromy.
Let $2g - 2 + n > 0$ and let $\mathcal{M}_{g,n}/\mathbb{Q}$ denote the stack with ordered marked points. Write

$$\Psi : \pi_1(\mathcal{M}_{g,n}) \longrightarrow \mathrm{Out}(\Delta_{g,n}), \qquad N_{g,n}^{\mathrm{arith}} = \mathrm{im}(\Psi), \qquad \Sigma_{g,n} = \mathrm{im}(\pi_1(\mathcal{M}_{g,n,\overline{\mathbb{Q}}})).$$

Then the canonical surjection

$$G_{\mathbb{Q}} \longrightarrow N_{g,n}^{\mathrm{arith}}/\Sigma_{g,n} \tag{D.17.1}$$

is an isomorphism. The same assertion holds with unordered marked points and the corresponding unordered geometric monodromy group. Here $\Sigma_{g,n}$ is the procongruence image of the orientation-preserving mapping class group, not an assumed copy of its entire profinite completion. This answers affirmatively the precise arithmetic-image question of §35, scans 502–505, including the proper types and $(1, 1)$ singled out on scan 504 as possibly "tout à fait faux".

Proof. The geometric image is closed and normal because it is the continuous image of the geometric kernel in the arithmetic stack extension. Hence the surjection (D.17.1) is well-defined without a rational base point on the stack. Choose a finite extension $L/\mathbb{Q}$ and an L-point of $\mathcal{M}_{g,n}$, equivalently a curve of the given type with ordered L-rational marked points. Such an L exists by taking a closed point of a nonempty finite type atlas. We do not assume the existence of a $\mathbb{Q}$-point for an arbitrary prescribed (g, n). The point gives a section over G_L whose universal outer representation is $\rho_{U/L}$. By Hoshi and Mochizuki (2011, Corollary 6.4),

$$\rho_{U/L}^{-1}(\Sigma_{g,n}) = 1.$$

Thus the kernel A of (D.17.1) intersects G_L trivially. It is finite, normal in $G_{\mathbb{Q}}$, and therefore trivial by D.14. This proves the ordered assertion.

For unordered marks the ordered geometric image is a normal subgroup of finite index in the unordered image. Over L the inverse image under $\rho_{U/L}$ of the latter group is finite, because its map to the finite quotient has trivial kernel by the ordered assertion of Corollary 6.4. It is also normal in G_L, since geometric monodromy is normal in arithmetic monodromy. Slimness forces it to be trivial. The same finite-normal-kernel argument over $\mathbb{Q}$ completes the proof. □

The distinction from the congruence subgroup problem is exact. Hoshi and Mochizuki (2011, Corollary 6.5) says that the full universal monodromy Ψ is injective if and only if its geometric restriction is injective. Thus identifying $\Sigma_{g,n}$ with the profinite mapping class group still requires the relevant congruence theorem. The quotient faithfulness (D.17.1) does not. Equally, replacing N^{arith} by a full normalizer is a different problem, retained separately in Part E. Quotienting by orientation-reversing mapping classes would also change the assertion; D.21 below gives the obstruction.

Statement D.18 (Proved). The direction of faithfulness under finite covers.
Let k have characteristic zero, let $V \to U$ be a finite étale morphism of geometrically connected hyperbolic curves over k, and assume that $\rho_U : G_k \to \mathrm{Out}(\Delta_U)$ is faithful. Then ρ_V is faithful. No rational point and no replacement of k by a finite extension are necessary. This proves the finite-cover assertion in §35, folios 253–253 verso, scans 499–500, in its stated characteristic-zero scope.

Proof. Choose compatible base points. Then $\Pi_V \subseteq \Pi_U$ is an open subgroup, maps onto G_k by geometric connectedness, and $\Delta_V = \Delta_U \cap \Pi_V$ is open in Δ_U. For $\sigma \in \ker \rho_V$ choose $e \in \Pi_V$ above σ. Its conjugation on Δ_V is conjugation by some $d \in \Delta_V$. Thus $d^{-1}e$ centralizes Δ_V. Its conjugation on Δ_U is an automorphism that is the identity on an open subgroup, hence is the identity by D.14–D.15. Therefore $\rho_U(\sigma) = 1$, whence $\sigma = 1$. □

This proof also explains the direction of the comparison map: an open geometric subgroup detects an automorphism of the larger group once the inner automorphism on the subgroup has been removed. In the manuscript's longer argument a finite residual kernel had to be eliminated; it is not enough merely to observe that this kernel acts through a finite group. Filling punctures is a surjection of geometric groups, so it is not the same argument; the proper faithfulness needed after filling is supplied by D.16 and D.17.

Statement D.19 (Proved). The tripod centralizer is the symmetric group.
For $U = \mathbb{P}^1_K \setminus \{0, 1, \infty\}$ over a sub-p-adic field K,

$$C_{\mathrm{Out}(\Delta_U)}(\rho_U(G_K)) = \mathrm{Aut}_K(U) \simeq \mathfrak{S}_3.$$

In particular the automorphism assertion of §34, scans 485–486, does not require assuming a chosen arithmetic normalizer in addition to commuting with the arithmetic action.

Proof. An outer automorphism a commuting with ρ_U induces, by D.1, an isomorphism of the arithmetic extension over G_K. D.4 and its Isom consequence identify it with a unique automorphism of U. Conversely every such scheme automorphism commutes with the Galois action. An automorphism extends to $\mathbb{P}^1$ and permutes $\{0, 1, \infty\}$; three distinct points determine a fractional linear transformation, and all six permutations are defined over $\mathbb{Q}$. This gives exactly $\mathfrak{S}_3$. Its injectivity on the outer group follows from the same Isom theorem. The alternatives $\mathfrak{S}_3 \mid \mathcal{S}_3$ in scan 485 are two notations for this six-element group; the mathematics does not decide the handwritten font. □

4.5 Small types and the exact remaining qualifications

Statement D.20 (Refuted). Dropping hyperbolicity from the faithfulness assertion.
The uniform assertion of D.16 becomes false if any of $(0,0)$, $(0,1)$, $(0,2)$, $(1,0)$ is admitted. The

geometric centres and the counterexamples are as follows:

(g,n)	$\Delta_{g,n}$	$Z(\Delta_{g,n})$
$(0,0),(0,1)$	1	1
$(0,2)$	$\widehat{\mathbb{Z}}$	$\widehat{\mathbb{Z}}$
$(1,0)$	$\widehat{\mathbb{Z}}^{,2}$	$\widehat{\mathbb{Z}}^{,2}$

In contrast $(0,3)$ and $(1,1)$ are both hyperbolic with geometric group $\widehat{F}_2$, but have different peripheral structures.

Proof. Over an algebraically closed field of characteristic zero, comparison with the sphere, plane, punctured plane, and torus gives the displayed groups. Thus the first two outer actions over $\mathbb{Q}$ are trivial. For $\mathbb{G}_m/\mathbb{Q}$ the outer action is the cyclotomic character, whose abelian image cannot be faithful on $G_{\mathbb{Q}}$. Explicitly the splitting field of x^3-2 has Galois group $\mathfrak{S}_3$: Eisenstein's criterion at 2 proves irreducibility and the discriminant is -108, a nonsquare. Hence $G_{\mathbb{Q}}$ has nontrivial commutator subgroup, killed by its cyclotomic character.

For $(1,0)$ take $E: y^2=x^3-x$ over $K=\mathbb{Q}(i)$. The endomorphism $[i](x,y)=(-x,iy)$ is defined over K. Complex uniformization makes the integral lattice a free rank-one $\mathbb{Z}[i]$-module, so the full Tate module is free rank one over $\widehat{\mathbb{Z}}[i]$. Equivalently one can choose a basis on which $[i]$ has matrix $J=\left(\begin{smallmatrix}0&-1\\1&0\end{smallmatrix}\right)$. All Galois matrices commute with J. Solving $AJ=JA$ over any commutative ring gives $A=aI+bJ$; its invertible solutions commute with each other. Thus the entire adelic representation has abelian image. On the other hand the $\mathfrak{S}_3$ splitting field of x^3-2 intersects $\mathbb{Q}(i)$ trivially: its only quadratic subfield is $\mathbb{Q}(\sqrt{-3})$. Its compositum with K is therefore an $\mathfrak{S}_3$-extension of K. Consequently G_K is nonabelian and the representation on $\Delta_E=\widehat{\mathbb{Z}}^{,2}$ is not faithful. The code verifies the discriminant, the finite commutator calculation, and the centralizer calculation modulo several integers, including 2 and 3; the ring calculation above is the proof for all moduli.

For $(0,3)$ the peripheral generators are $x, y, (xy)^{-1}$, each primitive in abelianization. For $(1,1)$ the only peripheral subgroup is generated by $[x,y]$, which has zero image in abelianization. A bare identification of their free profinite groups is therefore not an identification of curves with their loop data. □

Statement D.21 (Proved). Orientation, real fields, and residue characteristics.
The subgroup in D.17 must be the geometric orientation-preserving group. If a full extended mapping class group containing orientation reversal is used instead, the analogous faithfulness assertion fails already at the tripod. The geometric centre statements remain valid over $\mathbb{R}$, but the number-field Galois slimness statement does not extend to $G_{\mathbb{R}}$. None of D.4, D.8, D.16, or D.17 has a residue-characteristic exclusion at 2 or 3 in its stated characteristic-zero setting.

Proof. Complex conjugation on the complex tripod is an orientation-reversing homeomorphism preserving its punctures. Its outer class belongs to the image of the extended mapping class group and is nontrivial: on the abelianization $\widehat{\mathbb{Z}}^{,2}(1)$ it acts by -1. Therefore quotienting by that extended image kills a nontrivial element of $G_{\mathbb{Q}}$. Moreover the extended image need not be normal in N^{arith}: at the unordered tripod its image under the isomorphism of D.17 is the subgroup generated by complex conjugation. If it were normal it would be a nontrivial finite normal subgroup of $G_{\mathbb{Q}}$, contradicting D.14. One cannot silently form an ordinary group quotient by it. Taking its normal closure would still kill complex conjugation.

Over $\mathbb{R}$ the arithmetic quotient is C_2, whose centre is itself. The correct real section map for a hyperbolic curve is $\pi_0(U(\mathbb{R}))\simeq \mathrm{Sec}(U/\mathbb{R})$, by Bresciani and Vistoli (2020, Introduction, Theorem); it is not a bijection from individual real points. The outer action of C_2 is nevertheless

faithful for a hyperbolic real curve. If the proper completion has positive genus, conjugation reverses its orientation and is anti-symplectic on H_1; identity action would contradict its nondegenerate intersection pairing over $\mathbb{Q}_\ell$. If the genus is zero and $n \geq 3$, its action on puncture homology is minus the permutation of the punctures. An involution on at least three punctures has at least two orbits, so the reduced permutation representation has a nonzero $+1$ eigenspace. The action on that subspace after the minus sign is nontrivial.

A smooth complex algebraic curve is orientable. The fundamental group of a nonorientable topological surface is therefore not the geometric group of one of these curves. A real curve can have a nonorientable quotient by conjugation; its arithmetic datum is the geometric oriented group together with the involution and its extension, not the fundamental group of the quotient surface alone. The orientation double cover does not identify these different arithmetic extensions.

All the local fields in D.4 and D.8 have characteristic zero. Horizontal ramification along a deleted point is consequently tame even if its ramification index is divisible by the residue characteristic; the analytic unit-root threshold in D.7 handles that divisibility. The proof of D.8 explicitly uses the correct dyadic Hensel bound. The universal-monodromy and full profinite theorems in D.16–D.17 are also characteristic-zero statements. In §36 the orbifold orders 2 and 3 do prevent transporting an argument through tame coarse moduli in characteristics $2, 3$; this requires retaining the stack and is not an exception to the theorems over $\mathbb{Q}$ or $\mathbb{Q}_p$ stated here. □

Statement D.22 (Proved). Uniqueness of a Galois-invariant tripod loop structure. Let K be a number field and equip $\Delta = \pi_1((\mathbb{P}^1 \setminus \{0, 1, \infty\})_{\overline{K}})$ with its usual outer G_K-action. Among loop structures of type $(0, 3)$, meaning the three conjugacy classes $\overline{\langle a \rangle}$, $\overline{\langle b \rangle}$, $\overline{\langle (ab)^{-1} \rangle}$ for a profinite free basis (a, b), the usual loop structure is the unique one invariant under this action. This is a precise instance of the first sentence of the conjecture in §34, scan 483, independent of its false normalizer explanation.

Proof. Let such a triple be invariant. After a finite extension L/K, each conjugacy class is individually fixed. For any one of its topological generators z, the normalizer of $I = \overline{\langle z \rangle}$ in Π_{U_L} projects onto G_L. Its conjugation action on z is $z \mapsto z^{\lambda(h)}$ for some $\lambda(h) \in \widehat{\mathbb{Z}}^\times$. The true Galois action on Δ^{ab} is scalar multiplication by χ_K, since all three actual punctures are rational. The image of z in Δ^{ab} is a primitive vector: this is evident for a, b and for $-a - b$. Thus $\lambda(h)[z] = \chi_K(p(h))[z]$ implies $\lambda(h) = \chi_K(p(h))$. In particular the geometric part of the normalizer acts trivially. D.3, applied over L, identifies I with one of the actual inertia subgroups: it is contained in one and is maximal procyclic, being a free-basis factor or its automorphic image. The three candidate conjugacy classes are distinct, as their abelianized lines are distinct. Hence they are exactly the three actual classes. Finite base extension does not change the geometric inertia subgroups, so the result holds over K. □

Statement D.23 (Open). What invariant loop data and admissible moduli splittings still require.

For an arbitrary affine hyperbolic U/K over a number field, consider the assertion that every profinite surface loop structure of its type on Δ_U which is invariant under $\rho_U(G_K)$ equals the geometric cuspidal loop structure. The cyclotomic version is D.3 and the tripod case is D.22. We have not established this bare invariance assertion in the remaining types, and do not identify it with the false assertion of D.2. The first type not covered by our elementary argument is $(1, 1)$: its peripheral commutator has zero abelianization, so the primitive-vector argument forcing the cyclotomic exponent does not apply. Precisely the missing implication in

that argument is that invariance of an alternative peripheral subgroup forces the normalizer action required by D.3 after an open restriction of G_K.

Independently, the proposed description of admissible partial splittings of $N_{g,n}$ in §34, folios 247–247 verso, scans 487–488, is not a consequence of the Hom theorem or the proper section manuscript. Once N^{arith} is identified with an arithmetic moduli-stack fundamental group in a type where the congruence kernel vanishes, these splittings concern points of that stack. Classifying all of them as curve-moduli points is a moduli-stack section problem; neither D.5 nor D.7 supplies that classification. In particular the empty list of extra conditions at $(1, 1)$ in scan 487 is not proved sufficient by the 2026 proper-curve result. D.10 is the exact consequence obtained on the specified high-genus modular-curve open subgroup. In higher genus, even the identification with the full arithmetic stack fundamental group must retain the congruence-kernel qualification of D.17.

These Open labels state the exact assertions left unproved by this audit; they are not claims that every formulation of inertia recognition or every special section theorem is unknown in the literature. Our searches and checks used Nakamura's cyclotomic criterion, Mochizuki's relative Hom theorem, the full Hoshi–Mochizuki monodromy theorem, and the two new proof packs. An assertion left conditional or unverified has not been refuted merely because our proof stops at an identified implication.

5 The fundamental conjecture: three different comparison problems

5.1 The change of definition in §29

We use the conventions of Part A. In particular, $\pi_{g,n}$ is the *profinite* surface group, $P_{g,n}$ is the pure orientation-preserving mapping class group, and $\Sigma_{g,n}$ is the closed image of the full orientation-preserving mapping class group, with permutations of the marked points allowed, in $O_{g,n} = \mathrm{Out}_{\mathrm{lac}}(\pi_{g,n})$. The suffix "lac" retains the conjugacy classes of peripheral procyclic subgroups and their common cyclotomic exponent. All normalizers, images, quotients and commutator closures in this section are topological. An identification with an automorphism group is an identification of continuous automorphisms. Choices of a geometric marking identify fundamental groups; changing paths conjugates the resulting actions. The distinction between a canonical outer action and a chosen genuine action will be retained.

Our notation	Definition	Manuscript source
$N_{g,n}^{\mathrm{norm}}$	$N_{O_{g,n}}(\Sigma_{g,n})$	§26, fol. 164, scan 324, (14)
$\Gamma_{g,n}^{\mathrm{norm}}$	$N_{g,n}^{\mathrm{norm}}/\Sigma_{g,n}$	same passage; fol. 168, scan 332, (27)
$M_{g,n}^{\mathrm{arith}}$	closed image of the arithmetic universal-curve fundamental group in $\mathrm{Aut}_{\mathrm{lac}}(\pi_{g,n})$	§29, fols. 187–188, scans 370–371, (9)–(13)
$N_{g,n}^{\mathrm{arith}}$	exterior image of $M_{g,n}^{\mathrm{arith}}$	same passage
$\Gamma_{g,n}^{\mathrm{arith}}$	$N_{g,n}^{\mathrm{arith}}/\Sigma_{g,n}$	same passage
$G_{\mathbb{Q}}$	$\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$	his $\mathbb{I}\Gamma_{\mathbb{Q}}$

The symbols N and $\mathbb{I}\Gamma$ do not keep their definitions across these passages. Scan 370 announces the change to an arithmetic construction; on scan 371 we read the instruction to denote the *image* of the arithmetic homomorphism by M, and the arrow from $G_{\mathbb{Q}}$ to the resulting quotient is explicitly called surjective. The latter conjecture asks for injectivity. We checked these two scans, including the words surrounding the formulas. The missing orientation superscript in one occurrence on scan 371 is recorded in the readings register; it does not license identifying the two normalizers.

Statement E.1 (Proved). The early and revised conjectures are inequivalent statements.

For every hyperbolic type $2g - 2 + n > 0$ there are continuous homomorphisms

$$G_{\mathbb{Q}} \twoheadrightarrow \Gamma_{g,n}^{\mathrm{arith}} \hookrightarrow \Gamma_{g,n}^{\mathrm{norm}}.$$

The first is surjective by definition. Its injectivity is the revised conjecture of §29, fol. 188, scan 371. The conjecture of §26, fol. 168, scan 332, asks that the composite be an isomorphism; it additionally asserts that every element of the normalizer quotient is arithmetic. The stronger assertion $N_{g,n}^{\mathrm{norm}} = O_{g,n}$ appearing in that passage is a further assertion, not an equivalent formulation of either one.

Proof. The geometric subgroup is normal in the arithmetic fundamental group. Its image $\Sigma_{g,n}$ is consequently normalized by the arithmetic image. Thus $N_{g,n}^{\mathrm{arith}} \subseteq N_{g,n}^{\mathrm{norm}}$, and both quotients are by the same closed subgroup. The exact sequence of the arithmetic moduli stack gives the first surjection. Surjectivity of the composite is exactly equality of the two images modulo $\Sigma_{g,n}$, equivalently equality of the two groups N. No assertion about the rest of $O_{g,n}$ follows. This

also explains why the discussion of admissible splittings on §33, fol. 242, scan 479, cannot be read using the original normalizer definition without an additional argument. □

Statement E.2 (Known). The revised conjecture is true for every hyperbolic type. The natural map

$$G_{\mathbb{Q}} \xrightarrow{\sim} \Gamma_{g,n}^{\mathrm{arith}}$$

is an isomorphism of profinite groups for every hyperbolic (g, n). It is compatible with the arithmetic maps between types wherever those maps are defined. This includes $(0, 3)$, $(0, 4)$, $(0, 5)$, $(1, 1)$ and $(1, 2)$, as well as proper curves of genus at least two. It does not require the congruence subgroup property in unknown genera.

Proof. Part D, Statement D.17, gives the precise application of Hoshi and Mochizuki (2011, Corollary 6.4): the arithmetic Galois image has trivial intersection with the geometric universal-monodromy image. After a finite field extension admitting a curve of the specified type, this proves injectivity on an open subgroup of $G_{\mathbb{Q}}$. The remaining kernel is a finite normal subgroup of the slim group $G_{\mathbb{Q}}$, and is therefore trivial. Statement E.1 supplies surjectivity. The use of the actual image Σ, rather than an unproved identification of that image with the full profinite completion, is essential. Naturality follows from the arithmetic fundamental-group diagrams. A continuous bijection between profinite groups is a homeomorphism. □

For $(0, 3)$, the arithmetic moduli stack with unlabelled points is $B\mathfrak{S}_3$ over $\mathbb{Q}$; the labelled stack is $\operatorname{Spec}\mathbb{Q}$. Belyi faithfulness supplies the required injectivity here: the three-branch-point theorem is Belyi (1980, §4, Theorem 4), and faithfulness is its unnumbered corollary on printed page 256. Consequently the impression recorded on scan 479 is correct for the *revised* group. Lead 1 is false as an identification of that group with an independently defined $\widehat{\mathrm{GT}}$: surjectivity onto the revised arithmetic quotient is already built into its definition.

Statement E.3 (Known). The abelianization of the revised group. For every hyperbolic type,

$$(\Gamma_{g,n}^{\mathrm{arith}})^{\mathrm{ab}} = \Gamma_{g,n}^{\mathrm{arith}} / \overline{[\Gamma_{g,n}^{\mathrm{arith}}, \Gamma_{g,n}^{\mathrm{arith}}]} \xrightarrow{\sim} \widehat{\mathbb{Z}}^{\times}$$

is the cyclotomic character. Thus the consequence anticipated in §26–§29 (in particular scan 340) is a theorem for the revised groups.

Proof. Apply E.2 and the Kronecker–Weber theorem: $\mathbb{Q}^{\mathrm{ab}} = \bigcup_m \mathbb{Q}(\mu_m)$. The restriction map identifies the indicated quotient with $\varprojlim_m (\mathbb{Z}/m\mathbb{Z})^{\times}$. The image of inertia generators in the universal curve is acted on by this same character. This argument says nothing about additional abelian quotients of a larger normalizer group. □

5.2 The genus-zero comparison, including the isolated tripodal case

We write $\widehat{F}_2 = \langle x, y \rangle$, $z = (xy)^{-1}$. A profinite Grothendieck–Teichmüller element is represented by (λ, f) with $\lambda \in \widehat{\mathbb{Z}}^{\times}$ and $f \in \overline{[\widehat{F}_2, \widehat{F}_2]}$, such that

$$x \longmapsto x^{\lambda}, \qquad y \longmapsto f^{-1} y^{\lambda} f$$

defines an automorphism and satisfies Drinfeld's symmetry, hexagon and pentagon equations. The automorphism requirement is part of the definition in the profinite setting. We denote by $\widehat{\mathrm{GT}}_0$ the group with symmetry and hexagon only. The pentagon is a condition in the fundamental group of $\mathcal{M}_{0,5}$, not an extra relation in $\pi_1(\mathcal{M}_{0,4})$ with no reference to its embeddings. Our convention is that $P_{0,m} = \pi_1(\mathcal{M}_{0,m}(\mathbb{C}))$.

Statement E.4 (Known). The exact two-level theorem.
Let $\mathrm{Out}^{\#}(\widehat{P}_{0,m})$ consist of outer automorphisms which send each standard boundary generator x_{ij} to a conjugate of x_{ij}^{λ} for a single $\lambda \in \widehat{\mathbb{Z}}^{\times}$, and which commute with the outer relabelling action of $\mathfrak{S}_m$. Then

$$\mathrm{Out}^{\#}(\widehat{P}_{0,4}) \cong \widehat{\mathrm{GT}}_0, \qquad \mathrm{Out}^{\#}(\widehat{P}_{0,m}) \cong \widehat{\mathrm{GT}} \quad (m \geq 5).$$

The forgetful maps identify the latter groups for all $m \geq 5$, and the map to the four-point group is the usual injection $\widehat{\mathrm{GT}} \hookrightarrow \widehat{\mathrm{GT}}_0$.

Proof. These are precisely the Main Theorem (a),(b) and its Corollary in Harbater and Schneps (2000); the four-point identification is their Theorem 2, and the five-point identification their Theorem 3. Their Proposition 7 proves the pentagon extension criterion, and Proposition 8 proves injectivity of forgetting. The common exponent and the relabelling centralizer occur in their definition of $\mathrm{Out}^{\#}$ and cannot be dropped from the statement. □

There is also a version with chosen genuine braid-group automorphisms: the objects are $\widehat{B}_j$ and the subgroups generated by $\sigma_1^2, \sigma_2, \ldots, \sigma_{j-1}$, with the standard inclusions and doubling maps. Automorphisms act trivially on the permutation quotients and commute with those maps. The tower through three strands has automorphism group $\widehat{\mathrm{GT}}_0$; through four strands, and at every further finite truncation, it has $\widehat{\mathrm{GT}}$. These are the tower statements recalled and used in Harbater and Schneps (2000, §0.3). They specify a concrete meaning of "two levels". Arbitrary independent outer automorphisms of two groups are not such a tower action.

Statement E.5 (Known). The intrinsic five-point and higher automorphism theorem.
For $m \geq 5$,

$$\mathrm{Out}(\widehat{P}_{0,m}) \cong \widehat{\mathrm{GT}} \times \mathfrak{S}_m.$$

The symmetric factor is the geometric relabelling action; the other factor preserves all generalized fibre subgroups. Both factors and their commutation are part of the theorem.

Proof. Take $(g, r) = (0, 3)$ and $n = m - 3 \geq 2$ in Hoshi et al. (2022, Corollaries 2.6(i),(iii) and 2.8(i),(ii)). Their configuration group Π_n is $\widehat{P}_{0,m}$; their Out^{gF} is the factor displayed here. Their Definition 2.7 identifies GT^{Σ} with the classical $\widehat{\mathrm{GT}}$ when the set of primes Σ is all primes. In particular the result does not presuppose that a given automorphism preserves inertia. It reconstructs the relevant distinguished subgroups. □

Statement E.6 (Proved). A normalizer lemma for the universal curve.
Let π be a centre-free profinite group, let $S^! \triangleleft S \subseteq \mathrm{Out}(\pi)$ be closed subgroups, and let E be the inverse image of $S^!$ in $\mathrm{Aut}(\pi)$. If $S \subseteq N \subseteq \mathrm{Out}(\pi)$ and N normalizes S and $S^!$, conjugation induces an injection

$$N/S^! \hookrightarrow \mathrm{Out}(E)$$

whose image normalizes the image of $S/S^!$.

Proof. Choose a representative $a \in \mathrm{Aut}(\pi)$ of an element of N. Conjugation by a preserves E. Replacing a by an inner automorphism of π changes this automorphism of E by an inner automorphism of E, since $\mathrm{Inn}(\pi) \subseteq E$. Suppose conjugation by a on E equals conjugation by $e \in E$. Then $e^{-1}a$ centralizes $\mathrm{Inn}(\pi)$. For every $u \in \pi$ we have $\mathrm{inn}((e^{-1}a)(u)) = \mathrm{inn}(u)$; centre-freeness gives $(e^{-1}a)(u) = u$. Hence $a = e$, and its exterior class belongs to $S^!$. This proves injectivity. The normalizer assertion follows by conjugating the subgroup S before passage to the quotient. □

Statement E.7 (Proved). The early normalizer quotient in genus zero.
With the full permutation convention of §19 and §26,

$$\Gamma_{0,3}^{\mathrm{norm}} \cong \widehat{\mathrm{GT}}_0, \qquad \Gamma_{0,n}^{\mathrm{norm}} \cong \widehat{\mathrm{GT}} \quad (n \geq 4).$$

The identifications respect the arithmetic embeddings after the usual geometric markings. The arithmetic subgroup in every case is $\Gamma_{0,n}^{\mathrm{arith}} \cong G_{\mathbb{Q}}$.

Proof. First suppose $n \geq 4$. The profinite Birman sequence, and genus-zero congruence subgroup property from Parts B and C, identify the inverse image of $\Sigma_{0,n}^!$ in $\mathrm{Aut}(\pi_{0,n})$ with $E = \widehat{P}_{0,n+1}$. A normalizer element preserves $\Sigma^!$ because it preserves the peripheral permutation map. E.6 embeds $N^{\mathrm{norm}}/\Sigma^!$ into $\mathrm{Out}(E) = \widehat{\mathrm{GT}} \times \mathfrak{S}_{n+1}$. The image of $\Sigma/\Sigma^!$ is the point stabilizer $\mathfrak{S}_n \subset \mathfrak{S}_{n+1}$. Its normalizer is itself: its unique fixed letter must be preserved by any normalizing permutation. Consequently N^{norm}/Σ embeds in $\widehat{\mathrm{GT}}$.

Conversely, the standard genus-zero tower action of $\widehat{\mathrm{GT}}$ preserves the kernel of $\widehat{P}_{0,n+1} \to \widehat{P}_{0,n}$ and commutes, outerly, with the $\mathfrak{S}_n$ action. That kernel is $\pi_{0,n}$. Restriction therefore produces an exterior automorphism of $\pi_{0,n}$ preserving its peripheral classes and normalizing $\Sigma_{0,n}$. An inner change of a representative of the automorphism of E changes this restriction by $\Sigma^!$, and hence has no effect modulo Σ. Applying E.6 shows that this construction is inverse to the injection.

For $n = 3$, $\Sigma_{0,3} \cong \mathfrak{S}_3$ and $\Sigma_{0,3}^! = 1$. Every automorphism of $\mathfrak{S}_3$ is inner. Thus its normalizer in $O_{0,3}$ is the product of $\mathfrak{S}_3$ with its centralizer, with trivial intersection. The centralizer acts trivially on the three peripheral labels. Under $\pi_{0,3} \cong \widehat{P}_{0,4}$, it is exactly $\mathrm{Out}^{\#}(\widehat{P}_{0,4})$: the relabelling action of $\mathfrak{S}_4$ on four-point moduli has image $\mathfrak{S}_3$. Now apply E.4. Naturality of the arithmetic monodromy gives the last assertions. □

Replacing the full group $\Sigma_{0,3}$ by its pure subgroup 1 changes the problem: the pure normalizer quotient is then the entire label-preserving inertia automorphism group of $\widehat{F}_2$. It cannot be called $\widehat{\mathrm{GT}}$ merely because the same free group occurs. This distinction is especially relevant to retrospective accounts which suppress the exclamation-mark convention of §19.

Statement E.8 (Open). The remaining profinite arithmetic equality.
The assertion

$$G_{\mathbb{Q}} \longrightarrow \widehat{\mathrm{GT}} \quad \text{is surjective}$$

is the extra arithmetic assertion needed to identify the early normalizer quotients in types $(0,4)$ and $(0,5)$ with $G_{\mathbb{Q}}$. The map is a continuous closed embedding by Belyi faithfulness and the Galois compatibility of the tower. The isolated $(0,3)$ assertion instead asks for surjectivity onto $\widehat{\mathrm{GT}}_0$; it contains, in addition, the requirement that every such automorphism satisfy the pentagon. We do not infer either surjectivity from E.2.

Our source audit did not verify a primary proof of a *strict* profinite inclusion $\widehat{\mathrm{GT}} \subsetneq \widehat{\mathrm{GT}}_0$. Some historical discussions assert such strictness; we have not substituted a rational or a finite-shadow calculation for the missing profinite counterexample. Accordingly we record the isolated-level equality with its precise additional condition, rather than declare it false on that basis. This is a limitation of this audit, not a claim that no relevant theorem exists. The invalid step in Lead 1 is already settled independently by E.1–E.2.

5.3 The two elliptic types and the meaning of an all-genus tower

Statement E.9 (Known). The two-point elliptic automorphism theorem.
There is an isomorphism

$$P_{1,2} \cong B_4/Z(B_4), \qquad \mathrm{Out}(\widehat{P}_{1,2}) \cong \widehat{\mathrm{GT}}.$$

Here $P_{1,2}$ fixes both marked points individually. The assertion is Minamide and Nakamura (2022, Theorem B, Theorem 4.3 and Corollary C). It does not identify the outer automorphism group of $\widehat{\mathrm{SL}_2(\mathbb{Z})}$ with $\widehat{\mathrm{GT}}$, nor the unrestricted group with permuted elliptic markings with its pure counterpart. For the full group in type $(1,2)$ there is a central involution exchanging the two punctures. The inertia-preserving outer group and the unrestricted outer group are respectively $\widehat{\mathrm{GT}}$ and $C_2 \times \widehat{\mathrm{GT}}$ in Boggi (2026a, Proposition 3.9).

Statement E.10 (Proved). The one-point elliptic normalizer.
There is an isomorphism

$$\Gamma_{1,1}^{\mathrm{norm}} \cong \widehat{\mathrm{GT}},$$

compatible with the arithmetic inclusion. This statement concerns the normalizer in $\mathrm{Out}_{\mathrm{lac}}(\pi_{1,1})$, not all outer automorphisms of the elliptic mapping class group.

Proof. The profinite Birman sequence and genus-one congruence theorem identify the inverse image of $\Sigma_{1,1}$ in $\mathrm{Aut}(\pi_{1,1})$ with $E = \widehat{P}_{1,2}$. E.6 and E.9 give an injection of the asserted normalizer quotient into $\widehat{\mathrm{GT}}$.

We verify the converse, including the peripheral condition. Put $\zeta_j = (\sigma_1 \cdots \sigma_{j-1})^j$. The homomorphism $\beta : B_4 \to B_3$, given by $\sigma_1, \sigma_2, \sigma_3 \mapsto \sigma_1, \sigma_2, \sigma_1$, sends ζ_4 to ζ_3^2. It induces the elliptic forgetful map

$$E = \widehat{B_4/\langle\zeta_4\rangle} \longrightarrow \widehat{B_3/\langle\zeta_3^2\rangle} = \widehat{\mathrm{SL}_2(\mathbb{Z})},$$

with kernel $\pi_{1,1}$. Compose with the quotient to $\widehat{B_3/\langle\zeta_3\rangle}$. Its kernel K is $C_2 \mathbin{\widehat{*}} C_2 \mathbin{\widehat{*}} C_2$, and it is characteristic in E; these are the Cardano–Ferrari calculation and Proposition 3.2(iii) of Minamide and Nakamura (2022). The subgroup $\pi_{1,1}$ is the kernel of the unique homomorphism $K \to C_2$ which is nonzero on each conjugacy class of involutions. Every finite-order element of this profinite free product is conjugate into a factor. Thus this kernel is characteristic in K, hence in E.

Under the Birman identification its peripheral element is, up to inverse and conjugacy, $(\sigma_1\sigma_2)^6 = \zeta_3^2$ in E: this is the two-chain relation, in which the boundary of a one-holed torus also bounds the disc containing the two marked points. The standard Grothendieck–Teichmüller braid action sends ζ_3 to ζ_3^λ and ζ_4 to ζ_4^λ; consequently it preserves this peripheral subgroup with exponent λ. Restriction to the characteristic subgroup $\pi_{1,1}$ therefore defines an element of $O_{1,1}$ normalizing $\Sigma_{1,1}$. The ambiguity from an inner automorphism of E is exactly $\Sigma_{1,1}$ after exterior restriction. E.6 shows that restriction and conjugation are inverse. This proves the claim. The full-twist formulas used here are Drinfeld's standard braid action, also stated explicitly in Lochak et al. (2026, Theorem 2.1). □

Statement E.11 (Proved). An exact reduction for the two-point normalizer.
Put $E = \widehat{P}_{1,3}$, and let $\pi = \ker(E \to \widehat{P}_{1,2})$ be the kernel of forgetting the third point. Let $A \subseteq \mathrm{Out}(E)$ consist of the classes which preserve π, whose restrictions preserve its two peripheral conjugacy classes with a common exponent, and which normalize the outer $\mathfrak{S}_2$ action permuting the first two marks. Then

$$\Gamma_{1,2}^{\mathrm{norm}} \cong A/\mathfrak{S}_2.$$

In particular $G_{\mathbb{Q}} \cong \Gamma_{1,2}^{\mathrm{arith}} \hookrightarrow A/\mathfrak{S}_2$.

Proof. Use E.6 with $S^! = \Sigma^!_{1,2}$ and $S = \Sigma_{1,2}$. The pullback of $S^!$ in $\mathrm{Aut}(\pi)$ is E by the Birman sequence and the genus-one congruence theorem. Conversely, a representative b of a class in A restricts to an automorphism a of π. For $e \in E$ the identity $b(eue^{-1}) = b(e)a(u)b(e)^{-1}$, $u \in \pi$, implies that b on $E \subseteq \mathrm{Aut}(\pi)$ is conjugation by a. The normalizer and peripheral conditions are therefore exactly the conditions that its outer class belong to $N^{\mathrm{norm}}_{1,2}$. Changing b by an inner automorphism changes a by $\Sigma^!$. Quotienting by the remaining permutations proves the formula. □

This is an exact reduction, not an identification of the unrestricted group $A/\mathfrak{S}_2$ with $\widehat{\mathrm{GT}}$. The theorem E.9 concerns the base group $P_{1,2}$; the normalizer problem also retains its universal curve, whose group is $P_{1,3}$. Applying E.9 to the wrong group would erase precisely this distinction.

Statement E.12 (Known). Refined actions with quilt and clutching data.
Let $\mathcal{G}_{\mathrm{NS}}$ denote the subgroup called $\mathbb{\Gamma}$ in Nakamura and Schneps (2000, Definition 1.1), defined by their additional relations (III′) and (IV), with their Kummer cocycle ρ_2. Then

$$G_{\mathbb{Q}} \hookrightarrow \mathcal{G}_{\mathrm{NS}} \hookrightarrow \widehat{\mathrm{GT}}.$$

The first inclusion and subgroup assertion are their Theorem 1.2. Their Theorems 1.3–1.4 give genuine actions after choosing a quilted pants decomposition, canonical outer actions after forgetting that choice, and compatibility with inclusions of subsurfaces cut out by pants. The change-of-quilt ambiguity is an explicitly described inner automorphism. The group Λ of Hatcher et al. (2000) is a related subgroup of the $\lambda = 1$ group, defined using the genus-one relation (R); it is not the full arithmetic Galois group. Their Theorem A (Lemma 7 of §3) proves the subgroup assertion; Theorem B (Theorem 4 of §3) gives the pants-dependent genuine action and choice-independent outer action; Theorem C (Theorem 5 of §3) embeds it in the compatible outer automorphism group of their full subsurface tower. These statements require the specified extra compatibility data. No equality between these named refined subgroups and $\widehat{\mathrm{GT}}$ follows from E.9 or E.10.

Statement E.13 (Known). The current outer-tower results and their precise scope.
We distinguish two functors on orientable hyperbolic surfaces with boundary and embeddings which may fill punctures:

$$S \longmapsto \widehat{P(S, \partial S)}, \qquad S \longmapsto \check{P}(S, \partial S).$$

Their arrows are the induced continuous homomorphisms modulo inner automorphisms of the target. For these specific functors, Boggi (2026a, §2.8, Theorem 6.23 and Corollary 6.24) proves

$$\mathrm{Aut}(\check{\mathfrak{T}}^{\mathrm{out}}) \cong \widehat{\mathrm{GT}}, \qquad \widehat{\mathrm{GT}} \hookrightarrow \mathrm{Aut}(\widehat{\mathfrak{T}}^{\mathrm{out}}).$$

We use the explicitly dated version 6, 22 April 2026, as a preprint theorem; we did not audit its proof in full. Equality in the second display remains its Conjecture 6.21. The full profinite congruence subgroup property would imply that equality.

These are statements about natural automorphisms of a group-valued outer functor. They are not, without a comparison theorem, statements about all base-point paths, quilt moves, finite-cover maps or all degeneration bimodules in a finer tower. In particular they do not silently remove the named relations in E.12. The independent Lochak et al. (2026, Theorem 1.1) gives explicit actions of all $\widehat{\mathrm{GT}}$ on the full profinite groups in types $(g, 0)$ and $(g, 1)$; its introduction explicitly distinguishes these actions from actions respecting the additional Lego rules.

The functorial actions in E.13 restrict to the universal-curve kernels and preserve their peripheral twists. Since genus one has the congruence subgroup property, E.11 consequently

yields an inclusion $\widehat{\mathrm{GT}} \hookrightarrow \Gamma_{1,2}^{\mathrm{norm}}$ for that action. The missing reverse inclusion in our reduction is the assertion that every element of A preserves the nonseparating inertia classes in $E = \widehat{P}_{1,3}$. The inertia-preserving automorphism theorem does not itself prove this automatic-inertia assertion.

Statement E.14 (Open). The remaining automatic-inertia step in our elliptic reduction.

Every $a \in A$ from E.11 preserves the conjugacy classes of the closed procyclic subgroups generated by nonseparating Dehn twists in $\widehat{P}_{1,3}$. If this holds, the inertia-preserving theorem Boggi (2026a, Theorem A(ii)), together with the normalizer calculation for $\mathfrak{S}_2 \subset \mathfrak{S}_3$, gives $\Gamma_{1,2}^{\mathrm{norm}} \cong \widehat{\mathrm{GT}}$. We leave this as the precise unproved step in this audit; we do not claim that the general automatic-inertia problem has been settled.

For clarity, the comparisons obtained here are as follows. Entries describe the groups, not arbitrary choices of representatives of outer actions.

Type	Revised arithmetic quotient	Early normalizer quotient	Related tower theorem
$(0,3)$	$G_{\mathbb{Q}}$	$\widehat{\mathrm{GT}}_0$	pentagon extension required
$(0,4)$	$G_{\mathbb{Q}}$	$\widehat{\mathrm{GT}}$	universal curve has type $(0,5)$
$(0,5)$	$G_{\mathbb{Q}}$	$\widehat{\mathrm{GT}}$	stable genus-zero tower
$(1,1)$	$G_{\mathbb{Q}}$	$\widehat{\mathrm{GT}}$	elliptic universal curve and E.10
$(1,2)$	$G_{\mathbb{Q}}$	$A/\mathfrak{S}_2$ as in E.11	contains $\widehat{\mathrm{GT}}$ via E.13; E.14 unresolved here

The expected type-changing isomorphisms for the revised quotients are all E.2. For normalizer quotients, the genus-zero stable comparisons and the comparison with $(1,1)$ above are group-theoretic theorems, whereas equality with the arithmetic subgroup still requires E.8. These are three separate assertions.

5.4 Arithmetic reconstruction uses more morphisms

Statement E.15 (Known). What Pop's theorem reconstructs.

Let $U_0 = \mathbb{P}^1 \setminus \{0, 1, \infty\}$ and let $\mathcal{V}_{0,\mathrm{bir}}$ contain U_0, a cofinal system of open subsets of $\mathcal{M}_{0,5}$, their inclusions and the restrictions of the three functions $t_1, t_2, t_2 - t_1$ to U_0 wherever defined. Then the canonical map from $G_{\mathbb{Q}}$ to the compatible outer automorphism group of the geometric fundamental-group functor of this system is an isomorphism. This is the application on pages 3–4 of Pop (2018), from Theorem 2.7(1); its abelian-by-central pro-ℓ counterpart is Theorem 2.6(1).

Proof. The category satisfies Hypothesis (H) in Definition/Remark 2.2: the variety has dimension greater than one, the designated functions generate its function field over the algebraic closure, and the specified birational system is rigid. Theorem 2.7 identifies the automorphisms of the tame fundamental-group functor; in characteristic zero this is the full geometric *profinite* functor. The category has infinitely many localization objects even though only three rational functions are designated. Removing those objects destroys the stated hypothesis. Thus this theorem cannot be specialized to just $(\mathcal{M}_{0,4}, \mathcal{M}_{0,5})$ while keeping its conclusion. □

Statement E.16 (Known). The 2025 combinatorial reconstruction is not arithmetic surjectivity onto GT.

Hoshi et al. (2025, Theorem C(ii)) reconstruct the $\widehat{\mathrm{GT}}$-conjugacy class of $G_{\mathbb{Q}}$ as the maximal closed subgroups satisfying their AA conditions. The construction uses the abstract configuration group $\pi_1(\mathrm{Conf}_2(U_{0,\overline{\mathbb{Q}}}))$. The additional COF, RGC, arithmetic Belyi compatibility and ampleness requirements are indispensable hypotheses, defined in their Definitions 3.3 and 5.12. Their Theorem A constructs field operations; Theorem B proves an injectivity assertion under additional hypotheses. None states that all of $\widehat{\mathrm{GT}}$ has these properties.

We located this result through Collas (2026, §4), and then opened the original, dated 14 March 2025. A characterization of a conjugacy class of arithmetic subgroups answers one formulation of scan 479's request for an algebraic description. It is not the equality in E.8, and does not classify all admissible splittings of the moduli extensions.

Statement E.17 (Proved). The modular consequence of the claimed section theorem.

Assume the section theorem for $X_1(N)$ claimed in Corollary 8.5 of the 2026 manuscript in family 019, with $g(X_1(N)) \geq 2$. Let a section of the arithmetic extension for type $(1,1)$ lift to the open subgroup belonging to the fine modular curve $Y_1(N)$. Its composite with $\pi_1(Y_1(N)) \to \pi_1(X_1(N))$ is, up to geometric conjugacy, the section of a rational cusp of $X_1(N)$.

Proof. The composite is a section because both maps are over $G_{\mathbb{Q}}$. The asserted proper-curve section theorem makes it the section of a rational point. Mazur's theorem implies that all rational points of $X_1(N)$ are cusps when its genus is positive; see Mazur (1977, Introduction, Theorems (7),(7')), and the detailed conditional argument in Part D, Statement D.10. The level assumption here also removes the generic elliptic stabilizer, so that the subgroup is the fundamental group of a fine curve, not its coarse stack quotient. □

This does not show that the original affine section is conjugate into a cusp decomposition group. Nor does it show nonexistence: tangential sections at rational cusps supply affine sections. The unrefereed proper-curve theorem does not treat affine curves, moduli stacks or all splittings of $N_{1,1}^{\mathrm{arith}} \to G_{\mathbb{Q}}$. The hypothesis and the status of family 019 are retained in the results register; no unconditional section theorem is asserted here.

Statement E.18 (Known). Arithmetic rigidity is a different automorphism problem.

For a number field k, an odd prime ℓ, and $m \geq 4$, Nakamura (1994, Theorem A) identifies the G_k-compatible automorphisms of the geometrically pro-ℓ arithmetic fundamental group of $\mathcal{M}_{0,m}$, modulo geometric inner automorphisms, with $\mathrm{Aut}_k(\mathcal{M}_{0,m})$. The centralizer of the geometric outer Galois image is $\mathfrak{S}_3$ at $m = 4$, and $\mathfrak{S}_m$ at $m \geq 5$. This has an *odd-prime* hypothesis; it is not a theorem that the unrestricted outer automorphism group is finite. The arithmetic compatibility condition cannot be deleted to deduce E.8. His Theorem B is the corresponding Lie-algebra centralizer statement for $m \geq 5$. This is the precise rigidity result used here, rather than an unspecified assertion of "Galois rigidity".

5.5 The rational comparison and the maps to the pro-ℓ tower

For a field k of characteristic zero, unipotent completion means the universal pro-unipotent affine k-group receiving the discrete group, or its continuous ℓ-adic variant when $k = \mathbb{Q}_\ell$. It is not the discrete fundamental group retained by arbitrary locally constant $D(\mathbb{Q})$-coefficients, and it is not the entire pro-algebraic completion. In the braid tower we complete the pure braid groups and their path torsors, retaining the permutations and the cabling, deletion and reassociation operations. Completing the whole mapping class group indiscriminately would

define a different object. The grading below is positive word weight; motivic cohomological weight is its negative double.

Let $\mathfrak{t}_j$ be the infinitesimal braid Lie algebra on the $t_{ab} = t_{ba}$, with $[t_{ab}, t_{cd}] = 0$ for distinct indices and $[t_{ab}, t_{ac} + t_{bc}] = 0$. Let $W = \bigoplus_{w \geq 1} W_w \subseteq \mathrm{Lie}_{\mathbb{Q}}\langle x, y\rangle$ be the solution space

$$\begin{aligned} \psi(x,y) + \psi(y,x) &= 0, \\ \psi(x,y) + \psi(y,-x-y) + \psi(-x-y,x) &= 0, \\ \psi(t_{12}, t_{23}+t_{24}) + \psi(t_{13}+t_{23}, t_{34}) &= \psi(t_{23}, t_{34}) + \psi(t_{12}+t_{13}, t_{24}+t_{34}) + \psi(t_{12}, t_{23}). \end{aligned}$$

The third equation lies in $\mathfrak{t}_4$. The bracket is

$$\{\psi, \varphi\} = D_\psi \varphi - D_\varphi \psi + [\psi, \varphi], \qquad D_\psi(x) = 0, \quad D_\psi(y) = [y, \psi].$$

The ordinary commutator on the free Lie algebra is not this bracket.

Statement E.19 (Known). The precise rational source, target and established injection.

The weight-completed Lie algebra of GRT_1 is $\widehat{W} = \prod_w W_w$ with the Ihara bracket, and $\mathrm{GRT} = \mathbb{G}_m \ltimes \mathrm{GRT}_1$. It is the graded automorphism group for the completed parenthesized chord tower; the corresponding filtered braid tower has automorphism group $\mathrm{GT}(k)$. A choice of a k-associator identifies these two groups, noncanonically. The symmetry and hexagon occur at the four-point level and the pentagon at the five-point level; the same operations determine the action at every genus-zero level.

For the de Rham fibre functor on $\mathrm{MT}(\mathbb{Z})$, let $G_{\mathrm{MT}} = \mathbb{G}_m \ltimes U_{\mathrm{MT}}$ and let $\mathfrak{u}^{\mathrm{gr}}_{\mathrm{MT}}$ denote its weight-graded unipotent Lie algebra, reindexed by positive word weight. There is an injective graded Lie homomorphism

$$j : \mathfrak{u}^{\mathrm{gr}}_{\mathrm{MT}} \cong \mathrm{Lie}_{\mathbb{Q}}\langle e_3, e_5, e_7, \ldots\rangle \hookrightarrow W.$$

It integrates after weight completion to a monomorphism of pro-unipotent groups. The generators themselves are not canonical.

Proof. Drinfeld's definitions in §§5–6 and Propositions 5.1 and 5.5 give the associator torsor and the two actions (Drinfeld, 1990). The structure-preserving categorical formulation is Bar-Natan (1998, Proposition 3.4, Theorems 1–3, Corollary 4.1); its operations must be retained when invoking that formulation. Differentiating gives exactly the three equations displayed above. Brown (2012, Theorem 1.1 and its stated consequence, Conjecture 1) proves faithfulness of the mixed Tate action. The containment in the Grothendieck–Teichmüller solution Lie algebra is stated explicitly in Brown (2021, §1.4.2), using Brown's theorem and Drinfeld's equations. The free odd-weight source is the structure of $\mathrm{MT}(\mathbb{Z})$ recalled in Brown (2012, §1.1). In characteristic zero, exponential and logarithm give the equivalence between finite-dimensional nilpotent Lie algebras and unipotent groups; passing through finite weight quotients gives the completed assertion. □

Statement E.20 (Proved). The exact conditional rational counterpart.

Assume the all-weight bound of family 008, Corollary 5.4:

$$\dim_{\mathbb{Q}} W_w \leq \dim_{\mathbb{Q}} \mathrm{Lie}\langle e_3, e_5, \ldots\rangle_w \quad \text{for every } w.$$

Then the map j of E.19 is an isomorphism in every weight and after weight completion. It gives $U_{\mathrm{MT}} \cong \mathrm{GRT}_1$ and, with the matching Tate character and grading, $G_{\mathrm{MT}} \cong \mathrm{GRT}$. Via a chosen associator this is the analogous statement for GT on the unipotent genus-zero braid tower.

Proof. The injection j_w has a source of the dimension on the right. The assumed opposite inequality forces equality and surjectivity. Every weight piece is finite dimensional, so the product of these isomorphisms is an isomorphism for the complete weight filtration; the bracket in each weight involves finitely many lower weights. Exponentiation on the finite nilpotent quotients and inverse limit give the group isomorphism. The scalar actions agree by their weight definitions (with the inverse character if the negative motivic-degree convention is used), so they extend over $\mathbb{G}_m$. □

The conclusion is *claimed (OpenAI, 2026, unrefereed)* when Corollary 5.4 is supplied by that manuscript. The implication just proved is unconditional logic; its all-weight premise is not certified by our bounded calculation or by reading a comparator statement.

Statement E.21 (Known). The realization chain and what its arrows do.
Fix a prime ℓ and compatible tangential markings. Write GT_ℓ for the group of invertible pairs satisfying the pro-ℓ versions of Drinfeld's equations, with $\lambda \in \mathbb{Z}_\ell^\times$. This is generally not a pro-ℓ group, because its cyclotomic quotient need not be one. There are natural maps

$$G_{\mathbb{Q}} \hookrightarrow \widehat{\mathrm{GT}} \longrightarrow \mathrm{GT}_\ell \hookrightarrow \mathrm{GT}(\mathbb{Q}_\ell).$$

The first arrow is a continuous closed embedding. The second is continuous and is not injective; we assert no surjectivity. The third is injective and continuous for the topology on finite weight quotients. Neither its image nor the image of $G_{\mathbb{Q}}$ is surjective, topologically dense or open in $\mathrm{GT}(\mathbb{Q}_\ell)$ with this inverse-limit topology. Surjectivity or openness of the arithmetic image *inside* GT_ℓ is not concluded here.

Proof. Drinfeld constructs the completion comparison maps in §4 (Drinfeld, 1990); the continuous pro-ℓ to unipotent comparison is also Hain and Matsumoto (2003, Theorem A.6 and Corollary A.10). Faithful Magnus expansion of the free pro-ℓ group gives injectivity of its map to the characteristic-zero completed free unipotent group, hence injectivity of the pair map. The arithmetic pro-ℓ action factors through the Galois group $G_{\mathbb{Q},\{\ell\}}$ of extensions unramified outside ℓ; this is the factorization recalled and used in Hain and Matsumoto (2003, Introduction and §8.4). Nontrivial inertia at a prime different from ℓ therefore lies in the kernel of the composite. Belyi injectivity in the first arrow forces a nontrivial kernel in the second.

The pro-ℓ solution group is compact. Its scalar image lies in $\mathbb{Z}_\ell^\times$, a proper closed subset of $\mathbb{Q}_\ell^\times$, proving nonsurjectivity and failure of density in the full rational group. Its image is compact and hence closed. It cannot be open: any neighbourhood of the identity in the inverse-limit weight topology contains the kernel of a sufficiently large finite weight projection. By Brown's independent odd-weight elements, such a kernel contains a closed one-parameter subgroup isomorphic to the additive group $\mathbb{Q}_\ell$ in an arbitrarily large weight, and is not compact. A compact subgroup cannot contain that closed noncompact subgroup. The same argument applies to the compact arithmetic image. □

Hain–Matsumoto is a statement about weighted completion and the *image* of an arithmetic action, not faithfulness of $G_{\mathbb{Q}} \to \mathrm{GT}_\ell$. More precisely their Theorem 7.3 computes the free graded Lie algebra of the weighted completion of $G_{\mathbb{Q},\{\ell\}}$ with respect to the cyclotomic character; this initially includes weight one. Their Theorem 8.4 identifies the graded Lie image with the rationalized graded image of the Galois action. Lemma 8.5 eliminates weight one in the outer action; Remark 8.6 identifies the nonzero odd abelian classes by Soulé's characters. Their Theorem 9.8 gives the constrained mixed Tate factorization. Brown's faithfulness then supplies the free motivic image, while E.20, conditionally on the upper bound, says that this image exhausts $W \otimes \mathbb{Q}_\ell$. The resulting density is *Zariski density in each finite-dimensional algebraic quotient*. It is compatible with the preceding failure of topological density in the full group

of $\mathbb{Q}_\ell$-points. No argument here divides by 2 or 3 in a profinite inertia group; the all-prime Hain–Matsumoto construction and the separate characteristic-two lattice calculation must not be confused with Nakamura's odd-prime theorem E.18.

Statement E.22 (Proved). The needed profinite analogue of the upper bound.
For the closed embedding $\iota : G_{\mathbb{Q}} \hookrightarrow \widehat{\mathrm{GT}}$, the following are equivalent:

1. ι is surjective;
2. its image is topologically dense;
3. for every open normal subgroup V of $\widehat{\mathrm{GT}}$, the map $G_{\mathbb{Q}} \to \widehat{\mathrm{GT}}/V$ is surjective.

This finite-quotient exhaustion, not a rational dimension inequality, is the required profinite counterpart of Corollary 5.4.

Proof. A compact image in a Hausdorff profinite group is closed. Density is therefore equivalent to equality. Density implies surjectivity on every finite discrete quotient. Conversely, if a closed subgroup H is proper, a point outside H has an open neighbourhood disjoint from H; shrinking to a coset of an open normal subgroup V gives a coset missed by H in the finite quotient. This contradicts the third condition. □

Statement E.23 (Refuted). Equality of unipotent completions does not imply profinite equality.
The inference "a subgroup induces an isomorphism of characteristic-zero unipotent completions, therefore it is the whole profinite group" is false, already for $1 \subset C_2$.

Proof. Every finite-order element of a characteristic-zero unipotent group is the identity. Indeed, write it as $\exp X$ in a faithful upper unitriangular representation; $(\exp X)^r = 1$ implies $rX = 0$, hence $X = 0$. Therefore every homomorphism $C_2 \to U$ is trivial for every unipotent U, and the unipotent completions of 1 and C_2 are both trivial. Their profinite completions are themselves, and differ. The supplied program checks the two-element multiplication table and its associativity; the torsion-free argument is the proof of the completion assertion. This counterexample does not say that the arithmetic inclusion is proper. It refutes only the proposed transfer principle. □

Statement E.24 (Proved). Conditional abelianizations of GRT.
Assume the freeness assertion of family 008, Theorem 1.1. Then

$$W^{\mathrm{ab}} \cong \bigoplus_{k\geq 1} \mathbb{Q} e_{2k+1}, \qquad (\mathrm{GRT}_1)^{\mathrm{ab}} \cong \prod_{k\geq 1} \mathbb{G}_a(2k+1), \qquad (\mathrm{GRT})^{\mathrm{ab}} \cong \mathbb{G}_m.$$

The middle and last abelianizations are in the category of affine pro-algebraic groups, equivalently by closed commutator subgroups. The last formula, with the scalar character, is the rational analogue of Grothendieck's cyclotomic abelianization statement; the middle formula is not.

Proof. The abelianization of a free graded Lie algebra is its generating space. Passing to the weight completion replaces the direct sum by the product, and exponentiation identifies an abelian unipotent group with its additive Lie algebra. In the full semidirect product, every positive-weight abelian piece has commutator action $(h-1)v = (2^{\mathrm{wt}(v)} - 1)v$ for the scalar $h = 2$. The coefficient is nonzero and invertible over $\mathbb{Q}$. Thus all unipotent weight pieces vanish in the abelianization. This argument can be applied successively in every finite weight quotient;

inverse limit gives the closed-commutator assertion. The remaining quotient is the already abelian group $\mathbb{G}_m$. No assertion about the abstract, nonclosed derived subgroup of an infinite product is being made. □

Statement E.25 (Known). The arithmetic meaning of the odd generators.
On the motivic side the odd generating lines are dual to

$$\mathrm{Ext}^1_{\mathrm{MT}(\mathbb{Z})}(\mathbb{Q}(0),\mathbb{Q}(2k+1)) \cong K_{4k+1}(\mathbb{Z})\otimes\mathbb{Q}, \qquad k\geq 1.$$

Each line has dimension one. Their ℓ-adic realizations are measured by the nonzero Soulé characters in weight $2k+1$. In particular they are abelian classes in the unipotent or cyclotomic-kernel Lie algebra, not extra ordinary abelian characters of $G_{\mathbb{Q}}$ contrary to Kronecker–Weber.

Proof. This is the mixed Tate description recalled in Brown (2012, §1.1), together with Hain and Matsumoto (2003, Theorem 7.3 and Remark 8.6). The $G_{\mathbb{Q}}$-equivariant abelianization in the latter is $\bigoplus_{k\geq 1}\mathbb{Q}_\ell(2k+1)$. Taking the full semidirect product with the cyclotomic torus kills these positive-weight coinvariants, exactly as in E.24. □

5.6 Audit of the new all-weight upper bound

Statement E.26 (Open). The imported 2026 claim and the extent of our verification.
Family 008, OpenAI (2026a, Theorem 1.1 and Corollary 5.4), claims that W is free on one generator in every odd weight at least three. We read §§2–5, including the all-weight reduction, and verified the implication E.20 using the established Brown lower bound. The following audit records the actual checks and their limits. We have not independently certified every all-weight rewriting and deletion argument, so our mathematical use of the upper bound remains conditional. "Open" here records that outstanding verification in this deliverable; it does not erase the announced proofs described below.

Integral reduction (§2). The use of $R=\mathbb{Z}_{(2)}$ is appropriate. In a fixed weight the free ordinary Lie lattice is finite free by its integral Hall–Lyndon basis. The solution lattice $L_w = W_w \cap \mathrm{Lie}_R\langle x,y\rangle_w$ is saturated: if $2a\in L_w$, then $a\in W_w$ because the equations are rational linear, and a already lies in the ambient lattice. Hence its reduction embeds in the ambient reduction, and $\dim_{\mathbb{F}_2}\overline{L}_w = \dim_{\mathbb{Q}} W_w$. We followed this proof of Proposition 2.5. It does *not* assert that $\overline{L}_w$ is the entire mod-two solution space. The degree-one pentagon leaves $-\alpha t_{12}-\beta t_{34}$, and the degree-two three-cycle gives $3[x,y]$; thus Lemma 2.1 indeed gives $W_1=W_2=0$.

Lemma 2.3 uses the standard free-fibre decomposition of the infinitesimal braid Lie algebra; its explicit derivations give the required integral torsion-freeness of the targets. We followed the presentation comparison. For Lemma 2.4 we followed the reduction of the pentagon in the free fibre, the centralizer argument, and the linearization giving the special-derivation identity and its transverse derivative. Its division by the nonzero integer $w+1$ takes place over $\mathbb{Q}$ *before* reducing the saturated lattice; it is not a division by an even integer in $\mathbb{F}_2$. This distinction is necessary for the stated proof to make sense.

Filtered comparison (§3). The substitution $A=x^2$, $C=[x,y]$, $B=y$ is made in the associative algebra in characteristic two, with the number of occurrences of B as filtration. The prefix words xx, xy, y give the asserted injective associative encoding. The source and target commutator tables have a common count-preserving part: correction terms raise the count in

the braid source and lower it in the cyclotomic arrangement target. The ordering rule replaces a base letter followed by a fibre letter by the opposite order plus its commutator, decreasing the stated inversion measure. The proof needs independent normal words in the *source*; mere spanning in the target is enough. We checked this direction of the argument and selected entries of the tables (3.6)–(3.12). We did not rederive every entry or mechanize the complete induction. Proposition 3.2 is therefore still an imported all-weight step, not a consequence of our finite computation.

Deletion representation and leading-term injectivity (§4). The nine type-B_3 hyperplanes are first defined over $\mathbb{Q}$. The plus and minus hyperplanes remain distinct symbols after mod-two reduction; merging them geometrically in characteristic two would change the arrangement Lie algebra. The deletion operator at a letter has coefficient equal to the valuation on its right edge minus the valuation on its left edge, with equal-letter differences assigned zero. We followed Proposition 4.1's logarithmic flatness and residue argument over the integers. Our independent implementation of those edge valuations tests all 13 codimension-two flats, every incident distinguished hyperplane, and every word of length at most five in the three-letter alphabet. All 13,104 integer commutator instances vanish. This checks signs, hyperplane labels and the repeated-letter convention in that range. It does not prove the representation in unbounded length.

We followed how Lemma 4.3 uses the minimal number of A's, terminal deletion and reversal to constrain a leading polynomial after $A = 0$. Proposition 4.4 then uses the reduced special identity to compare coefficients, separating the cases of at least two A/C letters and the small boundary cases. In particular the $r = 2$ expression $ABB + BBA$ is not the uniform coefficient sum; the proof cannot discard that case by an informal characteristic-two symmetry argument. We checked the displayed reductions and case division, but did not independently reconstruct the general coefficient-exchange induction.

The image bound (§5). The even-index bound is an application of free-Lie elimination. Lemma 5.2 uses the Ore-extension calculation with $D(B) = C$, $D(C) = 0$, $D^2 = 0$, giving invariance under $B \mapsto B + s(C)$. In Proposition 5.3 the Vandermonde variables are independent indeterminates; the argument takes place over a polynomial ring, not by evaluating a supposedly invertible matrix on the two elements of $\mathbb{F}_2$. Divisibility by the successive differences in the encoding excludes the index-zero elimination generator. Lazard elimination then leaves the ordinary free Lie algebra generated by $g_k = \mathrm{ad}_C^k(B)$ for $k \geq 1$, whose weights are $2k + 1$. We followed the polynomial divisibility and ordinary-Lie retraction steps. The distinction between ordinary Lie elements and all primitive elements in characteristic two is consistently necessary. Combining this image bound with the injective projection and the finite filtration yields Corollary 5.4 exactly as stated. Proposition 5.5's leading-bracket calculation is not needed for E.20, since Brown already provides the compatible Lie injection.

What this audit establishes. The integral dimension transfer, low-weight vanishing, conditional dimension squeeze, completion passage, abelianization calculation and bounded deletion check have been independently checked here. The unbounded commutator-table/normal-form comparison of §3 and the coefficient-detection induction of §4 remain the principal steps not fully rederived by us. We did not use the alternative lower-bound construction in §§6–8, its regularized-transport proposition, or the graph-complex corollary. They are claimed, steps not checked by us. The program additionally computes, from the free-Lie generating series, the predicted weight dimensions through 30. Those numbers describe the proposed bound; they are not a computation of $\dim W_w$.

Statement E.27 (Refuted). The attached comparator is not a checked Lean proof. The claim that the attached file `lean/DeligneDrinfeld.lean` itself provides a kernel-checked proof of Theorem 1.1 is false.

Proof. We read the supplied file. It defines the solution equations, Ihara bracket and the claimed graded and completed statement, and ends with `theorem main : MainStatement := by sorry`. The scope note describes a formalization elsewhere, but that description does not remove the placeholder in this comparator. We did not fetch, build or audit a separate Lean proof repository. Thus this attachment confirms the shape of the target statement only. This is a refutation of a certification claim, not of the mathematical theorem. □

Statement E.28 (Known). The status search finds subsequent proof claims. As of our search on 11 November 2026, Willwacher (2026, Theorem 1.1) presents a simplified proof of the Deligne–Drinfeld conclusion following family 008; the opened version is arXiv:2610.15592v4, dated 28 October 2026. We read its principal statements and reduction, not a complete independent proof audit. It is an additional primary preprint source, so it would be inaccurate to report that no subsequent mathematical check or proof treatment exists. Its existence does not turn the attached comparator into a checked formal proof, and it does not address the profinite equality E.8. Our conditional labels for use of the supplied upper bound remain explicit.

5.7 Consequences for the requested categorical formulation

The group acting on a completed degeneration category in Part F must be specified by its retained operations. For the marked genus-zero system with common inertia exponent, relabellings, hole-filling and clutching, its restriction is the group in E.4. A proof that these restrictions determine and lift category automorphisms is additional work; it is not part of the wording of the Harbater–Schneps theorem. Part F supplies that comparison. Once it is supplied, replacing each pure braid group and tangential path torsor by its unipotent completion and retaining cabling and reassociation gives the filtered GT-tower, or, after an associator, the graded GRT-tower in E.19. Its motivic arithmetic equality is precisely the conditional theorem E.20. The revised profinite arithmetic quotient is already $G_{\mathbb{Q}}$ by E.2; equality with the full geometric tower automorphism group still asks E.8.

Neither the two excluded low-dimensional elliptic and rational nonhyperbolic types nor orientation-reversing mapping classes can be inserted into these assertions without changing the hypotheses. In particular, quotienting by orientation reversal would kill complex conjugation in a real model. The orientation-positive geometric subgroup in E.1–E.2 is therefore essential. Stack stabilizers at elliptic curves with extra automorphisms, including those of orders 4 and 6, are retained throughout; passing to a coarse j-line would not preserve the fundamental-group extensions used above.

6 F. One stratified object, its completion, and its automorphisms

6.1 F.1. Which category is being completed

Statement F.1 (Known). The input category, with its variance.
Let $n \geq 3$, $G_n = P_{0,n}$, and let $\mathcal{C}_n$ be the set of isotopy classes of essential multicurves on the labelled punctured sphere, including the empty multicurve. For $\sigma \in \mathcal{C}_n$ put $\Delta_\sigma = \langle t_c : c \in \sigma \rangle \simeq \mathbb{Z}^{|\sigma|}$. The exit category is

$$\operatorname{Ob} C_n = \mathcal{C}_n, \qquad \operatorname{Hom}_{C_n}(\sigma, \tau) = \{g \in G_n : g\sigma \supseteq \tau\}/\Delta_\sigma, \qquad [h] \circ [g] = [hg]. \tag{F.1}$$

Its nerve is the exit-path ∞-category of the boundary-stratified $\overline{\mathscr{M}}_{0,n}(\mathbb{C})$. Put $T_n = C_n^{\mathrm{op}}$. Thus arrows in C_n smooth nodes, arrows in T_n degenerate, and constructible coefficients are covariant functors on C_n, or presheaves on T_n. This is Jansen (2023, Construction 3.18, Proposition 3.23, Corollary 6.6, and Theorem 6.7), with the pure labelled genus-zero convention of A.

Verification of the input used. We use Jansen's identification, not a new proof of the topology of the stratification. Its group-theoretic composition is well-defined: if $\tau \subseteq g\sigma$, then $g^{-1}\Delta_\tau g \subseteq \Delta_\sigma$. This is exactly the identity needed to replace h by hd and g by ge without changing the composite coset. The stabilizer quotient is

$$\operatorname{Aut}_{C_n}(\sigma) = \operatorname{Stab}_{G_n}(\sigma)/\Delta_\sigma \simeq \prod_v P_{0,\mathrm{val}(v)}.$$

There are finitely many orbits, indexed by stable labelled trees. We choose one representative σ_s per tree s and henceforth work with this finite-object skeleton C_n^0. This changes the category by equivalence, not by groupoid completion. We retain its actual composition maps. For $n = 4$ there are four objects and for $n = 5$ there are twenty-six, with the Hom-sets specified in STRATA; the combinatorial counts and forgetting map are independently checked by the accompanying program. □

Statement F.2 (Proved). A strict, universal finite-category completion.
For a category C with a finite specified object set, a finite congruence R is an equivalence relation of finite index on each $\operatorname{Hom}_C(x, y)$ that respects identities and composition. Let C/R retain the same objects and the quotient Hom-sets. The poset of these congruences, ordered by refinement, is cofiltered, and

$$C^{\mathrm{fin}} = (C/R)_R \in \mathrm{Pro}(\mathrm{FinCat}), \qquad C/R' \longrightarrow C/R \quad (R' \subseteq R) \tag{F.2}$$

is our strict profinite completion. Every transition functor is the identity on objects and surjective on each Hom-set. Its materialization $|C^{\mathrm{fin}}| = \varprojlim_R C/R$ is a category object in profinite sets with a finite discrete object set. Continuous functors from it to a finite category are precisely the functors represented by the pro-object. Continuous natural isomorphisms are retained, and can be computed at this level. In particular their automorphism groups agree.

Proof. Intersection of two finite congruences is again a finite congruence: its classes inject into the product of the two finite quotient sets. Thus the system and its transitions are explicit. Composition on the inverse-limit Hom-sets is the inverse limit of the finite composition maps, and is continuous and associative. A functor from C to a finite category has a finite kernel congruence, including when different objects have the same image, so factors through a quotient in (F.2).

Conversely a continuous map from a compact profinite Hom-set to a finite set factors through finitely many quotient coordinates. There are only finitely many pairs of objects. Taking a

common refinement makes all Hom-maps factor at one stage. Because the projections are surjective on each Hom-set, identities and every composable pair lift from that stage; the functorial equations therefore hold already there. This proves the functor assertion. Apply the same argument to the finitely many components and naturality equations of a natural transformation. Equivalently,

$$\mathrm{Hom}_{\mathrm{Pro}}((C_i),(D_j)) = \varprojlim_j \varinjlim_i \mathrm{Hom}(C_i, D_j)$$

with the corresponding groupoid of functors and natural isomorphisms. Invertibility is imposed by an inverse with its two natural isomorphisms. Thus replacing this strict pro-system by its continuous materialization does not erase transitions; their equivalence follows from the stated finite-object and quotient hypotheses. It would not follow for an arbitrary pro-system presented only in a homotopy category. □

Statement F.3 (Refuted). Arbitrary natural transformations may not be inverted.
It is false that quotienting functors by arbitrary natural transformations is harmless for the completed exit category. The appropriate equivalence is natural *isomorphism*, not a zigzag of transformations whose components need not be invertible.

Proof. Let $[1]$ have objects $0, 1$ and its single nonidentity arrow $0 \to 1$. The functors $[1] \to \{*\}$ and $\{*\} \to [1]$ choosing 1 have composite naturally related to the identity by the arrow $0 \to 1$. They become inverse after arbitrary natural transformations are inverted. They are not equivalences of categories, since 0 and 1 are not isomorphic. Both categories are already finite, hence already complete under (F.2). Moreover $\mathrm{Fun}([1], \mathrm{Set})$ has three subobjects of its terminal object, whereas Set has two; the associated coefficient categories are different. The program enumerates these subobjects and checks the transformation. This gives a finite counterexample to the proposed inference. The CATHOT warning about stagewise groupoidification is therefore respected, not applied as an identification of the exit category itself with a groupoid. □

Statement F.4 (Proved). Finite representations, split representations, and all continuous representations.
For (F.2), finite-valued continuous functors satisfy

$$\mathrm{Fun}_{\mathrm{cts}}(|C^{\mathrm{fin}}|, \mathrm{FinSet}) \simeq \varinjlim_R \mathrm{Fun}(C/R, \mathrm{FinSet}). \tag{F.3}$$

For arbitrary sets, the analogous colimit denotes representations that factor through *one* finite stage. It need not contain every continuous representation. For a profinite group H, all discrete continuous H-sets form the filtered-colimit completion of the finite continuous H-sets. They are not the actions of the abstract group with its topology forgotten.

Proof. The first assertion is F.2 applied to the finite category of maps among the finitely many finite values of a given functor; it also treats its natural transformations. For the distinction, take $H = \widehat{\mathbb{Z}}$ and

$$S = \coprod_{m \geq 1} \mathbb{Z}/m!\mathbb{Z}.$$

Each orbit has open stabilizer, so this is a continuous action on a discrete set. A finite stage $\mathbb{Z}/N\mathbb{Z}$ cannot act through the given action on all summands, since that would force every $m!$ to divide N. On the other hand S is the filtered union of its finite invariant subsets. The abstract regular action of $\widehat{\mathbb{Z}}$ on itself with the discrete topology is not continuous, since its point

stabilizer is not open. This proves both distinctions. Finite representations in this paper always mean (F.3).

This verifies the precise relevant content of the attached GALOIS F.4: Hoyois (2018, notation preceding Definition 2.10, Theorem 3.13, and Corollary 3.20) distinguishes the pro-object functor category from its classifying topos and calls the stagewise objects *split*. The attached statement includes that distinction. We do not refute it; we refute the unqualified inference that every continuous set action factors through one finite quotient. No Kan–Thurston or rational-local-system claim is needed here. □

6.2 F.2. Transporters, closed stabilizers, and an explicit model

Statement F.5 (Known). The profinite curve complex and multitwist stabilizers. For genus zero and every $n \geq 3$, let $\widehat{G}_n$ be the full profinite completion. It equals the procongruence completion. The profinite curve complex is defined by embedding an unoriented simple curve in the space of conjugacy classes $\{\gamma, \gamma^{-1}\}$ in $\widehat{\Pi}_{0,n}$, taking the closure of simple curves, and then taking the closure of the discrete r-multicurves in its finite-subset space, for each r. Include the empty simplex. It is also the inverse limit of the finite orbit complexes over congruence levels. For a discrete multicurve σ, its profinite stabilizer is $\overline{H_\sigma}$, where $H_\sigma = \mathrm{Stab}_{G_n}(\sigma)$, and

$$1 \longrightarrow \overline{\Delta}_\sigma \simeq \widehat{\mathbb{Z}}^{|\sigma|} \longrightarrow \overline{H}_\sigma \longrightarrow \prod_v \widehat{P}_{0,\mathrm{val}(v)} \longrightarrow 1. \tag{F.4}$$

Furthermore $N_{\widehat{G}_n}(\overline{\Delta}_\sigma) = \overline{H}_\sigma$. The sources are Boggi (2020, Definitions 3.3–3.4, Theorems 3.5–3.6, Theorem 4.2, and Corollary 4.3), with the genus-zero congruence theorem. The trivial empty-multicurve case is interpreted separately.

Scope and hypotheses. Boggi's statements are initially about the procongruence group, with a general class of finite quotients. We take the class of all finite groups. Part B verifies equality with the full profinite topology in genus zero, so the substitution here is legitimate. Pure mapping classes on a labelled sphere cannot permute the edges or reverse the sides of a fixed multicurve, since each edge has a distinct labelled split. Thus the finite permutation group in the general stabilizer theorem is trivial, giving (F.4). The multitwist normalizer theorem gives the last equality. Distinct disjoint twists are independent, so no prime, including 2 or 3, is excluded from $\widehat{\mathbb{Z}}^{|\sigma|}$. This verifies the precise results needed below; it does not assume that an arbitrary subgroup of a residually finite group inherits its full profinite topology. □

Statement F.6 (Proved). The completion of a stabilizer really is the stabilizer. For the groups in F.5, the natural map $\widehat{H}_\sigma \to \overline{H}_\sigma$ is an isomorphism. Both H_σ and Δ_σ are closed in the profinite topology of G_n, and $C_n^0 \to |(C_n^0)^{\mathrm{fin}}|$ will be faithful.

Proof. Cutting the sphere along σ gives the discrete exact sequence

$$1 \to \Delta_\sigma \to H_\sigma \to Q_\sigma := \prod_v P_{0,\mathrm{val}(v)} \to 1.$$

Its completion is right exact. The natural homomorphism $\widehat{\Delta}_\sigma \to \widehat{H}_\sigma$ is injective because its composite into $\widehat{G}_n$ is the injection in (F.4). The kernel of $\widehat{H}_\sigma \to \widehat{Q}_\sigma$ is exactly the closed image of $\widehat{\Delta}_\sigma$: quotienting by it has the universal finite quotients of H_σ/Δ_σ. Comparison with (F.4), and equality of the component completions with their procongruence groups, gives an

isomorphism of both ends. A direct kernel-and-surjectivity argument, or the short five lemma, gives the asserted middle isomorphism.

The discrete curve complex injects into the profinite one by surface-group conjugacy separability, used in its construction. If $g \in G_n \cap \overline{H}_\sigma$, then $g\sigma = \sigma$ in the profinite complex, hence already in the discrete complex; thus $g \in H_\sigma$. This proves closedness of H_σ. Within H_σ, the subgroup Δ_σ is the kernel of the map to the residually finite group Q_σ, so is closed there. The full induced topology on H_σ just proved makes it closed also in G_n. Distinct cosets modulo Δ_σ can therefore be separated in a finite quotient of G_n. The transporter formula in the next statement proves faithfulness on every Hom-set. □

Statement F.7 (Proved). Finite quotient categories and the transporter completion agree.

For representatives σ_s of the stable labelled trees put $A_{st} = \{g \in G_n : g\sigma_s \supseteq \sigma_t\}$. For each finite-index normal subgroup $V \lhd G_n$ let $p_V : G_n \to G_n/V$, and define a finite category $C_{n,V}$ by

$$\operatorname{Ob} C_{n,V} = \{s\}, \qquad \operatorname{Hom}_{C_{n,V}}(s,t) = p_V(A_{st})/p_V(\Delta_{\sigma_s}), \quad [h][g] = [hg]. \tag{F.5}$$

If $V' \subseteq V$, its transition functor is induced by $G_n/V' \to G_n/V$. These quotient categories are cofinal among all finite congruence quotients of C_n^0. Thus

$$\widehat{C}_n := \varprojlim_V C_{n,V}, \qquad \operatorname{Hom}_{\widehat{C}_n}(s,t) = \overline{A_{st}}/\overline{\Delta}_{\sigma_s} \tag{F.6}$$

materializes the strict pro-object of F.2. In the full profinite multicurve model this is the candidate transporter formula from the prompt:

$$\operatorname{Hom}(\sigma,\tau) = \{g \in \widehat{G}_n : g\sigma \supseteq \tau\}/\overline{\Delta}_\sigma.$$

It is a category in profinite sets, and the discrete comparison is faithful for every $n \geq 3$.

Proof. The group-theoretic identity $g^{-1}\Delta_{\sigma_t} g \subseteq \Delta_{\sigma_s}$ in F.1 proves composition in (F.5) well-defined even after inserting elements of V, because V is normal. Associativity and identities descend from multiplication. The stated transitions are actual functors, not merely outer homomorphisms.

If A_{st} is nonempty, pure mapping classes preserve every split of a tree. Exactly one subset $\eta \subseteq \sigma_s$ has the splits of σ_t. Choose r with $r\eta = \sigma_t$. Then $A_{st} = H_{\sigma_t} r$. Hence its closure is $\overline{H}_{\sigma_t} r$, and the compact coset quotient gives (F.6). Conversely, a profinite transporter must send that same subset to σ_t, by the discrete finite set of split labels, so lies in this closed coset. This proves agreement with the proposed formula, including the empty Hom-sets.

To prove cofinality, let $F : C_n^0 \to D$ be a functor to a finite category. Its restriction to $\operatorname{Aut}(s) = H_{\sigma_s}/\Delta_{\sigma_s}$ has finite-index kernel. By F.6 this kernel pulls back to a subgroup of H_{σ_s} containing $V_s \cap H_{\sigma_s}$ for some finite-index normal $V_s \lhd G_n$. There are only finitely many s, so take $V = \bigcap_s V_s$. Suppose $g, g' \in A_{st}$ have the same class in (F.5). For some $d \in \Delta_{\sigma_s}$, $h = g'd^{-1}g^{-1} \in V \cap H_{\sigma_t}$. The restriction of F kills h, and d already represents the identity at s, so $F([g']) = F([g])$. Thus F factors through $C_{n,V}$. This is the missing cofinality assertion that could not be deduced from residual finiteness alone.

Each Hom in (F.6) is a closed subset of the profinite coset space $\widehat{G}_n/\overline{\Delta}_{\sigma_s}$, and multiplication is continuous. The full object space is the finite disjoint union $\coprod_s \widehat{G}_n/\overline{H}_{\sigma_s}$. Its transporter category is continuously equivalent to the skeleton just defined. To see that continuous choices are possible here, use the countable descending basis of open normal subgroups of the finitely generated profinite group and choose compatible transversals at successive finite quotients. Their inverse limits provide continuous transporters on each orbit. The choices are not canonical, but change the model by continuous natural equivalence. Faithfulness follows from the closedness of Δ_{σ_s} proved in F.6. □

Statement F.8 (Proved). Forgetting, clutching, and relabelling survive completion. Every specified forgetful functor $Q : C_{n+1} \to C_n$, and every clutching functor, extends to the strict pro-completions by its universal property. On the transporter model, forgetting is induced by the completed pure Birman homomorphism $\widehat{\phi} : \widehat{G}_{n+1} \twoheadrightarrow \widehat{G}_n$ and the continuous map on profinite multicurves. In degeneration variance it is $\widehat{Q}^{\mathrm{op}} : \widehat{T}_{n+1} \to \widehat{T}_n$. Relabelling remains a pseudo-action with its specified natural isomorphisms; choosing a skeleton does not manufacture a splitting of a mapping class permutation extension.

Proof. Pulling a finite congruence back along a functor gives a finite congruence. This defines the pro-morphism together with its comparison maps at all stages; composition is respected by functoriality of pullback. On a discrete multicurve forgetting drops peripheral components and identifies curves that become isotopic. The images of its twists lie in the multitwist subgroup of the resulting multicurve, which proves well-definedness on cosets. The continuous extension exists by F.7 and agrees on a dense subcategory, hence agrees with the universal extension. No arbitrary point of the profinite curve complex is assigned an image without this continuity argument.

For clutching, the discrete full replete subcategory consists of the objects whose tree contains the prescribed split. Cutting along it identifies the relevant stabilizer quotient and its transporter Hom-sets with the product categories of the components. Their full induced profinite topologies, proved in F.6, imply that completion is still fully faithful. One can also apply the finite congruence argument to this product: finite functors on finitely many factor Hom-sets are controlled by a common finite quotient. This proves the product identification in genus zero. In T-variance it is a cosieve; in C-variance it is a sieve.

A change of marking conjugates a group homomorphism by its transporter, which is a natural isomorphism of the corresponding functors. Retaining these isomorphisms and their cocycle identity proves pseudoequivariance for permutations and for repeated forgetting. It is stronger data than a list of homomorphisms modulo unrelated inner automorphisms. □

6.3 F.3. Geometric comparison and the Galois action

Statement F.9 (Known). Stratified Riemann existence supplies the geometric comparison. Let $X_n = \overline{\mathscr{M}}_{0,n}$ over $\mathbb{Q}$, stratified by its stable labelled trees. Fix $\overline{\mathbb{Q}} \hookrightarrow \mathbb{C}$. The finite boundary-constructible étalesheaves on $X_{n,\overline{\mathbb{Q}}}$, with their stalk functors at one chosen geometric point of each stratum, are equivalent to

$$\mathrm{Fun}_{\mathrm{cts}}(\widehat{C}_n, \mathrm{FinSet}), \tag{F.7}$$

with their evaluation functors. The finite-category exodromy object reconstructed from these data is therefore equivalent to the strict completion in F.7. At the full stratified ∞-categorical level, the precise theorem is

$$\mathrm{Gal}(X_{n,\mathbb{C}}/P_n) \simeq \mathrm{Exit}^{P_n}(X_n(\mathbb{C}))^{\wedge}_{\pi}, \tag{F.8}$$

where the right-hand completion is in profinite stratified spaces. Here P_n denotes the finite poset of strata, not the mapping class group $P_{g,n}$. This is Barwick et al. (2020, Proposition 12.6.4 and Corollaries 12.6.6–12.6.7). We take its finite-category, finite-set exodromy part in (F.7); we do not replace its full ∞-categorical completion by a limit of ordinary categories without a further comparison.

Precise deduction at the level used. Base extension between algebraically closed characteristic-zero fields preserves finite étalecoverings of the strata and their constructible gluing data. Apply

the analytic/étaleconstructible comparison of the cited proposition and Jansen's exodromy theorem F.1. Their restriction to finite stalks gives $\mathrm{Fun}(C_n^0, \mathrm{FinSet})$. By F.2 and F.7 every such functor factors through a finite quotient category, with its maps and natural transformations. This yields (F.7), including compatibility with stalks.

For a finite category, the natural transformations between evaluation functors on its finite covariant representations recover its Hom-sets by the covariant representables. Passing through the actual finite quotient system gives the corresponding profinite Hom-sets by inverse limit. Thus the finite representations *with their evaluation functors* recover precisely (F.6). This proves the claimed comparison rather than inferring it from equality of ordinary fundamental groups. Formula (F.8) is the stronger theorem at its own level. It is not a claim that the unstratified compactification is a $K(\pi, 1)$, nor that the full point-and-specialization category $\mathrm{Gal}(X)$ equals the object for one fixed stratification. The latter object is the boundary-constructible localization/spectrification denoted $\mathrm{Gal}(X/P_n)$ in the source. □

Statement F.10 (Proved). Continuous outer Galois action on the completed category.

Galois descent defines a coherent continuous action

$$BG_{\mathbb{Q}} \longrightarrow \mathrm{Aut}(\widehat{C}_n)$$

in the 2-category of profinite categories and continuous equivalences. On equivalence classes it gives a homomorphism $G_{\mathbb{Q}} \to \pi_0 \mathrm{Aut}(\widehat{C}_n)$. It is compatible with forgetting, clutching, and relabelling, and with the opposite category $\widehat{T}_n$. On a local normal-crossings inertia group at a rational boundary point it acts, modulo simultaneous geometric conjugation, by

$$t_e \longmapsto t_e^{\chi(\sigma)} \quad \text{for every node } e. \tag{F.9}$$

For $n \geq 4$ the outer action on $\widehat{C}_n$ is faithful.

Proof. Every $\sigma \in G_{\mathbb{Q}}$ acts on $X_{n,\overline{\mathbb{Q}}}$ over $\mathbb{Q}$ and hence, by inverse pullback $(\sigma^{-1})^*$, on finite boundary-constructible sheaves; this convention gives a left action. The natural associativity of pullback supplies the coherence for products of σ's. Each finite sheaf and morphism descends to a number field, so this action is continuous on the finite data. Transfer by (F.7), including the stalks and their transports, gives the displayed outer categorical action. Changing the chosen geometric paths changes it by continuous natural isomorphism. It does not alter its class in π_0.

The geometric maps forgetting or clutching markings and the relabellings are defined over $\mathbb{Q}$. Their pullbacks commute coherently with this descent action, proving compatibility. Locally at a rational boundary point there are smoothing coordinates q_e with divisor $q_e = 0$. In characteristic zero Kummer covers $q_e^{1/m}$ have inertia μ_m; Galois acts on their roots of unity by $\chi \bmod m$. Taking all m gives (F.9). If parameters, tangential points, or paths are changed, the common local inertia group is conjugated, which explains the stated ambiguity; there is no preferred global generator independent of these choices.

For $n = 4$, the open-stratum group is $\widehat{G}_4 \simeq \widehat{F}_2$ for $\mathbb{P}^1 \setminus \{0, 1, \infty\}$, and Belyi faithfulness, proved or cited precisely in D, detects $G_{\mathbb{Q}}$. For $n > 4$, forgetting to four markings is a surjective map on open-stratum groups and intertwines the Galois action. If the upstairs action were inner, its induced downstairs action would be inner, so faithfulness transfers in the needed direction. For $n = 3$ the labelled category is a point and this action is trivial. Simple connectedness of the compactification does not contradict these statements: it concerns its unstratified fundamental group, which is a different quotient of the category. □

Statement F.11 (Proved). The failure of smoothness persists profinitely.
For forgetting the fifth marking at the stratum $q_{12,34}$, the continuous lifting groupoid in the STRATA smoothness test is equivalent to $B\widehat{\mathbb{Z}}$. Its stabilizer is

$$\ker(\widehat{\phi}) \cap \overline{\Delta}_q = \{(a,-a) : a \in \widehat{\mathbb{Z}}\} = \overline{\langle t_1 t_2^{-1} \rangle}. \tag{F.10}$$

Thus the forgetful functor fails the analogous homotopy-smoothness test with continuous finite derived coefficients. Galois preserves this procyclic stabilizer and acts on it by the cyclotomic character, up to simultaneous conjugation.

Proof. The two distinct twists upstairs generate $\widehat{\mathbb{Z}}^2$ by F.5. Both map to the same primitive twist downstairs, so the exponent map is $(a,b) \mapsto a+b$. The only lift targets in the relevant test are open objects: a curve disappearing on forgetting 5 would have split $i5 \mid \{1,2,3,4\} \setminus \{i\}$, which occurs in neither of the two specified source splits. This is exactly the discrete split-label argument, and it persists on the profinite complex because the labels form a finite discrete invariant. The kernel $K = \ker \widehat{\phi}$ acts transitively on the lift objects, since $\widehat{\phi}(\overline{\Delta}_q)$ is the full downstairs twist group. Its stabilizer is (F.10), so the action groupoid is equivalent to $B\widehat{\mathbb{Z}}$.

For every prime ℓ, its continuous cohomology has $H^1_{\mathrm{cts}}(\widehat{\mathbb{Z}}, \mathbb{F}_\ell) = \mathrm{Hom}_{\mathrm{cts}}(\widehat{\mathbb{Z}}, \mathbb{F}_\ell) \simeq \mathbb{F}_\ell$. Consequently this pro-homotopy type is not terminal and the required homotopy-cofinality condition fails. We are not claiming that connectedness alone fails, or that ordinary set-valued colimits detect this H^1. At the rational normal-crossings point, (F.9) uses one exponent on both branches; hence $t_1 t_2^{-1} \mapsto (t_1 t_2^{-1})^{\chi(\sigma)}$ under a compatible local lift. This also proves the statement for the corresponding degeneration functor, with variance reversed. The program checks the kernel and its cyclotomic stability in every finite modulus $2 \le m \le 24$; the displayed algebra proves it in the inverse limit. □

6.4 F.4. Boundary-category automorphisms and the two levels

Statement F.12 (Proved). Restriction to the open stratum detects every categorical automorphism.
Let $n \ge 4$. A continuous autoequivalence of $\widehat{C}_n$ fixing the labelled stratum types is determined up to natural isomorphism by its outer action on $\widehat{G}_n = \mathrm{Aut}(u)$. Its image consists precisely of the outer automorphisms preserving the conjugacy classes of all boundary multitwist subgroups. Such an automorphism lifts if it preserves the associated profinite multicurve incidence. An inertia-preserving automorphism does preserve that incidence, by the commuting-twist characterization of simplices. These assertions hold for the continuous materialization and the strict pro-object alike.

Proof. For each object s choose the standard arrow $j_s : s \to u$ represented by 1. Its orbit is the left $\widehat{G}_n$-set $\mathrm{Hom}(s,u) = \widehat{G}_n / \overline{\Delta}_{\sigma_s}$. Its stabilizer therefore recovers the conjugacy class of $\overline{\Delta}_{\sigma_s}$. An autoequivalence must preserve these classes. Suppose first that its outer action at u is trivial. Conjugate the functor by a natural isomorphism at u to make the action literally the identity. Its map on $\mathrm{Hom}(s,u)$ is then an equivariant automorphism of a transitive profinite $\widehat{G}_n$-set, hence is given by an element of $N_{\widehat{G}_n}(\overline{\Delta}_{\sigma_s})/\overline{\Delta}_{\sigma_s}$. By F.5 this is exactly $\mathrm{Aut}(s)$. A natural change at s therefore makes the map on $\mathrm{Hom}(s,u)$ the identity. Postcomposition with j_t embeds $\mathrm{Hom}(s,t)$ into $\mathrm{Hom}(s,u)$ by the transporter formula. Thus the functor is the identity on every Hom-set. This proves injectivity after natural isomorphism.

Conversely let $\alpha \in \mathrm{Aut}(\widehat{G}_n)$ preserve the inertia subgroups and labelled strata. A profinite simple curve is determined by its inertia subgroup, and a finite set of distinct curves is a simplex precisely when the corresponding twist subgroups commute. The injectivity and commutation

assertions are Boggi and Funar (2023, Theorem 4.11, Corollary 4.12, and Proposition 7.2); the exact resulting extension to all inertia groups is also Boggi (2026a, Proposition 3.5). Hence α induces a continuous simplicial map f on profinite multicurves with $\alpha(\overline{\Delta}_\sigma) = \overline{\Delta}_{f\sigma}$ and $f(g\sigma) = \alpha(g)f(\sigma)$. The formula $(\sigma, [g], \tau) \mapsto (f\sigma, [\alpha(g)], f\tau)$ is consequently a well-defined continuous functor on the full transporter category. Its inverse is induced by α^{-1}. Transferring to the skeleton gives the desired autoequivalence, independent of the transporters up to natural isomorphism. Inner α give naturally isomorphic functors. The injection already proved gives uniqueness of this extension. F.2 proves agreement with pro-object automorphisms. □

Statement F.13 (Proved). The decorated two-level boundary tower has automorphism group $\widehat{\mathrm{GT}}$.

Let $\mathcal{T}^{\partial}_{4,5}$ consist of $\widehat{T}_4, \widehat{T}_5$, all five forgetting functors, clutching inclusions with their $\widehat{T}_3, \widehat{T}_4$ factors, and coherent relabelling pseudo-actions. Its automorphisms are continuous equivalences fixing labelled stratum types, preserving oriented boundary inertia with one exponent $\lambda \in \widehat{\mathbb{Z}}^\times$, and commuting with these operations through natural isomorphisms satisfying their coherence equations. Modulo coherent natural isomorphism,

$$\pi_0 \operatorname{Aut}(\mathcal{T}^{\partial}_{4,5}) \simeq \widehat{\mathrm{GT}}. \tag{F.11}$$

The same holds for the strict pro-objects. The coordinate identification uses the usual marking by oriented loops x, y around $0, 1$ and a compatible tangential base point. Changing the marking conjugates this description; the outer tower and arithmetic action are independent of these choices.

Proof. Restriction to the open objects is injective by F.12. At those objects the inertia and relabelling conditions are exactly the definition of $\mathrm{Out}^{\#}_n$ in Harbater and Schneps (2000, definition in the introduction and Main Theorem). Their theorem identifies $\mathrm{Out}^{\#}_4$ with $\widehat{\mathrm{GT}}_0$ and $\mathrm{Out}^{\#}_5$ with $\widehat{\mathrm{GT}}$; forgetting gives the standard injection from the latter to the former. Thus the five-point outer automorphism determines the compatible four-point one.

Conversely their $\widehat{\mathrm{GT}}$ action preserves inertia, labelled types, and forgetting homomorphisms. F.12 extends it uniquely, up to natural isomorphism, to both boundary categories. On a divisor d_{ij}, forgetting i or j restricts to the identity of its four-point factor after relabelling; its three-point factor is trivial. Compatibility with forgetting therefore identifies the induced action with the four-point action, giving clutching compatibility. At two-node strata the component factors are all three-point categories. F.12 already forces the crossing bimodules and their compositions.

The compatibility isomorphisms are coherent. A natural automorphism of the identity on $\widehat{C}_n$ has open component in $Z(\widehat{G}_n) = 1$; naturality along $j_s : s \to u$ and injectivity of postcomposition force every other component to be 1. For forgetting, surjectivity of the open homomorphism gives the same argument, since its image has centralizer $Z(\widehat{G}_4) = 1$. For clutching, the image is the full automorphism group of the target stratum, again centre-free or trivial. Thus compatibility isomorphisms in this two-level diagram are unique and their coherence equations hold. Relabellings are equivalences and satisfy the same uniqueness. Taking opposites preserves the result; F.2 compares the pro-object and continuous versions. □

Statement F.14 (Proved). The full genus-zero boundary tower and its arithmetic subgroup.

Let $\mathcal{T}^{\partial}_0$ contain all $n \geq 3$ with the same decorations and all forgetting and clutching functors. Restriction is an isomorphism

$$\pi_0 \operatorname{Aut}(\mathcal{T}^{\partial}_0) \xrightarrow{\sim} \pi_0 \operatorname{Aut}(\mathcal{T}^{\partial}_{4,5}) \simeq \widehat{\mathrm{GT}}. \tag{F.12}$$

F.10 gives the usual faithful arithmetic inclusion $G_{\mathbb{Q}} \hookrightarrow \widehat{\mathrm{GT}}$. Equating its image with all automorphisms is the open equality E.8. The categorical group thus identifies the early normalizer-type group E.7; the revised arithmetic-image group is the subgroup E.2.

Proof. The Harbater–Schneps Main Theorem and its stability corollaries identify $\mathrm{Out}_n^{\#}$ with $\widehat{\mathrm{GT}}$ for all $n \geq 5$, compatibly with forgetting. Their tower action supplies existence on open strata; F.12 extends it uniquely at every level. The group of a boundary stratum is $\prod_v \widehat{G}_{\mathrm{val}(v)}$. Each factor is recovered by forgetting all but its markings and one marking in every branch attached to it. This restricts to the corresponding projection. Forgetting compatibility therefore forces the product of factor actions and proves clutching compatibility. All these groups have trivial centres, including the trivial valence-three factors, so F.13 proves coherence. The extension from two levels is unique.

Arithmetic descent and the usual arithmetic action coincide on the four-point open stratum by fundamental-group comparison, hence everywhere by uniqueness. Faithfulness is F.10. E.8 is the additional surjectivity assertion; no profinite equality follows from a rational Lie-algebra calculation. □

6.5 F.5. Positive genus, isotropy, and the rational comparison

Statement F.15 (Proved). Completion for the elliptic types retains stack isotropy. For $(g, n) = (1, 1), (1, 2)$ take Jansen's labelled stable-curve category in exit variance, with a finite skeleton indexed by stable labelled dual graphs. Its strict completion F.2 is a category in profinite sets, faithful on discrete Hom-sets, and equivalent to the profinite-multicurve transporter category for $\widehat{P}_{g,n}$. At σ,

$$1 \longrightarrow \prod_v \widehat{P}_{g_v,n_v} \longrightarrow \mathrm{Aut}_{\widehat{C}_{g,n}}(\sigma) \longrightarrow A_\sigma \longrightarrow 1, \tag{F.13}$$

where A_σ is the finite image of the discrete stabilizer on the dual graph and half-edges, fixing the original marked legs. No splitting is chosen. These categories carry F.9–F.10's finite constructible-sheaf comparison and outer Galois action, now for moduli *stacks*.

Proof. Jansen's Corollary 6.6 and Construction 3.18 apply in all genera, retaining graph automorphisms, and give $\mathrm{Aut}(\sigma) = H_\sigma/\Delta_\sigma$. The finite-index subgroup of H_σ fixing components and half-edges has the cutting extension by Δ_σ with quotient $\prod_v P_{g_v,n_v}$. Full and procongruence completions agree in genus one and its genus-zero components. Boggi's stabilizer theorem and F.6's short-five-lemma argument identify this subgroup's completion with its closure. Passing through its finite-index extension does the same for H_σ and gives (F.13). Multitwist independence and conjugacy separability give closedness of H_σ, Δ_σ.

Here a transporter may be a finite union of cosets $H_{\sigma_t} r$, since different submulticurves of σ_s may have the type of σ_t. Closedness of H_{σ_t} supplies a finite quotient separating these finitely many cosets. Refine to it before the kernel argument in F.7. Equality modulo $V\Delta_{\sigma_s}$ then forces the same transporter component, since Δ_{σ_s} fixes all submulticurves. Hence $g'd^{-1}g^{-1} \in H_{\sigma_t}$ and the cofinality proof applies. This proves the transporter description and faithfulness, retaining graph symmetries.

Take an étaleatlas and its simplicial scheme of overlaps. Finite constructible sheaves satisfy descent. The analytic/étalecomparison on each overlap commutes with all face and degeneracy maps; its descent limit gives the stack comparison. This agrees with Barwick et al. (2020, Proposition 13.9.17 and Corollary 13.9.19). Galois pullback on the descent data supplies the coherent action and retains finite stabilizers. □

For $(1,1)$ the open group is $\widehat{\mathrm{SL}_2(\mathbb{Z})}$, with central involution. Its boundary graph has one rational vertex, a labelled leg and a loop: exchanging the half-edges gives $A_\sigma = C_2$, while $P_{0,3} = 1$. The boundary object's group is therefore C_2. For $(1,2)$ a rational tail carrying both markings leaves an elliptic component with one node, hence a factor $\widehat{P}_{1,1}$; a nonseparating node instead has a four-point rational normalization and an exchange of node branches. Formula (F.13) retains these stabilizers, including their 2- and 3-primary information.

Statement F.16 (Open). The precise remaining positive-genus automorphism comparison.

We have not established that the automorphism group of the two elliptic boundary categories, forgetting, and all boundary clutchings is $\widehat{\mathrm{GT}}$. The precise comparison asks whether restriction from the decorated procongruence outer tower E.13 to this diagram is an isomorphism, with identical oriented inertia and relabelling data. Injectivity requires detection at these levels; surjectivity requires extension of every such pair to the full decorated tower. We do not claim that existence of $\widehat{\mathrm{GT}}$ actions in positive genus remains open: E.12–E.13 give current results and their exact morphism collections.

Where our argument stops. Formal completion and descent do not compute this automorphism group. Since $\widehat{P}_{1,1}$ has centre C_2, F.13's uniqueness-of-coherence argument needs additional work. Elliptic presentations also include nonseparating inertia and handle relations, whereas the genus-zero proof uses labelled splits and separating twists. E.10–E.11 concern specified open groups, and do not alone settle coherence and extension across all boundary bimodules. One must compare the precise decorated towers, retaining the central involution. The smallest diagram not settled here is exactly $(1,1)$–$(1,2)$. The label *Open* records an unresolved comparison in this audit, not an assertion that no theorem exists elsewhere. □

Statement F.17 (Known). The rational operadic analogue and its exact limitation.

Over a characteristic-zero field k, define the rational analogue to be the parenthesized braid groupoids completed relative to their finite permutation quotients, with pure braid groups replaced by their Malcev prounipotent completions. Retain operadic composition, strand deletion and doubling, the braid symmetry, and the distinguished oriented inertia. Their structure-preserving automorphism group is $\mathrm{GT}(k)$. In the parenthesized chord-diagram realization it is $\mathrm{GRT}(k)$; an associator compares the realizations. The four- and five-point constraints are the symmetry equations and pentagon, and the identity-multiplier Lie algebra is the weight completion of $W = \mathfrak{grt}_1$. The exact sources are Drinfeld (1990, §§5–6, Propositions 5.1 and 5.5) and Bar-Natan (1998, Proposition 3.4, Theorems 1–3, Corollary 4.1).

The corollary supplied by the cited formulation. The algebraic completion is universal for homomorphisms from the pure groups to unipotent algebraic groups, while the discrete objects and finite permutation data are retained. The cited theorem computes automorphisms preserving the indicated operations, and identifies their equations on the two lowest nontrivial levels. It is the operadic, algebraically enriched analogue of the decorated tower automorphism calculation F.13. The change from braids to infinitesimal braids depends on an associator and is not canonical. Applying E.19–E.20 gives its motivic corollary: the mixed Tate group injects, and its equality with the full rational automorphism group follows conditionally from the all-weight upper bound in family 008. That last assertion remains explicitly *claimed (OpenAI, 2026, unrefereed)* with the audit in E.26–E.28.

This does not prove that termwise Malcev completion of every Hom-set in the boundary category produces those parenthesized groupoids. A boundary quotient G/Δ_σ is generally a homogeneous space, not a torsor for G, and the exit category has noninvertible smoothing arrows. The parenthesized braid categories are groupoids. For instance the arrow from a

boundary object to the open object of C_4 has no reverse arrow; completing its groups does not create one. Thus those bare categories cannot be identified. A further comparison of their *structure-preserving automorphism groups* would require an algebraically enriched boundary construction and a reconstruction proof analogous to F.12; it is not imported silently from the groupoid theorem. This is the exact scope of the rational corollary supplied here. Locally constant coefficients in $D(\mathbb{Q})$ do not by themselves implement Malcev completion, and neither rational construction yields $G_{\mathbb{Q}} = \widehat{\mathrm{GT}}$. □

Statement F.18 (Open). A stronger higher-completion comparison is not assumed. One can also ask whether the nerve of the strict finite-category pro-object F.7, as a pro-object of stratified spaces, equals the full profinite stratified ∞-category $\mathrm{Gal}(X_{n,\mathbb{C}}/P_n)$. This requires comparison for all finite homotopy types and their constructible gluing, beyond finite sets. We do not establish it here. The first nontrivial test is $n = 4$, whose boundary gluing retains a two-dimensional unstratified homotopy class even though its stratum groups are free.

Exact limitation. BGH's Corollary 12.6.7 identifies the right-hand object with profinite stratified completion of the *nerve* of C_n. F.2 instead takes strict quotient categories before nerves. Interchanging them requires cofinality or goodness for representations in all π-finite spaces. F.9's finite representables prove only the 0-truncated coefficient comparison. All automorphism statements F.12–F.14 concern the strict completion defined here, whose finite-set comparison is proved. We leave the stronger statement unresolved without asserting a counterexample. No unchecked contractibility assertion from STRATA F.10 is used. □

A Verification protocol and limits of the deliverable

The accompanying register has one row for every numbered statement. Its status attaches to the exact assertion, including its hypotheses. In particular a proved conditional implication does not certify its antecedent. A bibliography entry marked *verified* means that the original source was opened and the cited passage located; it does not mean that every theorem in that source has been re-proved. The report lists the portions actually read and the depth of verification. Mathematical support for a doubtful transcription is recorded separately from visual evidence, and neither is described as a final editorial reading.

A.1 The three attached 2026 proof families

Family	Exact use	Boundary of this audit
008	E.19–E.28; F.17	We checked the dimension argument conditional on the all-weight upper bound, read the mod-two lattice and deletion reductions, and supplied a bounded integral computation. The all-weight upper bound is not certified by that computation. The attached Lean comparator contains a placeholder; it is not a kernel-checked proof. Subsequent October preprints are recorded, with their own verification scope.
019	D.7–D.10; E.17	We followed the deduction from the completion claim to the local section statement, and the conditional global and modular consequences. The patching, analytic convergence and all-valuation uniformization inputs are not promoted to independently verified theorems. Affine and moduli-stack sections are not covered by the proper-curve theorem.
031	D.11–D.13	We prove preservation of the geometric kernel independently, then deduce the absolute curve Hom theorem conditionally from the claimed Galois open-Hom theorem. We separate that implication from the new local-to-global argument in the supplied preprint.

The report gives the precise first unresolved step for each imported claim. We found no verified counterexample to the main theorem of one of these three manuscripts. Failure to complete its proof audit is not a refutation. Conversely, publication of a later proof treatment is reported as evidence of a subsequent claim, without representing an unread argument as checked.

A.2 Computations and proof boundaries

The programs in `code/` use only the Python standard library. Their saved outputs include the finite category and transition checks for F, the stable-tree counts and forgetting map, the finite inertia-kernel calculations, the permutation and finite-linear-group counterexamples of B/C, the finite section/normalizer checks of D, and the bounded integral deletion calculation of E. The infinite assertions are proved in the text or explicitly imported from the specified primary theorem. A finite enumeration never substitutes for density, convergence, an all-weight inequality, or a statement about all open subgroups.

The genus-zero categorical theorem concerns actual pro-objects of finite categories and their continuous materializations, with the stated decorations and natural *isomorphisms*. It is not a theorem about an unspecified inverse limit in a category obtained by inverting all natural

transformations. Its comparison with finite constructible sheaves includes the stalk functors. F.18 records the additional higher-completion comparison which the finite-set argument alone does not prove.

A.3 What remains unresolved

The profinite arithmetic equality $G_{\mathbb{Q}} = \widehat{\mathrm{GT}}$ remains open. The higher-genus congruence kernel remains in B.3. Other exact unrestricted comparisons left unresolved in this audit are B.7, B.14, C.8, D.23, E.14, F.16 and F.18. Their wording distinguishes an established open problem from a comparison for which our search and argument did not supply a theorem. The 2026-dependent claims are separated in D.7, D.13 and E.26. All these statements remain in the results register; no question has been silently dropped after a reduction to a special case.

We claim no priority for the arguments, counterexamples, constructions, or reformulations in this paper. The research report records the search scope before any possible inference of novelty. This paper, its code and its source audit are AI-produced by GPT-6 Astra (OpenAI, ChatGPT Pro mode).

References

Amini, O., Baker, M., Brugallé, E., and Rabinoff, J. (2015). Lifting harmonic morphisms I: Metrized complexes and Berkovich skeleta. *Research in the Mathematical Sciences*, 2(7). verified: primary text opened; Definitions 2.21 and 4.31 and Theorem 7.7 checked, including tameness, branch points, and prescribed local covers Source: https://arxiv.org/html/1303.4812v3.

Baker, M. and Rabinoff, J. (2015). The skeleton of the Jacobian, the Jacobian of the skeleton, and lifting meromorphic functions from tropical to algebraic curves. *International Mathematics Research Notices*, 2015(16):7436–7472. verified: primary preprint opened; Proposition 6.1, Corollary 6.6, and Remark 6.7 checked. Divisor lifting does not by itself impose every simultaneous analytic requirement of 019 Source: https://doi.org/10.1093/imrn/rnu168.

Bar-Natan, D. (1998). On associators and the Grothendieck–Teichmüller group, I. *Selecta Mathematica. New Series*, 4(2):183–212. verified: Primary author text opened: Proposition 3.4, Theorems 1–3 and Corollary 4.1; the author file carries minor later updates. Source: https://doi.org/10.1007/s000290050029.

Barwick, C., Glasman, S., and Haine, P. J. (2020). Exodromy. arXiv:1807.03281, book draft of 21 August 2020. verified: supplied source and original PDF opened; Proposition 12.6.4, Corollaries 12.6.6–12.6.7, and Section 13.9 checked Source: https://www.maths.ed.ac.uk/~cbarwick/papers/exodromy_book.pdf.

Belyi, G. V. (1980). On Galois extensions of a maximal cyclotomic field. *Mathematics of the USSR-Izvestiya*, 14(2):247–256. verified: attached English primary translation opened; Theorem 4 on p.255 and the unnumbered corollary on p.256 checked. Russian original: Izvestiya Akademii Nauk SSSR, Seriya Matematicheskaya 43(2), 267–276 (1979) Additional audit: English translation of the 1979 original. Attached primary text checked: section 4, Theorem 4 and the unnumbered corollary on page 256. Source: https://doi.org/10.1070/IM1980v014n02ABEH001106.

Boggi, M. (2009). The congruence subgroup property for the hyperelliptic modular group: The open surface case. *Hiroshima Mathematical Journal*, 39:351–362. Verified: attached arXiv v6 opened; full profinite Birman sequence in Section 2, Proposition 2.6, Theorems 2.5, 2.8 and 3.5 checked. Main theorem concerns punctured surfaces; the closed case is not imported from this paper Source: https://arxiv.org/pdf/0803.3841.

Boggi, M. (2020). Congruence topologies on the mapping class group. *Journal of Algebra*, 546:518–552. Verified: primary PDF and publisher metadata opened. Theorem 1.4(ii), Proposition 2.4, Theorems 3.5–3.6, 4.2, 4.4, Corollaries 4.3, 4.5, 4.7 and Remark 4.8 checked. The date printed on the retrieved arXiv PDF is July 12, 2021; the journal publication is 2020 Additional audit: verified: original PDF opened; Theorem 1.4, Definitions 3.3–3.4, Theorems 3.5–3.6 and 4.2, Corollaries 4.3 and 4.7 checked Source: https://doi.org/10.1016/j.jalgebra.2019.11.004.

Boggi, M. (2026a). Automorphisms of profinite mapping class groups. arXiv:2011.15075v6, 22 April 2026. Version 6, 22 April 2026. Primary full text opened; statements, definitions and the cited portions checked, not a complete independent proof audit. Theorem A, Proposition 3.9, section 2.8, Conjecture 6.21, Theorem 6.23 and Corollary 6.24. Additional audit: verified: original PDF and version record opened; Proposition 3.5 and the exact tower statements cited in Part E checked. Preprint Source: https://doi.org/10.48550/arXiv.2011.15075.

Boggi, M. (2026b). A congruence subgroup property for symmetric mapping class groups. *São Paulo Journal of Mathematical Sciences*, 20(35). Verified: primary publisher full text

opened, published September 23, 2026. Introduction and Theorem A/Corollary B checked for current congruence status and scope. General high-genus CSP is not claimed Source: https://doi.org/10.1007/s40863-026-00558-y.

Boggi, M. and Funar, L. (2023). Automorphisms of the procongruence curve and pants complexes. *Journal of Topology*, 16:936–989. verified: original PDF opened; Theorem 4.11, Corollary 4.12, Proposition 7.2 checked Source: https://doi.org/10.1112/topo.12306.

Boggi, M. and Zalesskii, P. (2019). A restricted Magnus property for profinite surface groups. *Transactions of the American Mathematical Society*, 371(1):729–753. Verified: primary arXiv v2 opened. Definition 1.3, Lemma 3.1(ii), Proposition 3.6 and Theorem 3.7 checked. Proposition 3.6 and its proof give the normalizer of every nonzero profinite power of an algebraically simple element. March 27, 2021 printed on the retrieved PDF is not its publication year Source: https://doi.org/10.1090/tran/7311.

Bresciani, G. and Vistoli, A. (2020). An elementary approach to Stix's proof of the real section conjecture. arXiv:2012.06278v2, December 18, 2020. verified: complete primary note opened; introductory Theorem identifies sections with real connected components for the curves used here Source: https://arxiv.org/pdf/2012.06278.

Brown, F. (2012). Mixed Tate motives over $\mathbb{Z}$. *Annals of Mathematics*, 175(2):949–976. verified: Attached primary text checked: Theorem 1.1, the stated consequence Conjecture 1, and the free motivic Lie algebra in section 1.1. Source: https://doi.org/10.4007/annals.2012.175.2.10.

Brown, F. (2021). Depth-graded motivic multiple zeta values. *Compositio Mathematica*, 157(3):529–572. verified: Published primary full text opened, section 1.4.2: explicit motivic Lie inclusion into the Grothendieck–Teichmuller Lie algebra. Source: https://doi.org/10.1112/S0010437X20007654.

Collas, B. (2026). Anabelian perspectives in Galois–Teichmüller theory. verified: Version 2, 5 November 2026. Attached survey read, especially sections 2–4; used to locate original results, which were opened separately. Source: https://arxiv.org/abs/2603.02848v2.

Drinfeld, V. G. (1990). On quasitriangular quasi-Hopf algebras and on a group that is closely connected with $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$. *Algebra i Analiz*, 2(4):149–181. verified: English translation: Leningrad Mathematical Journal, 2(4), 829–860 (1991). Attached original checked for sections 4–6, especially Propositions 5.1 and 5.5; categorical conventions cross-checked with Bar-Natan (1998).

Farb, B. and Margalit, D. (2012). *A Primer on Mapping Class Groups*, volume 49 of *Princeton Mathematical Series*. Princeton University Press. verified: book PDF opened; Theorems 4.6, 8.1, 8.8 checked Source: https://pagine.dm.unipi.it/~a019210/Farb%20Magalit_Primer%20on%20Teichmuller%20theory.pdf.

Hain, R. and Matsumoto, M. (2003). Weighted completion of Galois groups and Galois actions on the fundamental group of $\mathbb{P}^1 - \{0, 1, \infty\}$. *Compositio Mathematica*, 139(2):119–167. verified: Attached primary preprint checked: Theorems 7.3, 8.4, 9.8, Lemma 8.5, Remark 8.6, Theorem A.6 and Corollary A.10. Publication metadata checked separately. Source: https://doi.org/10.1023/B:COMP.0000005077.42732.93.

Harari, D. and Stix, J. (2012). Descent obstruction and fundamental exact sequence. In Stix, J., editor, *The Arithmetic of Fundamental Groups—PIA 2010*, volume 2 of *Contributions in Mathematical and Computational Sciences*, pages 147–166. Springer. verified: author preprint of February 11, 2011 opened; Theorem 2.1 and Remark 2.2(1), in that preprint's numbering,

checked with U=1 and S empty Source: https://www.imo.universite-paris-saclay.fr/~david.harari/articles/finidesc.pdf.

Harbater, D. and Schneps, L. (2000). Fundamental groups of moduli and the Grothendieck–Teichmüller group. *Transactions of the American Mathematical Society*, 352(7):3117–3148. Attached primary text checked: Main Theorem (a),(b), Theorems 2–3, Propositions 7–8, and the tower conventions in section 0.3. Additional audit: verified: supplied original PDF opened; definitions and Main Theorem on pp. 3117–3119 checked Source: https://doi.org/10.1090/S0002-9947-00-02347-3.

Hatcher, A., Lochak, P., and Schneps, L. (2000). On the Teichmüller tower of mapping class groups. *Journal für die reine und angewandte Mathematik*, 521:1–24. verified: Attached primary PDF rendered and visually checked on pages 3–6: Theorems A–C, with internal references Lemma 7 and Theorems 4–5 of section 3. Source: https://webusers.imj-prg.fr/~pierre.lochak/textes/crelle.pdf.

Hoshi, Y. (2025). Homomorphisms of global solvably closed Galois groups compatible with cyclotomic characters. *Tohoku Mathematical Journal*, 77(1):17–32. verified: revised primary preprint opened; Theorem 3.4 checked. Cyclotomic compatibility is an explicit hypothesis Source: https://doi.org/10.2748/tmj.20230414.

Hoshi, Y., Minamide, A., and Mochizuki, S. (2022). Group-theoreticity of numerical invariants and distinguished subgroups of configuration space groups. *Kodai Mathematical Journal*, 45(3):295–348. verified: Attached primary text checked: Corollaries 2.6 and 2.8 and Definition 2.7. Source: https://doi.org/10.2996/kmj45301.

Hoshi, Y. and Mochizuki, S. (2011). On the combinatorial anabelian geometry of nodally nondegenerate outer representations. *Hiroshima Mathematical Journal*, 41(3):275–342. verified: attached primary full text opened; Corollaries 6.3–6.5 on PDF pages 64–67 checked, including the orientation-preserving geometric monodromy image and the distinction between geometric and total injectivity Additional audit: Attached primary text checked in Part D: Corollaries 6.4 and 6.5; Part E uses the explicit argument D.17. Source: https://doi.org/10.32917/hmj/1323700038.

Hoshi, Y. and Mochizuki, S. (2012). Topics surrounding the combinatorial anabelian geometry of hyperbolic curves I: Inertia groups and profinite Dehn twists. In Nakamura, H., Pop, F., Schneps, L., and Tamagawa, A., editors, *Galois–Teichmüller Theory and Arithmetic Geometry*, volume 63 of *Advanced Studies in Pure Mathematics*, pages 659–811. Mathematical Society of Japan. Verified: attached primary paper and primary publication metadata opened. Theorem A and Corollary 1.9 checked; no assertion that arbitrary single-surface automorphisms are FC-admissible is imported Source: https://doi.org/10.2969/aspm/06310659.

Hoshi, Y., Mochizuki, S., and Tsujimura, S. (2025). Combinatorial construction of the absolute Galois group of the field of rational numbers. *Journal of Mathematical Sciences, the University of Tokyo*, 32(1):1–125. verified: Primary text dated 14 March 2025 opened: Theorems A–C and Definitions 3.3 and 5.12 checked for the precise scope used; no complete independent proof audit claimed. Source: https://www.kurims.kyoto-u.ac.jp/~motizuki/Combinatorial%20absolute%20Galois%20groups.pdf.

Hoyois, M. (2018). Higher Galois theory. *Journal of Pure and Applied Algebra*, 222(7):1859–1877. verified: original PDF opened; split representations before Definition 2.10, Theorem 3.13 and Corollary 3.20 checked Source: https://doi.org/10.1016/j.jpaa.2017.08.010.

Jansen, M. Ø. (2023). Moduli stack of stable curves from a stratified homotopy viewpoint. arXiv:2308.09551v2. verified: supplied original PDF opened; Construction 3.18, Proposition

3.23, Corollary 6.6, Theorem 6.7 checked. We use the stable condition stated in the definitions Source: https://doi.org/10.48550/arXiv.2308.09551.

Kerckhoff, S. P. (1983). The Nielsen realization problem. *Annals of Mathematics*, 117(2):235–265. Verified: journal metadata and primary paper scan opened. Theorems 5 and 7 and the general Fuchsian discussion on pages 257–258 checked; Theorem 7 is the allowable finite-extension result used here Source: https://doi.org/10.2307/2007076.

Koenigsmann, J. (2005). On the 'section conjecture' in anabelian geometry. *Journal für die reine und angewandte Mathematik*, 588:221–235. verified: primary preprint opened; Observation 1.3, Definition 1.4, Proposition 2.4(2), and its henselianity conclusion checked Source: https://doi.org/10.1515/crll.2005.2005.588.221.

Lochak, P., Nakamura, H., and Schneps, L. (2026). The Grothendieck–Teichmüller group $\widehat{\mathrm{GT}}$ acts on the genus g mapping class group with 0 or 1 marked point. verified: Version 1, 12 February 2026. Primary text opened: Theorems 1.1 and 2.1, including full-twist formulas and the distinction from the Lego compatibility rules. Source: https://arxiv.org/html/2602.12462v1.

Matsumoto, M. (1996). Galois representations on profinite braid groups on curves. *Journal für die reine und angewandte Mathematik*, 474:169–219. verified: attached primary scan opened; Theorem 2.1 and Conjecture 2.1 on printed p.191 explicitly checked: number-field curves with nonabelian geometric fundamental group, with the theorem proving the affine case.

Mazur, B. (1977). Modular curves and the Eisenstein ideal. *Publications Mathématiques de l'IHÉS*, 47:33–186. verified: attached primary paper opened; Introduction Theorem (7), p.35, states that positive-genus $X_1(N)$ has only cuspidal rational points; Theorems (7') and (8) distinguish cyclic torsion orders and the full torsion classification Additional audit: Primary introduction checked in Part D: Theorem (7), printed page 35, and Theorems (7') and (8). Source: https://doi.org/10.1007/BF02684339.

Minamide, A. and Nakamura, H. (2022). The automorphism groups of the profinite braid groups. *American Journal of Mathematics*, 144(5):1159–1176. verified: Primary full text opened and checked: Theorem B, Theorem 4.3, Corollary C, and Proposition 3.2(iii). Source: https://arxiv.org/html/1904.06749v2.

Mochizuki, S. (1999). The local pro-p anabelian geometry of curves. *Inventiones Mathematicae*, 138:319–423. verified: attached primary full text opened; Theorem A, Theorem 16.5 and following Remark, Theorem C and Theorem 19.1 checked.

Mochizuki, S. (2006). Global solvably closed anabelian geometry. *Mathematical Journal of Okayama University*, 48:57–71. verified: published primary text opened. Published Theorem 2.4, Corollary 2.2, Lemma 3.2, Proposition 3.3, Theorem 3.1 correspond to author-preprint 1.5, 1.3, 2.2, 2.4, 2.1. Printed pages are 57–71; repository metadata instead lists 57–72 Source: https://doi.org/10.18926/mjou/33348.

Mochizuki, S. (2010). On the combinatorial cuspidalization of hyperbolic curves. *Osaka Journal of Mathematics*, 47(3):651–715. Verified: primary author PDF and university publisher version opened. Definition 1.1(ii), Theorem 4.1, Corollary 4.2 and their scope checked. Later bounds in Hoshi–Mochizuki Theorem 6.1 are used in C.9 Source: https://doi.org/10.18910/12138.

Mochizuki, S. and Tsujimura, S. (2024). Resolution of nonsingularities, point-theoreticity, and metric-admissibility for p-adic hyperbolic curves. RIMS Preprint 1974, revised version cited by proof pack 019. verified: primary 108-page version dated June 16, 2023 opened; Theorem A and Proposition 2.4(iv) checked. The attached 019 bibliography dates its cited revision

2024; that version difference is retained Source: https://www.kurims.kyoto-u.ac.jp/preprint/file/RIMS1974.pdf.

Nakamura, H. (1994). Galois rigidity of pure sphere braid groups and profinite calculus. *Journal of Mathematical Sciences, the University of Tokyo*, 1:71–136. verified: primary full text opened; Theorem (2.1.1), Lemma (2.1.2), Corollaries (1.3.3)–(1.3.4) checked Additional audit: Attached primary text checked: Theorems A and B, including the odd-prime and Galois-compatibility hypotheses. Source: https://www.ms.u-tokyo.ac.jp/journal/pdf/jms010103.pdf.

Nakamura, H. and Schneps, L. (2000). On a subgroup of the Grothendieck–Teichmüller group acting on the tower of profinite Teichmüller modular groups. *Inventiones Mathematicae*, 141:503–560. verified: Attached primary text checked: Definition 1.1 and Theorems 1.1–1.4. Source: https://webusers.imj-prg.fr/~leila.schneps/newGT.pdf.

OpenAI (2026a). The Deligne–Drinfeld conjecture. verified: 23 September 2026. User-supplied family 008; unrefereed manuscript. Sections 2–5 read and partially audited in E.26. Attached Lean file is a statement comparator with a sorry placeholder, not a checked proof.

OpenAI (2026b). Open homomorphisms of global solvably closed Galois groups. Attached unrefereed preprint, October 5, 2026, proof pack 031. verified as a source: complete Sections 1–6 read, including all new lemmas and propositions; cited primary local and cyclotomic theorems opened. This label does not certify the new proof. The uniform arithmetic separation estimates needed for the unrestricted Theorem 1.1 have not been independently established in this audit.

OpenAI (2026c). The p-adic section conjecture. Attached unrefereed preprint, October 6, 2026, proof pack 019. verified as a source: complete main preprint read and its used external results checked. This label does not certify the new proof. Theorem 1.2's uniform specified-embedding completion assertion remains unverified in this audit; the implications in Sections 7 and 8.1–8.2 are proved conditionally in D.8–D.10. Corollary 8.4 is not used.

Paris, L. (2009). Residual p properties of mapping class groups and surface groups. *Transactions of the American Mathematical Society*, 361(5):2487–2507. Verified: primary arXiv text opened. Definition of Property A and its attribution to Grossman in Section 1, Theorem 3.2 for free groups, and Theorem 4.1 (=Theorem 1.7) for closed surface groups checked. Grossman's original proof is not claimed to have been checked independently Source: https://doi.org/10.1090/S0002-9947-08-04573-X.

Pop, F. (2018). Finite tripod variants of I/OM on Ihara's question/Oda–Matsumoto conjecture. verified: Attached author manuscript dated 24 December 2018. Definition/Remark 2.2, Definition 2.5, Theorems 2.6–2.7, and the application on pages 3–4 checked. The birational localization objects are essential.

Pop, F. and Stix, J. (2017). Arithmetic in the fundamental group of a p-adic curve: On the p-adic section conjecture for curves. *Journal für die reine und angewandte Mathematik*, 725:1–40. verified: author preprint of August 11, 2011 opened; Theorem 26 and Section 6.1 checked. The numbering used in D refers to this primary preprint Source: https://doi.org/10.1515/crelle-2014-0077.

Stix, J. (2015). On the birational section conjecture with local conditions. *Inventiones Mathematicae*, 199(1):239–265. verified: author preprint of February 27, 2014 opened; Corollary 5 and the finite-support hypothesis checked Source: https://doi.org/10.1007/s00222-014-0514-3.

Stoll, M. (2007). Finite descent obstructions and rational points on curves. *Algebra & Number Theory*, 1(4):349–391. verified: revised author primary paper opened; Corollary 8.8 and the modular-curve application checked together with the 2017 erratum Source: https://doi.org/10.2140/ant.2007.1.349.

Stoll, M. (2017). Errata: Finite descent obstructions and rational points on curves. Online erratum, March 29, 2017. verified: complete three-page primary erratum opened; omitted complex factors, disconnected torsor correction, and correction concerning real places checked Source: https://msp.org/ant/2007/1-4/ant-v1-n4-x01-Errata-FiniteDescent.pdf.

The Stacks Project Authors (2026). The Stacks project: Zariski's main theorem. Verified: Lemmas 37.43.2 and 37.43.3, Tags 02LR and 05K0, opened November 11, 2026; used for the normalization/open-immersion basis of the complete curve's full small etale site Source: https://stacks.math.columbia.edu/tag/05K0.

Uchida, K. (1979). Isomorphisms of Galois groups of solvably closed Galois extensions. *Tohoku Mathematical Journal*, 31(3):359–362. verified: all four pages of the primary paper opened; main theorem and Lemma 3 checked Source: https://www.jstage.jst.go.jp/article/tmj1949/31/3/31_3_359/_pdf/-char/en.

Uchida, K. (1981). Homomorphisms of Galois groups of solvably closed Galois extensions. *Journal of the Mathematical Society of Japan*, 33(4):595–604. verified: primary paper opened; local three-case alternative, Proposition 1, Proposition 2, Theorem 1 and Theorem 2 checked. The unrestricted conjecture is on p.595 Source: https://doi.org/10.2969/jmsj/03340595.

Willwacher, T. (2026). The Deligne–Drinfeld conjecture after OpenAI. verified: Version 4, 28 October 2026. Primary preprint opened; Theorem 1.1 and principal reductions read. Complete all-weight proof not independently certified in this audit. Source: https://arxiv.org/html/2610.15592v4.